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Quantifying Gender Citation Imbalance in Computer Science Across Citation-Derived Network Representations

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Abstract

Citation-based indicators play an important role in scholarly evaluation, yet previous research has shown that citation recognition can differ systematically across gender groups. This thesis investigates whether gender citation imbalance in Computer Science can be robustly quantified when citation opportunity is defined using multiple structural, topical, temporal, and network-based representations.

The analysis uses a large bibliographic corpus containing 1.54 million Computer Science papers, of which 1.08 million have known inferred gender for both the first and last authors. Papers are classified into MM, MW, WM, and WW first-last author gender categories. Citation imbalance is measured using a common observed/expected framework across direct citation baselines, taxonomy-based topical relevance specifications, bibliographic-coupling and co-citation neighborhoods, generalized linear models, citation-lag analyses, and randomized reference models. Results are reported for both citations to Computer Science from the broader corpus and citations occurring entirely within Computer Science.

The estimated magnitude of citation imbalance varies substantially across baseline definitions. Temporal conditioning substantially reduces the aggregate differences, while topical, venue, network, and prior-visibility adjustments further modify the resulting expected-citation ratios. Nevertheless, several directional patterns remain stable. WW papers receive fewer citations than expected under all 19 principal specifications in both citation slices, while MM papers are above expectation under nearly all specifications. MW and WM papers are generally below expectation but are more sensitive to the choice of null model. Bibliographic-coupling baselines preserve comparatively larger residual deviations, whereas co-citation-based specifications produce estimates closer to parity, although co-citation is substantially more selective in coverage and is inherently future-informed.

The findings show that gender citation imbalance in Computer Science can be quantified robustly in directional terms, but not as a single baseline-independent effect size. More broadly, the thesis demonstrates that conclusions about citation disparity depend on how expected citation opportunity and scholarly comparability are defined. The results are descriptive rather than causal and remain conditional on name-based gender inference, metadata coverage, and the observed corpus-internal citation network.

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1. Introduction

1.1 Motivation

Gender inequality in academia extends beyond differences in participation and publication output. Previous research has documented disparities in authorship, scholarly visibility, and citation impact across disciplines and countries (1; 2). Citations are particularly relevant in this context because they are widely used as indicators of scholarly influence and contribute to research evaluation, reputation, and career advancement (3; 4). If recognition through citations is distributed unevenly across gender groups, citation-based indicators may therefore reproduce or amplify inequalities already present in the academic system.

A substantial literature has documented gender differences in citation outcomes. Studies in international relations and political science have reported lower citation rates for research authored by women (5; 6), while large-scale analyses have identified gender differences in scientific authorship and citation-related outcomes across fields (1; 7). Similar patterns have also been observed within individual scientific disciplines. In astronomy, Caplar et al. (8) found a citation disadvantage associated with female authorship after accounting for several observable paper characteristics.

More recent studies illustrate why citation imbalance should also be examined within the specific structural context of a discipline. In neuroscience, Dworkin et al. (9) found that papers with women as first and last authors were underrepresented in reference lists relative to expected citation rates, while papers with men in both positions were overrepresented. They further showed that citation patterns varied with characteristics of the citing authors and publication context. In contemporary physics, Teich et al. (10) similarly reported under-citation of woman-authored papers and over-citation of man-authored papers relative to expected rates, with the magnitude of the imbalance varying according to who was citing, where the citation occurred, and the citing authors' proximity to the cited research. Together, these studies show that gender citation imbalance is not necessarily uniform across scholarly contexts and motivate discipline-specific analyses that account explicitly for the structure of citation opportunity.

Computer Science is a particularly relevant setting for such an analysis. The field has experienced rapid growth in publication volume and substantial changes in gender composition over time, while also containing heterogeneous subfields with different publication and citation practices. Its publication system additionally gives a prominent role to conference proceedings alongside journals, creating substantial variation in venue structure and visibility. Previous work has identified gendered citation imbalance in Computer Science, first in conference publications (11) and, more recently, across both conferences and journals (12). The present thesis builds on this evidence by asking whether the same broad patterns remain visible when citation opportunity is defined using multiple structural, topical, temporal, and citation-derived network representations, rather than through direct-citation reference models alone.

A further motivation arises from how relationships between scientific papers are represented. Gender-citation studies generally evaluate citation outcomes using direct citation links, where one paper explicitly cites another. Bibliometric research, however, distinguishes direct citation from other citation-derived representations of scholarly relatedness. Bibliographic coupling connects papers that cite overlapping prior work, whereas co-citation connects papers that are cited together by subsequent literature (13; 14). These representations encode different aspects of the scholarly relationship between papers and should therefore not be assumed to provide interchangeable comparison structures.

Comparative bibliometric studies support this distinction. Boyack and Klavans (15) compared direct citation, bibliographic coupling, and co-citation as approaches to representing research fronts and found substantive differences in the structures produced by the three methods. Kleminski et al. (16) examined the same citation-network representations in Computer Science and showed that they provide complementary information for identifying scientific topics. The distinction between these representations is therefore already established in the bibliometric literature; the

contribution of this thesis is not to introduce them as new network constructions.

Instead, this thesis applies these established representations to the measurement of gender citation imbalance. Direct citation captures explicit citation reception, bibliographic coupling defines comparison neighborhoods through shared research foundations, and co-citation defines retrospective neighborhoods through later community association. These representations are combined with taxonomy-based topical relevance, citation-lag analysis, and multiple expected-citation baselines to examine whether conclusions about gender citation imbalance depend on how scholarly similarity and citation opportunity are defined.

The central motivation for this multi-representation approach is therefore robustness rather than the selection of one uniquely correct citation network. If similar observed-versus-expected patterns remain visible across different representations and expected-citation models, the conclusion is less dependent on one particular operationalization of citation opportunity. Conversely, if the estimated magnitude or direction changes across representations, that variation reveals which conclusions are sensitive to the structural assumptions embedded in the baseline. This provides a broader basis for quantifying gender citation imbalance in Computer Science than relying on direct citation counts alone.

1.2 Problem Statement

Despite the central role of citations in allocating scholarly credit, it remains unclear how to measure gender-based citation imbalance in a way that is both (i) appropriate for the structural complexity of Computer Science and (ii) robust to the choice of how citations are represented. Simple comparisons of raw citation counts are insufficient because citation outcomes are strongly shaped by factors such as publication time, venue stratification, subfield and topic composition, and citation-network position. Without accounting for these factors, apparent gender differences may be either overstated or masked.

Moreover, citations are not a single homogeneous mechanism. A direct citation edge captures one form of credit transfer, but other citation-derived relationships capture different scholarly processes. Co-citation patterns reflect how communities group papers into intellectual neighborhoods and can contribute to canon formation. Bibliographic coupling reflects shared research foundations through overlapping prior work. Citation-lag analysis indicates whether work persists in long-term scientific memory. Topic- or taxonomy-based relevance weighting can distinguish citations that are central to the citing paper’s intellectual context from weaker cross-area references. If gender imbalance differs across these representations, then conclusions drawn from a single network view may be incomplete or misleading.

The problem addressed in this thesis is therefore twofold: (1) to operationalize and quantify gender-based citation imbalance for Computer Science papers under appropriate controls and expected-citation baselines, and (2) to determine how sensitive the resulting imbalance estimates are to modeling citations through different citation-derived network representations. This requires a scalable methodology capable of constructing and analyzing large citation graphs, defining comparable imbalance metrics across representations, and evaluating robustness under realistic data limitations such as imperfect gender attribution, missing metadata, and uneven network coverage.

1.3 Research Questions

1.3.1 Main Research Question

Can gender citation imbalance in Computer Science be robustly quantified using multiple citation-derived network representations and topical similarity signals?

Scope. The empirical analysis focuses on papers classified as *Computer Science*. Cross-field citation links are retained where required to capture the broader observed citation context.

1.3.2 Subquestions

The following subquestions operationalize the main research question by testing whether gender citation imbalance can be measured consistently across data definitions, citation representations, and robustness specifications.

1. **Operationalization feasibility:** Can author gender be operationalized at the paper level using first and last author categories, and what proportion of the Computer Science corpus is excluded from known-gender analyses due to gender inference limitations?
2. **Direct citation evidence:** Can gender citation imbalance be identified in the direct citation setting after accounting for time, venue, and topical structure, both in citation reception and in descriptive citing-side patterns?
3. **Topical relevance sensitivity:** To what extent does the measurement of gender citation imbalance change when citations are restricted or weighted by topical relevance using the OpenAlex taxonomy, distinguishing core in-area citations from cross-area citations?
4. **Co-citation consistency:** Can gender citation imbalance also be identified within co-citation neighborhoods, which reflect later community association and may capture aspects of canon formation?
5. **Bibliographic coupling consistency:** Can gender citation imbalance also be identified within bibliographic-coupling neighborhoods, which represent shared research foundations and prior work?
6. **Long-memory consistency:** Can gender citation imbalance be identified across citation-lag buckets, indicating whether papers in different gender categories differ in short-term attention and longer-term scientific memory?
7. **Robustness across methods:** To what extent are the resulting estimates of gender citation imbalance stable across citation slices, network representations, expected-citation baselines, GLM specifications, and randomized robustness checks?

1.4 Contributions

This thesis makes the following contributions:

- **Multi-representation perspective on citation imbalance:** It formulates gender-based citation imbalance in Computer Science across multiple citation-derived representations, including direct citation, co-citation, bibliographic coupling, citation-lag structure, and taxonomy-based topical relevance.
- **Taxonomy-weighted citation relevance:** It uses the OpenAlex taxonomy to weight citation edges by topical proximity, distinguishing core in-area citations from cross-area citations without relying on text-based similarity.
- **Expected-citation baselines and comparison:** It designs and compares multiple expected-citation baselines, including time-, venue-, subfield-, network-neighborhood-, GLM-, and randomized-reference baselines, to quantify observed-versus-expected citation outcomes.

- **Scalable network construction pipeline:** It implements a computational pipeline suitable for large-scale bibliometric data, combining final filtered direct citation edges with sparse top- K bibliographic-coupling and co-citation neighborhoods.
- **Reproducible empirical study in Computer Science:** It provides a reproducible analysis workflow and derived artifacts that support auditing of preprocessing choices and replication of the reported results.

1.5 Scope and Definitions

1.5.1 Computer Science Scope Definition

The empirical focus of this thesis is the *Computer Science* literature. A paper is included in the Computer Science subset if its OpenAlex taxonomy assigns it to the field *Computer Science*. Descriptive analyses and imbalance estimates are reported for this Computer Science subset unless stated otherwise.

Citation relationships, however, are not always restricted to within-field links. To preserve citation context, the analysis distinguishes two main citation views: (i) All→CS and (ii) CS→CS. The All→CS slice is the main citation-reception setting, because it captures observed citations to Computer Science papers from the broader available corpus. The CS→CS slice is used as a stricter within-field specification.

Publication time is operationalized using decade bins derived from publication year. These bins are used to describe temporal composition, condition expected-citation baselines, and analyze citation-lag and long-memory patterns.

1.5.2 Gender Operationalization and Unknown Labels

This thesis operationalizes gender at the paper level using name-based inference for the first and last authors. The resulting labels are not observed self-identified gender; they are inferred first-last author gender categories used for large-scale citation analysis. The primary known-gender categories are *MM*, *MW*, *WM*, and *WW*.

Papers with unknown first or last author gender are retained in coverage diagnostics but excluded from the main known-gender observed/expected analyses. Unknown assignments are not assumed to be random: they may vary across language, region, naming convention, time period, and metadata quality. For this reason, unknown-gender coverage is reported descriptively and discussed as a limitation rather than treated as a separate substantive gender category.

Implementation details, including the probability thresholding rule and the construction of first-last author categories, are provided in Chapter 3, especially Section 3.3.

1.5.3 Citation Data and Filtering

Citations are represented as directed edges between papers in the corpus. An edge (i, j) means that paper i cites paper j . The analysis uses corpus-internal citation links, meaning that a reference contributes an edge only when the cited work is also present in the dataset.

The final analysis is based on a filtered citation graph. The filtering procedure removes duplicate citation edges, paper-identifier self-links, author-overlap self-citations, time-aware coauthor-neighborhood citations, and isolated papers after filtering. These filtering steps are part of the final preprocessing and network construction pipeline, not a separate before/after sensitivity analysis. The detailed construction of the final filtered edge set is defined in Chapter 4, especially Section 4.4.1.

1.6 Thesis Structure

The remainder of this thesis is organized as follows.

Chapter 2 reviews prior work on gender disparities in citations, citation-network analysis, topical similarity, expected-citation baselines, and methodological limitations of gender inference.

Chapter 3 describes the dataset, the definition of the Computer Science corpus, the gender attribution procedure, and preprocessing steps.

Chapter 4 presents the methodological framework, including final filtered network construction, observed/expected baselines, GLM expected-citation models, citation-lag analysis, and randomized robustness checks.

Chapter 5 provides exploratory data analysis of the Computer Science corpus, including metadata coverage, authorship, references, venues, taxonomy composition, citation structure, and network diagnostics.

Chapter 6 reports the main empirical findings across direct citation baselines, topical and temporal analyses, network-based baselines, GLM models, randomized robustness checks, and cross-baseline comparison.

Chapter 7 discusses the interpretation of the findings, comparison with related work, methodological implications, limitations, and future work.

Chapter 8 concludes by answering the research questions and summarizing the thesis contributions.

The accompanying analysis code and reproducibility materials are provided in the GitHub repository listed in Appendix A.1.

2. Background & Related Work

2.1 Gender Inequality and Citation Imbalance in Academia

Gender disparities in academia occur at multiple stages of scholarly production and recognition. They are visible not only in participation and authorship, but also in the distribution of scientific visibility and impact. Large-scale bibliometric studies have documented persistent differences in publication patterns, authorship positions, collaboration, and citation outcomes between men and women across disciplines (1; 2). These inequalities are particularly relevant in citation-based research evaluation because citations contribute to widely used measures of scholarly impact, even though their interpretation as direct measures of research quality is subject to important limitations (3; 4).

Citation disparities therefore form one component of a broader system of unequal scholarly recognition. At the same time, observed citation differences cannot automatically be interpreted as evidence of differential treatment. Citation outcomes depend strongly on characteristics such as publication period, research field, productivity, venue, collaboration structure, and prior visibility. Gender differences in the distribution of these characteristics can produce differences in raw citation counts even in the absence of differences within otherwise comparable scholarly contexts. Consequently, research on gender citation imbalance increasingly evaluates observed citation patterns relative to an expected citation distribution rather than relying solely on unconditional citation totals.

2.1.1 Gender Disparities in Scholarly Production and Recognition

Early large-scale bibliometric analyses established that the representation of women in scholarly publishing differs considerably across disciplines, countries, and authorship positions. Larivière et al. (1), using a large international publication dataset, documented substantial global gender disparities in scientific authorship and citation impact. West et al. (2) similarly showed that authorship patterns vary systematically by gender across disciplines and authorship positions. Together, these studies indicate that the underlying population of published research is itself gender-structured before citation behaviour is considered.

This distinction between participation and recognition is important for citation analysis. A group that represents a smaller share of the published literature would also be expected to receive a smaller share of citations under a simple proportional model. Raw citation shares therefore cannot distinguish between differences arising from publication representation and differences in citation reception relative to that representation. The appropriate empirical question is instead whether papers associated with different gender categories receive more or fewer citations than would be expected given a relevant comparison population.

Citation accumulation can also be affected by behaviours that operate after publication. Self-citation is one example. King et al. (7) documented substantial gender differences in self-citation behaviour across fields and over time, with men citing their own work more frequently than women. Because early citation accumulation can contribute to subsequent visibility, such differences illustrate how citation inequalities may develop through cumulative processes rather than through a single citation decision.

These findings provide a broader context for the present thesis. Gender citation imbalance should not be understood as an isolated property of reference lists, but as one component of a scholarly system in which publication representation, visibility, collaboration, and citation accumulation are interconnected. This also means that citation disparities must be studied relative to appropriately defined opportunities for receiving citations.

2.1.2 Gender Citation Imbalance across Disciplines

Gender differences in citation outcomes have been identified in multiple research communities. In international relations, Maliniak et al. (5) reported a gender citation gap in which work authored by women received fewer citations than work authored by men. Dion et al. (6) subsequently documented gendered citation patterns across political science and social-science methodology fields. Evidence of citation differences has also been reported in the natural sciences. Caplar et al. (8), for example, identified a citation disadvantage associated with female authorship in astronomy after considering several observable characteristics of publications.

Particularly relevant to the present study are recent analyses in neuroscience and physics because they explicitly compare observed citation behaviour with citation rates expected from the composition and characteristics of the available literature. Dworkin et al. (9) examined reference lists in neuroscience and found that papers with men as both first and last authors were cited more frequently than expected, while papers with women in these positions were cited less frequently than expected. Their analysis also showed that the magnitude of citation imbalance varied with characteristics of the citing papers and authors. Citation imbalance was therefore not uniform across all citing contexts.

Teich et al. (10) reported a related pattern in contemporary physics. Papers authored by women were under-cited relative to expectation, whereas papers authored by men were over-cited. Importantly, the magnitude of the imbalance varied across citing contexts and according to characteristics of the relationship between citing and cited researchers. The physics results therefore reinforce the conclusion from neuroscience that aggregate citation imbalance can conceal substantial contextual heterogeneity.

The neuroscience and physics studies are methodologically important for the present thesis for two reasons. First, they demonstrate that expected citation rates provide a more informative reference point than raw citation shares because the available literature differs in gender composition and other structural characteristics. Second, they show that citation imbalance can depend on the scholarly context within which citations occur. An overall discipline-level estimate may therefore be insufficient if citation behaviour differs across research areas, publication periods, or other forms of scholarly proximity.

The accumulated evidence across political science, astronomy, neuroscience, and physics suggests that gender citation disparities are observable across substantially different academic environments. However, differences between disciplines in publication systems, citation rates, collaboration patterns, and gender representation make direct transfer of numerical estimates inappropriate. The relevant question for Computer Science is therefore not whether the exact citation gaps reported in another discipline are reproduced, but whether comparable deviations between observed and expected citation reception can be identified after accounting for the structural characteristics of the Computer Science literature.

2.1.3 Gender Citation Imbalance in Computer Science

Computer Science constitutes a distinct setting for citation-disparity analysis. Its publication system gives a prominent role to conference proceedings alongside journals, while its subfields differ considerably in publication volume, citation density, venue structure, and research practices. The field has also undergone substantial growth and changes in gender representation over time. These characteristics imply that Computer Science papers published in different periods, subfields, and venues may face substantially different citation opportunity structures.

Previous work provides direct evidence that gendered citation imbalance is relevant within Computer Science. Nakajima et al. (11) initially quantified gendered citation imbalance in Computer Science conference publications. More recently, Nakajima et al. (12) extended this analysis to both conference and journal papers. Their results showed that papers for which a woman is listed as

either the first or last author receive fewer citations than expected, with the imbalance being more pronounced for conference papers and particularly for papers published at top-tier conferences.

The newer study also examined structural properties associated with these citation patterns. Using a family of direct-citation reference models, Nakajima et al. found that citation homophily with respect to characteristics such as country of affiliation, research topic, and venue rank was strongly associated with the observed gendered imbalance, whereas additionally preserving heterogeneity in the number of citations received per paper produced comparatively little change. Their matched-pair analyses further identified associations between gendered citing patterns and characteristics of the citing papers, including author prominence and the local coauthorship networks of the first and last authors (12).

The present thesis builds on this evidence but addresses a different question. Rather than focusing primarily on which properties of the direct citation network are associated with gendered citation imbalance, it asks how robust the measured imbalance is to alternative definitions of scholarly comparability and citation opportunity. The empirical analysis distinguishes between citations received by Computer Science papers from the broader observed corpus (All→CS) and citations occurring entirely within Computer Science (CS→CS), and evaluates expected citation reception under multiple structural, topical, temporal, and network-based specifications.

In particular, publication period, subfield, topical proximity, venue structure, prior citation visibility, and citation-derived network position provide alternative forms of comparison. Bibliographic coupling and co-citation additionally allow papers to be compared according to shared research foundations and later community association rather than only through the direct citation graph. These representations are complemented by GLM-based expected-citation models, citation-lag analyses, and randomized reference specifications.

The contribution relative to recent Computer Science work is therefore primarily methodological and comparative rather than an extension from conference to journal publications. The objective is to determine whether the broad directional patterns identified using direct-citation reference models remain visible when citation opportunity is represented in substantially different ways, including through non-direct citation representations such as bibliographic coupling and co-citation. A result that remains directionally stable across these alternatives provides evidence that the conclusion is not specific to one representation or normalization choice, while variation in magnitude reveals how strongly the measured imbalance depends on the definition of scholarly comparability.

2.1.4 Structural Explanations and Confounding Factors

Observed gender differences in citation reception can arise from multiple processes that operate simultaneously. These processes include differences in publication timing and career stage, productivity and collaboration, distribution across research areas, venue visibility, prior citation accumulation, and position within scholarly networks. Consequently, an observed citation difference cannot by itself identify the mechanism that produced it.

Publication time is a particularly important source of confounding. Older papers have had more time to accumulate citations, while the gender composition of the scientific literature has changed substantially over time. If women are more strongly represented among recent publications, an unconditional comparison with older, more male-dominated cohorts can produce citation differences partly because the groups have experienced different exposure periods. Temporal conditioning is therefore required before differences in citation reception can be meaningfully compared.

Research-area composition provides a second source of heterogeneity. Citation rates and referencing practices differ between disciplines, subfields, and topics, and gender representation can also vary across these areas (1; 6). If gender categories are distributed differently across high- and low-citation research areas, aggregate citation differences may partly reflect this topical composition. Similar considerations apply to venue structure: papers published in venues with different

audiences, visibility, or prestige may face different opportunities for subsequent citation.

Visibility can additionally accumulate over time. Papers that have already received citations may be easier to discover or may be perceived as more established contributions, producing cumulative differences in later citation opportunity. Gender differences in self-citation (7) and broader differences in scholarly visibility can therefore interact with subsequent citation accumulation. This makes prior citation history another relevant dimension when constructing expected citation baselines.

Finally, papers occupy different positions in citation-derived scholarly networks. Two papers published in the same period and broad research area may still differ substantially in the prior work on which they build or in how later literature associates them with other contributions. Structural metadata alone may therefore leave important dimensions of scholarly proximity unaccounted for. This motivates considering citation-derived neighborhood information in addition to conventional temporal, topical, and venue-based controls.

These factors do not provide mutually exclusive explanations for observed gender citation differences, nor does controlling for them produce a causal estimate of the effect of gender. Instead, they define alternative dimensions of citation opportunity against which observed citation reception can be compared. The methodological approach of this thesis therefore evaluates gender citation imbalance under multiple expected-citation specifications rather than treating any single adjustment as a complete model of the citation process.

2.2 Bibliometrics and Citation Networks

Bibliometrics studies scholarly communication through quantitative traces such as publications, references, and citations. Citation indexing made it possible to represent the scientific literature not only as a collection of documents, but also as a system of relations between them (17). This network perspective was developed further by Price (18), who described scientific papers as forming an interconnected citation structure through which knowledge accumulates over time.

Citation networks are useful because different relations between papers encode different forms of scholarly connection. A direct citation records an explicit reference from one paper to another. Bibliographic coupling connects papers that draw on overlapping prior literature, while co-citation connects papers that are subsequently cited together. Although all three representations are derived from references and citations, they should not be interpreted as interchangeable descriptions of the same relationship. They capture scholarly proximity at different stages of the citation process and therefore provide different views of the structure of scientific knowledge.

This distinction is central to the present thesis. Gender citation imbalance is usually measured on direct citation relations, but conclusions derived from one citation representation may depend on the notion of scholarly comparability implicit in that representation. Bibliographic coupling and co-citation provide alternative structures within which citation outcomes can be compared, making it possible to assess whether observed gender differences remain visible when scholarly proximity is defined in different ways.

2.2.1 Direct Citation Networks

The most direct representation of scholarly citation is a directed citation network. Each node represents a paper, and an edge

$$i \rightarrow j$$

indicates that paper i cites paper j . The direction of the edge therefore separates two distinct roles. Outgoing edges represent references made by a paper, whereas incoming edges represent citations received from other papers.

This distinction is particularly relevant for citation-disparity research. Incoming citations quantify one observable form of scholarly recognition, while outgoing references describe how citing papers allocate attention across the existing literature. These perspectives are related but are not equivalent. A group may differ in the number of citations its papers receive without exhibiting the same pattern in the composition of the references its papers make. Analyses of citation reception and citing-side behaviour should therefore be interpreted separately.

Direct citation networks also preserve the temporal ordering of citation events. A paper can normally cite only work that is already available, and citation probability changes substantially with the age of the cited work. Older publications have had more time to accumulate incoming citations, while citing papers tend to concentrate references on comparatively recent literature. Consequently, raw direct-citation counts are strongly affected by publication timing and citation age.

A further issue concerns self-citation and collaboration structure. Self-citations may occur when citing and cited papers share an author, and broader citation relationships may occur within established collaboration networks. Such edges can contribute to citation accumulation without representing the same degree of independent recognition as citations between unconnected authors. Previous bibliometric research has therefore examined the effects of self-citation and emphasized the importance of documenting how such edges are handled (19; 20; 7).

Despite these limitations, direct citation remains the most explicit representation of citation-based credit transfer because each edge corresponds to an observed reference from one paper to another. For this reason, the direct citation network forms the primary representation in this thesis, while the other citation-derived networks are used to define alternative structures of scholarly proximity.

2.2.2 Co-Citation and Bibliographic Coupling

Bibliographic coupling and co-citation extend citation-network analysis beyond explicit directed edges by connecting papers through shared citation context. The two constructions are closely related mathematically but differ substantially in their temporal interpretation and in the scholarly relation they represent.

Bibliographic coupling was introduced by Kessler (13). Two papers are bibliographically coupled when they cite one or more of the same references. If $R(a)$ and $R(b)$ denote the reference sets of papers a and b , a simple coupling strength can be expressed as

$$w_{bc}(a, b) = |R(a) \cap R(b)|.$$

A larger overlap indicates that the two papers draw more extensively on the same prior literature. Bibliographic coupling can therefore be interpreted as a representation of similarity in research foundations, background literature, or intellectual antecedents.

An important property of bibliographic coupling is that the relationship is determined by the reference lists of the papers themselves. Once both papers have been published and their reference lists are fixed, their coupling strength does not require subsequent citation activity to emerge. This makes bibliographic coupling particularly useful as a backward-looking representation of scholarly similarity.

Co-citation was introduced by Small (14). Two papers are co-cited when one or more later papers cite them together. If $C(a)$ and $C(b)$ denote the sets of papers citing a and b , co-citation strength can be expressed as

$$w_{co}(a, b) = |C(a) \cap C(b)|.$$

Whereas bibliographic coupling asks whether two papers rely on similar prior work, co-citation asks whether subsequent literature treats them as related enough to appear together in reference lists. Co-citation can therefore be interpreted as a representation of later community association or perceived intellectual proximity.

This distinction also creates an important temporal asymmetry. Bibliographic coupling is based on information available from the papers at publication, whereas co-citation develops through later citation activity. A newly published paper may have a well-defined bibliographic-coupling neighborhood immediately but cannot acquire a substantial co-citation neighborhood until later work begins citing it together with other papers. Co-citation is therefore inherently future-informed when used to characterize papers retrospectively.

The two representations consequently encode different forms of scholarly relatedness. Bibliographic coupling groups papers according to the intellectual resources on which they build, whereas co-citation groups papers according to how later scholarship associates them. Neither relation is equivalent to a direct citation: two papers may be strongly coupled or co-cited without one ever directly citing the other.

For disparity analysis, these differences provide alternative notions of which papers should be treated as comparable. A direct structural baseline may compare papers sharing a publication period, research area, or venue stratum. A bibliographic-coupling comparison instead asks how a paper performs relative to work with similar reference foundations, while a co-citation comparison places it among papers occupying a similar position in the intellectual associations generated by subsequent citation behaviour.

2.2.3 Comparative Evidence across Citation Representations

The differences between direct citation, bibliographic coupling, and co-citation have been examined explicitly in the bibliometric literature. Comparative studies show that the three representations recover related but non-identical structures from the same scholarly corpus, making the choice of citation relation consequential for how scientific organization is measured.

Boyack and Klavans (15) compared direct citation, bibliographic coupling, and co-citation as alternative approaches for representing research fronts. Their analysis demonstrated that the resulting structures differ across citation approaches rather than reproducing an identical organization of the literature. The study therefore provides empirical support for treating the three representations as distinct views of scholarly relatedness rather than as interchangeable network constructions.

Kleminski et al. (16) investigated the same distinction in Computer Science, comparing direct citation, bibliographic coupling, and co-citation in the identification of scientific topics. Their results likewise showed that the three representations provide complementary and partly non-overlapping information about the organization of the literature. The choice of citation-derived relation can therefore affect the structure recovered even when the underlying publication corpus is unchanged.

Previous comparative work has primarily used these representations to study science mapping, research fronts, communities, or topic structure. The present thesis builds on the established distinction between the representations but applies it to a different empirical problem: the measurement of gender citation imbalance.

The contribution is therefore not the introduction of bibliographic coupling or co-citation as new network representations. Instead, the thesis uses them as alternative definitions of scholarly context against which citation reception can be evaluated. Direct citation provides the observed transfer of citation attention, bibliographic coupling defines local comparison neighborhoods through shared prior work, and co-citation defines retrospective comparison neighborhoods through later

community association.

This distinction enables a form of robustness analysis that cannot be obtained from the direct citation graph alone. If gender-category deviations from expected citation reception remain similar when papers are compared across different citation-derived representations, the conclusion is less dependent on a single definition of scholarly proximity. Conversely, if the estimated imbalance changes substantially between direct, bibliographic-coupling, and co-citation representations, this indicates that the measurement is sensitive to which aspect of the citation system is incorporated into the comparison.

The role of these representations in the present thesis is therefore comparative rather than hierarchical. The objective is not to determine which network is the universally “best” representation of scientific structure. Each captures different information, and co-citation additionally relies on future citation activity that is not available at publication time. The empirical analysis instead uses the differences between representations to evaluate how strongly conclusions about gender citation imbalance depend on the definition of scholarly similarity and citation opportunity.

2.3 Topic Similarity and Citation Relevance

A recurring challenge in citation analysis is that citations vary in their substantive relationship to the citing work: some reflect close topical dependence, while others are cross-area, perfunctory, or primarily rhetorical. Measuring *citation relevance* is therefore closely connected to measuring topical similarity between citing and cited papers. In disparity settings, this distinction matters because imbalance may differ between citations occurring *within* a research area, which can be interpreted as core recognition, and citations that occur *across* areas, which can be interpreted as broader diffusion.

2.3.1 Text-Based Approaches

A standard family of approaches represents documents using textual content, such as title and abstract, and computes similarity in a vector space. In the vector-space model, documents are mapped to weighted term vectors, and similarity is commonly computed with cosine similarity (21). Modern embedding approaches follow the same general idea but learn dense representations; for example, Sentence-BERT produces sentence embeddings designed to be compared using cosine similarity (22).

In this thesis, text-based similarity is not used as the primary topical signal. Large bibliographic corpora frequently contain multilingual titles, missing abstracts, and heterogeneous preprocessing requirements, which can introduce systematic measurement error and reduce comparability across time and subfields. Instead, the thesis relies on taxonomy-based topical information provided by the bibliographic database.

2.3.2 Taxonomy-Based Approaches

Taxonomy-based approaches characterize topical structure using standardized category systems. In OpenAlex, papers are assigned *topics* and placed into a four-level hierarchy: domain, field, subfield, and topic (23). Topic assignments are generated automatically from available work information, and the highest-scoring topic is stored as the work’s primary topic (23). This provides a standardized representation at scale that supports aggregation and stratification over long time spans.

This hierarchy enables several operationalizations of citation relevance used in this thesis:

- **Exact-match relevance:** a citation is treated as topically close if citing and cited papers share the same topic, or more coarsely the same subfield, field, or domain.

- **Hierarchical proximity:** citations can be weighted by the deepest shared level in the hierarchy, with topic match treated as closer than subfield match, field match, or domain match.
- **Flow representations:** aggregating citations between taxonomy units yields taxonomy-level citation flows that summarize diffusion patterns across research areas.

By distinguishing in-area from cross-area citations, taxonomy-based similarity supports relevance-weighted citation analyses and complements the multi-representation design of this thesis.

2.4 Citation Normalization and Expected Citation Rates

Raw citation counts are not directly comparable across papers because citation opportunities differ systematically across publication periods, research fields, venues, and other dimensions of scholarly context. Older papers have had more time to accumulate citations, while citation densities and reference practices vary substantially across research areas. Citation-based comparison therefore commonly requires some form of normalization before differences between papers or groups can be interpreted (24; 25).

A central idea in citation normalization is to evaluate observed citation outcomes relative to an appropriate reference population. Depending on the research question, this reference population may be defined by publication year, field, topic, venue, or combinations of such characteristics. The resulting expected citation rate represents the citation outcome associated with a specified comparison structure rather than an unconditional corpus-wide benchmark.

2.4.1 Temporal and Field Normalization

Publication time is one of the most important determinants of citation opportunity. Papers can only accumulate citations after publication and citation rates typically change with paper age. Comparisons across publication cohorts therefore require temporal conditioning or age-sensitive normalization (26; 24).

Citation practices also differ substantially across disciplines and research areas. Fields vary in publication volume, reference-list length, citation density, and the typical speed of citation accumulation. Field-normalized indicators therefore compare papers against publications from similar scholarly contexts rather than against the literature as a whole (25; 27).

Venue structure can provide an additional dimension of citation opportunity. Papers published in venues with different levels of visibility, audience, or selectivity may face different opportunities for subsequent citation. Consequently, publication period, research area, and venue characteristics are commonly treated as relevant dimensions when constructing expected citation rates.

2.4.2 Expected Citations and Alternative Comparison Models

Expected citations are not uniquely defined. Different normalization strategies preserve different properties of the scholarly environment and can therefore assign different expectations to the same observed paper. A time-normalized comparison asks a different question from a field-normalized comparison, while a model that additionally incorporates venue or prior visibility defines a still narrower notion of citation opportunity.

This point is particularly important in disparity research. An observed difference between demographic groups may decrease, persist, or change direction depending on which structural characteristics are incorporated into the reference model. Robustness should therefore not be evaluated solely by whether one particular normalized estimate differs from parity, but also by whether the

direction of the result remains stable across reasonable alternative definitions of expected citation opportunity.

The present thesis follows this comparative perspective. The specific observed/expected framework, structural baselines, network-neighborhood expectations, regression models, and randomized null specifications used in the empirical analysis are defined in Chapter 4.

2.5 Ethical and Methodological Considerations

2.5.1 Name-Based Gender Inference

This thesis operationalizes gender at the paper level using name-based gender inference for first and last authors. This procedure introduces unavoidable limitations. First, inferred gender is probabilistic and can be incorrect due to ambiguity, transliteration, and cross-cultural variation in naming practices. Second, name-based systems typically operationalize a binary gender model and therefore do not represent non-binary identities; the analysis should be interpreted as measuring disparities across inferred binary categories rather than gender identity. Third, the unknown category is not missing at random: it may be more common for authors from regions or linguistic contexts with lower coverage or higher ambiguity, which can affect downstream estimates.

To mitigate overinterpretation, this thesis reports the share of papers with unknown first or last author gender and restricts the main observed/expected analyses to known first-last author categories. Unknown-gender coverage is interpreted as a data limitation rather than as a substantive fifth gender category. The implications of this restriction are revisited in the Discussion.

2.5.2 Coverage Bias and Missing Data

Large bibliographic datasets are incomplete and heterogeneous. Metadata such as venue information, venue rank, abstracts, references, or topic assignments may be missing or inconsistently recorded. Missingness can correlate with publication year, publication type, language, and indexing practices, potentially inducing systematic bias in both network construction and expected-citation baselines.

Accordingly, this thesis documents missingness and coverage for key variables and interprets results conditionally on the observed corpus, available metadata, known-gender subset, and final filtered citation graph. This is especially important for network-derived representations: bibliographic coupling depends on observed reference lists, while co-citation depends on later observed citing papers. Coverage differences are therefore not only data quality issues but also part of the interpretation of each baseline.

3. Data

This chapter describes the data used in this thesis and the preprocessing steps required to construct the analysis-ready corpus. The chapter first introduces the bibliographic data source and the metadata available for each paper. It then defines the full paper corpus, the Computer Science subset, and the citation slices used throughout the thesis. Next, it describes how author gender is operationalized from first and last author names. Finally, it summarizes the main preprocessing steps and data quality considerations that affect the interpretation of the later exploratory and results chapters.

3.1 Data Source and Schema

This thesis uses the large-scale bibliographic dataset released by Nakajima et al. (11). The dataset integrates metadata from several sources. OpenAlex is a fully open scholarly knowledge graph introduced as a successor to Microsoft Academic Graph (28). OpenAlex provides work identifiers, bibliographic metadata, topic taxonomy labels, and reference lists (29). DBLP provides venue identity and venue type information (30). Venue rank information is derived from CORE Conference Ranking for conferences (31) and SCImago Journal Rank quartiles for journals (32).

The dataset contains 3,720,575 papers and 22,908,275 internal citation links. An internal citation is a reference from one paper in the dataset to another paper that is also present in the dataset. This means that the citation graph used in this thesis is a corpus-internal graph: citations to works outside the dataset are not represented as edges. Consequently, citation counts and reference-list statistics should be interpreted as observed within-corpus citation behavior rather than as complete bibliographic histories.

Each paper record is identified by an OpenAlex work identifier stored in the field `id`. The main metadata fields used in this thesis are `title`, `pub_date`, `primary_topic`, `author_lst`, `ref_lst`, `venue_type`, `venue_name`, and `venue_rank`. The field `pub_date` stores the publication date in ISO format and is used to derive publication year and decade. The field `primary_topic` contains the OpenAlex topic assignment and its hierarchy: topic, subfield, field, and domain. These taxonomy labels are later used for descriptive statistics, topical citation flows, and expected-citation baselines.

Authorship information is stored in `author_lst`, an ordered list of authors as they appear on the publication. Each author entry includes at least a name and available identifiers. This ordering is important because the gender operationalization in this thesis is based on the first and last authors. The same list is also used to derive descriptive authorship variables, such as the number of authors per paper.

Reference information is stored in `ref_lst`. This field contains identifiers of cited works. In the network analyses, a reference contributes to the directed citation graph only when the cited identifier can be resolved to another paper in the dataset. Therefore, the directed citation edge (i, j) means that paper i cites paper j , with both papers present in the corpus. The detailed construction and filtering of citation edges is described in Chapter 4.

Venue information is represented by `venue_type`, `venue_name`, and `venue_rank`. The field `venue_type` distinguishes journals from conference proceedings where available. The field `venue_rank` stores external rank categories: CORE ranks such as A*, A, B, and C for conferences, and SCImago quartiles such as Q1-Q4 for journals. When no rank is available, the rank is recorded as missing or unavailable. Venue rank is used both descriptively and as a conditioning variable in some expected-citation baselines.

3.2 Corpus Definition and Analysis Slices

Let $\mathcal{P}$ denote the full set of papers in the dataset. The Computer Science subset is defined using the OpenAlex field label associated with each paper’s primary topic. For a paper $i \in \mathcal{P}$, let $\phi(i)$

denote its OpenAlex field label. The Computer Science corpus is defined as

$$\mathcal{P}_{CS} = \{i \in \mathcal{P} : \phi(i) = \text{Computer Science}\}. \quad (3.1)$$

This explicit distinction between the full corpus $\mathcal{P}$ and the Computer Science subset $\mathcal{P}_{CS}$ is required because later analyses compare multiple citation slices. The main citation-reception slice is All→CS: citing papers may come from the full observed corpus, while cited papers are restricted to Computer Science. This slice measures how Computer Science papers are cited by the broader observed dataset. A stricter robustness slice is CS→CS, where both citing and cited papers are restricted to Computer Science. This slice measures citation imbalance within Computer Science itself.

Formally, let $\mathcal{E}_{\text{final}}$ denote the final filtered directed citation edge set constructed in Chapter 4. The All→CS slice is defined as

$$\mathcal{E}_{\text{All} \rightarrow \text{CS}} = \{(i, j) \in \mathcal{E}_{\text{final}} : j \in \mathcal{P}_{CS}\}. \quad (3.2)$$

The CS→CS slice is defined as

$$\mathcal{E}_{\text{CS} \rightarrow \text{CS}} = \{(i, j) \in \mathcal{E}_{\text{final}} : i \in \mathcal{P}_{CS}, j \in \mathcal{P}_{CS}\}. \quad (3.3)$$

These two slices are used throughout the results chapter to distinguish broad citation reception from within-field citation behavior.

The thesis does not impose an additional manual time window beyond the temporal coverage of the dataset. Publication years are parsed from `pub_date`. For decade-based analyses, the publication decade of paper i is defined as

$$d(i) = 10 \left\lfloor \frac{y_i}{10} \right\rfloor, \quad (3.4)$$

where y_i is the publication year of paper i . Decade bins are used for descriptive analysis, citation-flow aggregation, and decade-conditioned expected-citation baselines.

3.3 Gender Attribution

This thesis operationalizes gender using a name-based inference pipeline applied to author names. The resulting labels are not interpreted as observed self-identified gender. They are inferred categories used to construct paper-level first-last author gender configurations. This distinction is important: the thesis studies citation patterns across inferred name-based author categories, not gender identity directly.

For each paper i , the ordered author list is denoted by

$$A_i = (a_{i1}, \dots, a_{in_i}). \quad (3.5)$$

The first author is a_{i1} and the last author is a_{in_i} . For single-author papers, the same author is both first and last author.

Author names are normalized before inference using lightweight cleaning steps such as trimming whitespace, removing obvious punctuation artifacts, and extracting the given-name component when possible. The inference pipeline then maps names to gender labels with an associated probability-like score. The implementation uses two thresholds to map the score p to a discrete label:

$$\hat{g}(p) = \begin{cases} W, & p \geq 0.7, \\ M, & p \leq 0.3, \\ U, & 0.3 < p < 0.7. \end{cases} \quad (3.6)$$

Here, M denotes inferred male, W denotes inferred female, and U denotes unknown. If the score is missing or the tool cannot return a confident assignment, the label is set to U .

The symmetric thresholds leave the central probability range unclassified rather than forcing ambiguous names into a binary category, thereby trading coverage for greater confidence in the retained assignments.

The first and last author labels of paper i are denoted by $g_i^{(F)}$ and $g_i^{(L)}$. These are combined into a paper-level first-last author gender category:

$$c_i = (g_i^{(F)}, g_i^{(L)}). \quad (3.7)$$

The full set of possible categories includes known and unknown combinations, such as MM , MW , WM , WW , MU , UM , WU , UW , and UU . The primary observed/expected analyses restrict attention to the known-gender categories

$$\mathcal{C}_K = MM, MW, WM, WW. \quad (3.8)$$

A paper is included in known-gender citation analyses only if both its first and last author labels are known. Papers with unknown first or last author gender are retained for coverage diagnostics but excluded from gender-specific observed/expected aggregation. This restriction is necessary because the outcome is aggregated by paper-level gender category. The implications of unknown gender coverage are treated as a data limitation and discussed again in Chapter 7.

The first-last author representation is a pragmatic paper-level operationalization. It is useful because first and last authorship positions often carry interpretable roles in many scientific fields, and because the representation is compact enough to use across large-scale citation and network analyses. However, it does not describe the gender composition of the full author team. The number of authors per paper can be used descriptively, but the main gender variable in this thesis is always based on first and last authors only.

3.4 Preprocessing and Derived Variables

The preprocessing pipeline converts the raw dataset into a consistent analysis-ready representation. The main goals are to normalize identifiers, remove duplicate records, derive variables used in later analyses, and prepare the metadata required for citation-network construction.

All OpenAlex work identifiers are normalized to a consistent string representation. Identifiers are stripped of whitespace and malformed or missing identifiers are excluded from operations that require a resolvable paper id. The same normalization is applied to reference identifiers in `ref_lst`, so that cited works can be matched reliably against the set of valid paper identifiers.

Duplicate paper records are removed by work identifier, keeping a single canonical record per `id`. This ensures that each node in the later citation graph corresponds to one unique paper. Edge-level filtering, such as citation-edge deduplication, paper-id self-link removal, author-overlap self-citation filtering, coauthor-neighborhood filtering, and isolated-node removal, is part of the network construction procedure described in Chapter 4.

Several derived variables are computed from the raw metadata. The publication year y_i and decade $d(i)$ are derived from `pub_date`. The number of authors is derived from the length of `author_lst`:

$$n_{\text{authors}}(i) = |A_i|. \quad (3.9)$$

This variable is used descriptively to characterize authorship and collaboration patterns. It is not used to infer the gender composition of all coauthors.

The reference-list length is derived from the number of recorded references:

$$n_{\text{refs}}(i) = |\text{Ref}(i)|. \quad (3.10)$$

This measures the number of references recorded for the paper in the dataset. It should not be confused with the number of final filtered outgoing citation edges, because the latter depends on additional filtering and corpus-internal matching.

The preprocessing also produces known-gender analysis subsets. For a paper set $\mathcal{S}$, the known-gender subset is

$$\mathcal{S}^K = \{i \in \mathcal{S} : c_i \in \mathcal{C}_K\}. \quad (3.11)$$

In the main citation-reception analyses, this restriction is applied to cited Computer Science papers because the thesis compares citation outcomes by the gender category of the cited paper.

3.5 Data Quality and Scope

The dataset is large and heterogeneous, so missingness and coverage affect different parts of the analysis in different ways. Publication dates are required for decade-based baselines and citation-lag analyses. Topic taxonomy labels are required for subfield conditioning and topical relevance analysis. Venue rank is required for venue-conditioned baselines. Authorship metadata is required for first-last author gender attribution. Reference lists are required for directed citation edges and bibliographic-coupling networks, while incoming citation information is required for co-citation networks.

For a metadata variable X , missingness in a paper set $\mathcal{S}$ can be summarized as

$$\text{miss}(X; \mathcal{S}) = \frac{\sum_{i \in \mathcal{S}} \mathbf{1}(X_i \text{ is missing})}{|\mathcal{S}|}. \quad (3.12)$$

The exploratory data analysis in Chapter 5 reports coverage patterns for the variables most relevant to the thesis, including publication year, taxonomy labels, venue information, reference-list availability, and first/last author gender coverage.

Unknown gender is treated as a coverage outcome rather than as a fifth substantive gender category in the main analyses. Papers with unknown first or last author gender are not assumed to be missing at random. Unknown rates may vary across time, regions, naming conventions, venues, and subfields. For this reason, unknown-gender coverage is reported descriptively and later discussed as a limitation.

The citation graph is also coverage-limited. Since only internal citation links are observed, a missing or empty `ref_list` does not necessarily mean that a paper had no references in its original bibliography. It means that no references were recorded or resolved as internal citations within the dataset. Similarly, network-derived representations such as bibliographic coupling and co-citation are only available for papers with sufficient reference or citation neighborhood information. These coverage constraints are reported where relevant in the EDA and Results chapters.

Overall, the data should be interpreted as a large-scale, corpus-internal representation of scholarly citation behavior. The scale of the dataset makes it suitable for robust descriptive and comparative analyses, but the results remain conditional on metadata coverage, name-based gender inference, OpenAlex taxonomy assignments, venue-rank availability, and the internal-citation scope of the corpus.

4. Methodology

4.1 Overview of the Empirical Strategy

This chapter describes the empirical strategy used to quantify gender citation imbalance in Computer Science across multiple citation-derived representations. The central methodological idea is to compare observed citation outcomes with expected citation outcomes under a sequence of baselines. Each baseline defines a different notion of what it means for papers to be comparable: some baselines compare papers within the same publication period, subfield, or venue stratum, while others compare papers within citation-derived neighborhoods such as bibliographic coupling or co-citation.

The purpose of using multiple baselines is not to estimate one single “true” level of imbalance. Rather, the goal is to test whether the direction and magnitude of the imbalance are stable across alternative definitions of expected citations. If the same gender category is below expected under several baselines, this provides stronger evidence that the result is not an artifact of one specific normalization choice. Conversely, if the magnitude changes substantially across baselines, this is itself informative: it indicates that the estimated imbalance depends on which structural properties are preserved in the comparison.

Raw citation counts are not directly comparable across papers. Citation outcomes are affected by structural factors such as publication year, citation age, subfield, topic, venue visibility, and network position. Older papers have had more time to accumulate citations; some subfields and topics cite more heavily than others; high-visibility venues tend to attract more attention; and papers located in dense citation neighborhoods may be more easily discovered and cited. Therefore, a raw comparison of citation counts across gender categories would risk mixing gender-related imbalance with differences in exposure and citation opportunity.

To address this problem, this thesis uses an observed/expected citation framework. For each baseline, the analysis first computes how many citations each gender category actually receives. It then computes how many citations the same category would be expected to receive if citation attention followed a specified comparison structure. The ratio between these two quantities is used as the main imbalance measure.

A key design choice is that gender is not used to construct the expected-citation baseline. Expected citations are computed from non-gender structural information, such as publication decade, subfield, topic, venue, reference-list structure, citation-network neighborhoods, or model-predicted citation counts. Gender is used only after the expected citation counts have been computed, in order to aggregate observed and expected citations by paper-level gender category. This separation is important because the goal is not to predict gender or to estimate a causal effect of gender. Instead, the goal is to ask whether papers in different gender categories receive more or fewer citations than would be expected given non-gender comparison information.

The empirical strategy proceeds in stages. First, the final filtered direct citation graph is constructed and used to define citation-reception and citing-side analyses. Second, direct citation baselines estimate expected citation counts under increasingly detailed structural controls. Third, topical relevance baselines test whether the estimated imbalance changes when citations are restricted or weighted according to the OpenAlex taxonomy. Fourth, citation-derived network baselines compare papers within bibliographic-coupling and co-citation neighborhoods. Fifth, generalized linear models provide regression-based expected-citation baselines that combine structural and network features in one model. Finally, citation-lag and randomized robustness analyses test whether the findings persist across long-memory citation flows and alternative null models.

Throughout the thesis, the main interpretation of robustness is directional rather than numerical. A result is treated as directionally robust if the same gender category remains below or above expected across multiple reasonable baselines. This does not require the exact observed/expected ratio to be identical across methods. Differences in magnitude are expected because each baseline

preserves different structural properties of the citation system.

4.2 Units of Analysis, Citation Slices, and Gender Categories

The primary unit of analysis is the paper. Let $\mathcal{P}$ denote the full set of papers in the dataset, indexed by OpenAlex work identifiers. Each paper $i \in \mathcal{P}$ is associated with bibliographic metadata, including publication year, venue information, author list, reference list, and OpenAlex taxonomy labels. Let $\mathcal{P}_{CS} \subseteq \mathcal{P}$ denote the subset of papers classified as Computer Science.

The final analysis is based on the filtered citation edge set $\mathcal{E}_{\text{final}}$, constructed in Section 4.4.1. An edge $(i, j) \in \mathcal{E}_{\text{final}}$ means that paper i cites paper j , after identifier normalization, edge deduplication, self-citation filtering, coauthor-neighborhood filtering, and isolated-paper removal. From this final edge set, the analysis uses two citation slices:

$$\mathcal{E}_{\text{All} \rightarrow \text{CS}} = \{(i, j) \in \mathcal{E}_{\text{final}} : j \in \mathcal{P}_{CS}\}, \quad (4.1)$$

$$\mathcal{E}_{\text{CS} \rightarrow \text{CS}} = \{(i, j) \in \mathcal{E}_{\text{final}} : i \in \mathcal{P}_{CS}, j \in \mathcal{P}_{CS}\}, \quad (4.2)$$

The main cited-side analyses use the All $\rightarrow$ CS slice, because it captures all observed citations received by Computer Science papers from within the available corpus. The CS $\rightarrow$ CS slice is used as a stricter within-field specification.

Gender is operationalized at the paper level using the inferred genders of the first and last authors. For each paper j , let c_j denote its first-last author gender category. In the known-gender specification, the category set is

$$\mathcal{C}_K = \{MM, MW, WM, WW\}. \quad (4.3)$$

Here, the first letter denotes the inferred gender of the first author and the second letter denotes the inferred gender of the last author. For example, WW denotes a paper whose first and last authors are both inferred as women, while MM denotes a paper whose first and last authors are both inferred as men.

Once the final analysis universe has been obtained, the known-gender Computer Science paper set is

$$\mathcal{P}_{CS}^K = \{j \in \mathcal{P}_{\text{final}} : j \in \mathcal{P}_{CS}, c_j \in \mathcal{C}_K\}. \quad (4.4)$$

Papers with unknown first or last author gender are excluded from gender-specific observed/expected aggregation. They are retained in coverage diagnostics and discussed as a limitation, but they are not treated as a fifth substantive gender category in the main analyses.

This paper-level operationalization is a pragmatic choice. It makes the gender variable comparable across direct citations, bibliographic coupling, co-citation, citation-lag analysis, and GLM-based expected-citation models. At the same time, the interpretation is limited to this operational definition and should not be read as a complete representation of all contributors to a paper.

4.3 Observed/Expected Citation Framework

The main imbalance measure in this thesis is the observed/expected ratio. For a gender category $g \in \mathcal{C}_K$, the observed count is the number of citations actually received by papers in that category. The expected count is the number of citations that the same category would receive under a specified baseline. The observed/expected ratio is defined as

$$\text{OE}_g = \frac{O_g}{E_g}, \quad (4.5)$$

where O_g is the observed citation count for category g , and E_g is the corresponding expected citation count under the baseline.

If $OE_g = 1$, the category receives approximately as many citations as expected. If $OE_g < 1$, the category receives fewer citations than expected. If $OE_g > 1$, the category receives more citations than expected. For example, $OE_{WW} = 0.80$ means that *WW* papers receive 80% of the citations expected under the baseline.

Let $\mathcal{E}^*$ denote a selected final filtered citation slice, such as $\mathcal{E}_{All \rightarrow CS}$ or $\mathcal{E}_{CS \rightarrow CS}$, restricted when necessary to edges whose cited paper belongs to $\mathcal{P}_{CS}^K$. The observed count for category g is

$$O_g = \sum_{(i,j) \in \mathcal{E}^*} \mathbf{1}(c_j = g). \quad (4.6)$$

This counts how many citation edges point to cited papers j whose paper-level gender category is g .

The expected count depends on the baseline. In general, each baseline assigns an expected contribution $p_{j,g}^{(b)}$ to category g for each cited paper j , where b indexes the baseline. The expected count for gender category g is

$$E_g^{(b)} = \sum_{(i,j) \in \mathcal{E}^*} p_{j,g}^{(b)}. \quad (4.7)$$

The value of $p_{j,g}^{(b)}$ differs by baseline. In a paper-share baseline, it is computed from the gender composition of structurally comparable papers. In a network-neighborhood baseline, it is computed from local citation-derived neighbors. In a GLM baseline, expected counts are obtained from model-predicted citation counts and then aggregated by the actual gender category of the cited paper.

This framework makes the different baselines comparable. Although each baseline defines expectation differently, all baselines produce the same type of output: observed and expected citation counts by gender category, summarized using observed/expected ratios.

The expected count is not intended to represent a causal counterfactual. The analysis does not claim that changing the gender category of a paper would cause its citation count to change by the observed difference. Instead, the expected count is a descriptive normalization device. It defines what citation reception would look like if citations followed a specified comparison structure. The resulting O/E ratio measures deviation from that structure.

4.4 Network Construction

All analyses in this thesis are built from a common final filtered paper-level citation edge list. Different citation-derived representations are then constructed from this edge list to capture different notions of scholarly relatedness. The direct citation network represents explicit citation links between citing and cited papers. Bibliographic coupling represents similarity in research foundations by connecting papers that share references. Co-citation represents later community association by connecting papers that are cited together by subsequent literature. Citation-lag and taxonomy-based annotations add temporal and topical structure to direct citation edges.

The purpose of constructing multiple representations is methodological rather than purely descriptive. A single direct citation edge only captures the fact that one paper cites another. It does not distinguish whether two papers belong to the same topical area, build on similar prior work, are treated as related by later literature, or remain visible over long time horizons. By transforming the same underlying citation data into several views, the analysis can test whether gender citation imbalance is specific to one representation or persists across different definitions of citation-based similarity and recognition.

Because the full pairwise construction of bibliographic coupling and co-citation networks is computationally infeasible at corpus scale, sparse top- K network approximations are used. For each paper, only a limited set of strongest neighbors is retained. This preserves the most relevant local comparison structure while keeping the networks tractable for downstream baselines. The resulting networks are not intended to represent every possible weak similarity between papers; rather, they define practical local neighborhoods for expected-citation comparisons.

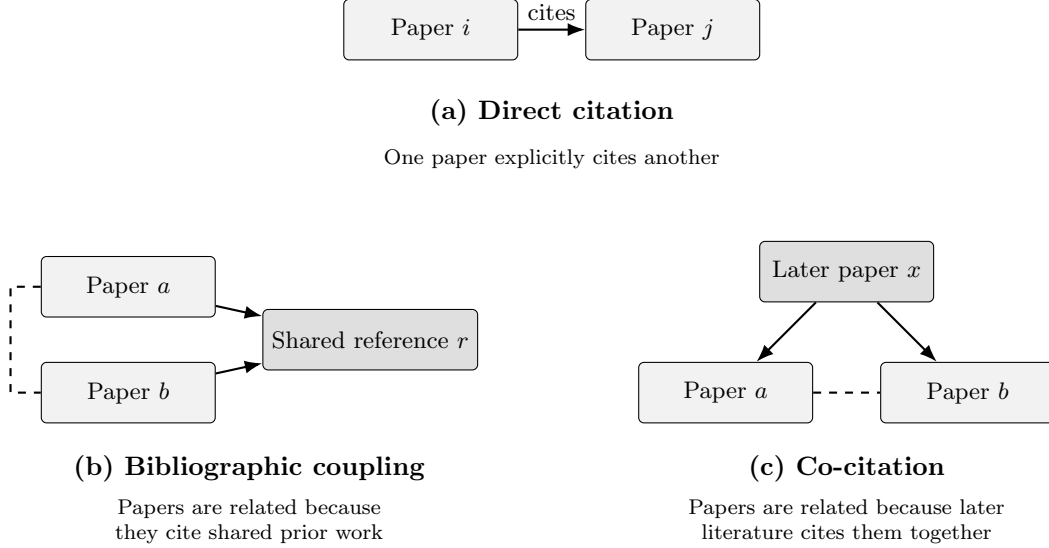


Figure 4.1: Schematic illustration of the three citation-derived network representations used in this thesis. Direct citation represents an explicit citation from a citing paper to a cited paper. Bibliographic coupling connects papers through references they share, representing similarity in their research foundations. Co-citation connects papers that are cited together by later literature, representing retrospective community association.

4.4.1 Direct Citation Network

The direct citation network is the primary representation from which citation reception and reference allocation outcomes are derived. It is a directed paper-level graph in which nodes represent papers and directed edges represent citations from one paper to another. The initial graph is

$$G_{\text{dir}} = (\mathcal{P}, \mathcal{E}_0), \quad (4.8)$$

where $\mathcal{E}_0$ is the initial set of directed internal citation edges.

Let $\text{Ref}(i)$ denote the set of referenced paper identifiers recorded for paper i . A citation edge is included only if the cited identifier can be resolved to another paper in the corpus:

$$\mathcal{E}_0 = \{(i, j) : i \in \mathcal{P}, j \in \text{Ref}(i), j \in \mathcal{P}\}. \quad (4.9)$$

This produces an internal citation network over the observed dataset. References to works outside the corpus are not represented as edges because the cited work is not available as a node in $\mathcal{P}$.

Before downstream analysis, duplicate citation edges are dropped so that each ordered pair of papers contributes at most one citation edge:

$$\mathcal{E}_1 = \text{unique}(\mathcal{E}_0). \quad (4.10)$$

Paper-identifier self-links are then removed:

$$\mathcal{E}_2 = \{(i, j) \in \mathcal{E}_1 : i \neq j\}. \quad (4.11)$$

These links correspond to cases where the citing and cited paper identifiers are identical.

The analysis also removes author-overlap self-citations. Let A_i denote the set of OpenAlex author identifiers associated with paper i . A citation is treated as an author-overlap self-citation if the citing and cited papers share at least one author:

$$\mathcal{E}_3 = \{(i, j) \in \mathcal{E}_2 : A_i \cap A_j = \emptyset\}. \quad (4.12)$$

This filter removes direct author self-citations at the paper-pair level.

Finally, the analysis removes time-aware coauthor-neighborhood citations. This filter excludes citations from papers written by prior coauthors of the cited paper's authors. For cited paper j and citing year y_i , define the prior coauthor neighborhood of j as

$$\mathcal{N}_{\text{coauth}}(j, y_i) = \{b \notin A_j : \exists h \in \mathcal{P}, y_h < y_i, A_h \cap A_j \neq \emptyset, b \in A_h\}. \quad (4.13)$$

A citation (i, j) is retained only if the citing paper contains no author from this prior coauthor neighborhood:

$$\mathcal{E}_4 = \{(i, j) \in \mathcal{E}_3 : A_i \cap \mathcal{N}_{\text{coauth}}(j, y_i) = \emptyset\}. \quad (4.14)$$

The condition $y_h < y_i$ makes the filter time-aware: only collaboration relationships that existed before the citing paper was published are used to define the prior coauthor neighborhood.

After these citation filters, papers that become isolated in the filtered citation graph are removed from the final analysis universe. The final paper set is

$$\mathcal{P}_{\text{final}} = \{p \in \mathcal{P} : \exists q \in \mathcal{P}, (p, q) \in \mathcal{E}_4 \text{ or } (q, p) \in \mathcal{E}_4\}. \quad (4.15)$$

The final directed citation edge set is

$$\mathcal{E}_{\text{final}} = \{(i, j) \in \mathcal{E}_4 : i \in \mathcal{P}_{\text{final}}, j \in \mathcal{P}_{\text{final}}\}. \quad (4.16)$$

For a paper j , incoming citations in the filtered direct citation network are defined as

$$d_{\text{in}}(j) = |\{i \in \mathcal{P}_{\text{final}} : (i, j) \in \mathcal{E}_{\text{final}}\}|. \quad (4.17)$$

For a paper i , outgoing internal references are defined as

$$d_{\text{out}}(i) = |\{j \in \mathcal{P}_{\text{final}} : (i, j) \in \mathcal{E}_{\text{final}}\}|. \quad (4.18)$$

Incoming citations are used to measure citation reception, while outgoing citations are used to characterize citing behavior and reference-list structure.

All subsequent citation slices and citation-derived network representations are defined from $\mathcal{E}_{\text{final}}$.

4.4.2 Bibliographic Coupling Network

The bibliographic-coupling network is a weighted paper-level network that connects papers with overlapping reference lists. Whereas the direct citation network represents explicit citation links, bibliographic coupling represents similarity in research foundations: two papers are treated as related if they cite some of the same prior work.

For each paper a , let $\text{Ref}_{\text{final}}(a)$ denote the set of papers cited by a in the final filtered citation graph:

$$\text{Ref}_{\text{final}}(a) = \{j \in \mathcal{P}_{\text{final}} : (a, j) \in \mathcal{E}_{\text{final}}\}. \quad (4.19)$$

Only papers with at least five internal references in the final filtered graph are used as source papers for bibliographic-coupling construction.

The raw bibliographic-coupling strength is the number of shared references:

$$w_{bc}(a, b) = |\text{Ref}_{\text{final}}(a) \cap \text{Ref}_{\text{final}}(b)|. \quad (4.20)$$

The Jaccard similarity between reference sets is also computed:

$$J_{bc}(a, b) = \frac{|\text{Ref}_{\text{final}}(a) \cap \text{Ref}_{\text{final}}(b)|}{|\text{Ref}_{\text{final}}(a) \cup \text{Ref}_{\text{final}}(b)|}. \quad (4.21)$$

In the implementation, Jaccard similarity is used for candidate generation and secondary ranking, while the main retained edge weight is the raw shared-reference count $w_{bc}(a, b)$.

The resulting bibliographic-coupling graph is an undirected weighted graph:

$$G_{bc} = (\mathcal{P}_{\text{final}}, \mathcal{E}_{bc}, w_{bc}). \quad (4.22)$$

A pair is retained only if it shares at least three references:

$$\{a, b\} \in \mathcal{E}_{bc} \iff w_{bc}(a, b) \geq 3. \quad (4.23)$$

Computing all pairwise overlaps between reference sets is infeasible at corpus scale. The implementation therefore uses MinHash locality-sensitive hashing to generate candidate pairs based on approximate Jaccard similarity. Each paper is represented by a MinHash signature of its final filtered reference set, using 128 permutations. Candidate neighbors are retrieved using an LSH threshold of 0.25, and exact overlap statistics are then computed for candidate pairs.

After candidate generation and exact overlap computation, the graph is sparsified into a top- K neighborhood representation. For each paper a , retained neighbors are sorted primarily by shared-reference count and secondarily by Jaccard similarity. The bibliographic-coupling neighborhood is defined as

$$\mathcal{N}_{bc}(a) = \text{TopK}_b \{b \in \mathcal{P}_{\text{final}} \setminus \{a\} : w_{bc}(a, b) \geq 3\}. \quad (4.24)$$

In the final implementation, $K = 20$. If fewer than 20 valid neighbors are available for a paper, all valid neighbors are retained. For bibliographic coupling, the LSH threshold of 0.25 and $K = 20$ were chosen as pragmatic sparsification settings that balance computational tractability with the retention of strong local similarity relationships.

Bibliographic coupling is backward-looking because it depends on reference lists that are fixed at publication time. Consequently, it provides a comparison structure based on shared prior work rather than later community recognition.

4.4.3 Co-citation Network

The co-citation network is a weighted paper-level network that connects papers which are cited together by later papers. Whereas bibliographic coupling links papers through shared references, co-citation links papers through shared citers. This means that co-citation captures how later literature groups papers together.

For each paper a , let $C_{\text{final}}(a)$ denote the set of papers that cite a in the final filtered citation graph:

$$C_{\text{final}}(a) = \{i \in \mathcal{P}_{\text{final}} : (i, a) \in \mathcal{E}_{\text{final}}\}. \quad (4.25)$$

Only papers with at least five citing papers in the final filtered graph are used as target papers for co-citation construction.

The raw co-citation strength is the number of shared citing papers:

$$w_{co}(a, b) = |C_{\text{final}}(a) \cap C_{\text{final}}(b)|. \quad (4.26)$$

The Jaccard similarity between citer sets is also computed:

$$J_{\text{co}}(a, b) = \frac{|C_{\text{final}}(a) \cap C_{\text{final}}(b)|}{|C_{\text{final}}(a) \cup C_{\text{final}}(b)|}. \quad (4.27)$$

Jaccard similarity is used for candidate generation and secondary ranking, while the main retained edge weight is the raw shared-citing-paper count $w_{\text{co}}(a, b)$.

The resulting co-citation graph is an undirected weighted graph:

$$G_{\text{co}} = (\mathcal{P}_{\text{final}}, \mathcal{E}_{\text{co}}, w_{\text{co}}). \quad (4.28)$$

A pair is retained only if it has at least five shared citing papers:

$$\{a, b\} \in \mathcal{E}_{\text{co}} \iff w_{\text{co}}(a, b) \geq 5. \quad (4.29)$$

Computing all pairwise co-citation overlaps is infeasible at corpus scale. The implementation therefore uses MinHash locality-sensitive hashing to generate candidate pairs based on approximate Jaccard similarity between citer sets. Each paper is represented by a MinHash signature of its final filtered citer set, using 128 permutations. Candidate neighbors are retrieved using an LSH threshold of 0.20, and exact co-citation overlap statistics are then computed for candidate pairs.

After candidate generation and exact overlap computation, the graph is sparsified into a top- K neighborhood representation. For each paper a , retained neighbors are sorted primarily by shared-citing-paper count and secondarily by Jaccard similarity. The co-citation neighborhood is defined as

$$\mathcal{N}_{\text{co}}(a) = \text{TopK}_b \{b \in \mathcal{P}_{\text{final}} \setminus \{a\} : w_{\text{co}}(a, b) \geq 5\}. \quad (4.30)$$

In the final implementation, $K = 30$. If fewer than 30 valid neighbors are available for a paper, all valid neighbors are retained. For co-citation, the lower LSH threshold of 0.20 and larger neighborhood size $K = 30$ were used to retain more candidate relationships in the substantially sparser and more selective co-citation representation.

A key interpretive caveat is that co-citation is future-informed. The co-citation neighborhood of a paper depends on later papers that cite it together with other papers. Therefore, co-citation-based results should not be interpreted as prospective predictions available at publication time. Instead, they are used as retrospective robustness checks.

4.4.4 Citation-Lag and Taxonomy Annotations

Citation-lag and taxonomy-based representations are treated as annotations of the final filtered direct citation graph rather than as separate paper-similarity networks. For a citation edge $(i, j) \in \mathcal{E}_{\text{final}}$, where paper i cites paper j , the citation lag is

$$\ell(i, j) = y_i - y_j. \quad (4.31)$$

Citation-lag analyses use this edge-level annotation to distinguish short-term citation attention from longer-term citation memory. The full observed/expected calculation for citation-lag buckets is defined later in Section 4.9.

The taxonomy-based representation uses the OpenAlex hierarchy associated with each paper. For paper i , let $\tau(i)$ denote its topic, $\sigma(i)$ its subfield, $\phi(i)$ its field, and $\delta(i)$ its domain. These labels are used both to define structural comparison sets and to assign edge-level topical relevance weights.

For weighted topical relevance analyses, each citation edge receives a weight based on the deepest shared level of the OpenAlex taxonomy:

$$\omega_{\text{tax}}(i, j) = \begin{cases} 1, & \tau(i) = \tau(j), \\ \alpha, & \sigma(i) = \sigma(j), \tau(i) \neq \tau(j), \\ \beta, & \phi(i) = \phi(j), \sigma(i) \neq \sigma(j), \\ \gamma, & \delta(i) = \delta(j), \phi(i) \neq \phi(j), \\ 0, & \text{otherwise.} \end{cases} \quad (4.32)$$

The constraints in the lower rows make the categories mutually exclusive. Thus, each citation is assigned exactly one relevance level: same-topic, same-subfield only, same-field only, same-domain only, or outside the shared taxonomy hierarchy. Subfield labels are also used to aggregate final filtered citation edges into subfield-level flow matrices for exploratory analysis.

4.5 Direct Citation Baselines

The direct citation baselines quantify citation imbalance on the final filtered direct citation graph. Their purpose is to compare the observed gender distribution of cited Computer Science papers with the gender distribution that would be expected under progressively more specific structural baselines. Unless stated otherwise, the direct baselines are computed on the All→CS slice; the CS→CS slice is used as a stricter within-field specification.

Each baseline defines a comparison set $\mathcal{R}_b(j)$ for the cited paper j , where b denotes the baseline. This comparison set contains known-gender CS papers that are treated as structurally comparable to j . The expected probability that a citation to a paper comparable to j would point to gender category g is

$$p_{j,g}^{(b)} = \frac{\sum_{q \in \mathcal{R}_b(j)} \mathbf{1}(c_q = g)}{|\mathcal{R}_b(j)|}. \quad (4.33)$$

The observed and expected counts are

$$O_g = \sum_{(i,j) \in \mathcal{E}^*} \mathbf{1}(c_j = g), \quad (4.34)$$

$$E_g^{(b)} = \sum_{(i,j) \in \mathcal{E}^*} p_{j,g}^{(b)}, \quad (4.35)$$

where $\mathcal{E}^*$ denotes the selected final filtered citation slice restricted to edges whose cited paper belongs to $\mathcal{P}_{CS}^K$. The observed/expected ratio for baseline b is

$$\text{OE}_g^{(b)} = \frac{O_g}{E_g^{(b)}}. \quad (4.36)$$

The four direct baselines differ only in how the comparison set $\mathcal{R}_b(j)$ is defined:

Baseline	Comparison set	Interpretation
D0	$\mathcal{R}_{D0}(j) = \mathcal{P}_{CS}^K$	overall known-gender CS paper share
D1	$\mathcal{R}_{D1}(j) = \{q \in \mathcal{P}_{CS}^K : d(q) = d(j)\}$	same publication decade
D2	$\mathcal{R}_{D2}(j) = \{q \in \mathcal{P}_{CS}^K : d(q) = d(j), \sigma(q) = \sigma(j)\}$	same decade and subfield
D3	$\mathcal{R}_{D3}(j) = \{q \in \mathcal{P}_{CS}^K : d(q) = d(j), v(q) = v(j)\}$	same decade and venue-rank stratum

D0 provides the broad corpus-level reference point: it asks whether gender categories receive citations in proportion to their overall presence among known-gender CS papers. D1 adds publication

decade and therefore adjusts for broad exposure-time differences and long-term changes in gender composition. D2 adds subfield composition, which controls for differences in citation environments across areas of Computer Science. D3 instead conditions on venue rank, which captures venue-level visibility and prestige effects.

D2 and D3 are kept as separate baselines rather than combined into a fully crossed decade-subfield-venue-rank baseline. A fully crossed baseline would create many small comparison cells, especially for smaller gender categories and less common venue-rank strata. Keeping D2 and D3 separate also makes the interpretation clearer: D2 isolates topical composition, while D3 isolates venue-rank visibility.

Together, D0-D3 provide a progression from broad paper availability to increasingly structured metadata-based expectations. If a gender category remains below expected across this sequence, the result remains directionally consistent after conditioning on overall representation, publication decade, subfield composition, and venue-rank structure.

4.6 Topical Relevance Baselines

The topical relevance baselines test whether the observed citation imbalance changes when citation edges are restricted or weighted according to the topical relationship between the citing and cited papers. The direct citation baselines define expected citations using structural properties of cited papers, whereas the topical relevance baselines focus on the citation edge itself: they ask whether the imbalance differs for citations that are topically close to the citing paper compared with citations that cross topical boundaries.

Let $\mathcal{E}^*$ denote the selected final filtered citation slice. The topical relevance baselines use the same observed/expected logic as the direct baselines, but the expectation is fixed to the D2-style decade-and-subfield baseline:

$$p_{j,g}^{(D2)} = \frac{\sum_{q \in \mathcal{R}_{D2}(j)} \mathbf{1}(c_q = g)}{|\mathcal{R}_{D2}(j)|}. \quad (4.37)$$

Using the same D2-style expectation across all topical relevance variants ensures that differences between variants are driven by the topical restriction or weighting of citation edges, rather than by changing the expected-citation baseline itself.

The hard topical categories are defined as follows:

$$\mathcal{E}_{\text{same topic}}^* = \{(i, j) \in \mathcal{E}^* : \tau(i) = \tau(j)\}, \quad (4.38)$$

$$\mathcal{E}_{\text{same subfield}}^* = \{(i, j) \in \mathcal{E}^* : \sigma(i) = \sigma(j)\}, \quad (4.39)$$

$$\mathcal{E}_{\text{cross topic, same subfield}}^* = \{(i, j) \in \mathcal{E}^* : \sigma(i) = \sigma(j), \tau(i) \neq \tau(j)\}, \quad (4.40)$$

$$\mathcal{E}_{\text{cross subfield}}^* = \{(i, j) \in \mathcal{E}^* : \sigma(i) \neq \sigma(j)\}. \quad (4.41)$$

For any topical edge subset $\mathcal{E}_\kappa^*$, where κ denotes the topical category, the observed and expected counts are

$$O_{g,\kappa} = \sum_{(i,j) \in \mathcal{E}_\kappa^*} \mathbf{1}(c_j = g), \quad (4.42)$$

$$E_{g,\kappa}^{(D2)} = \sum_{(i,j) \in \mathcal{E}_\kappa^*} p_{j,g}^{(D2)}. \quad (4.43)$$

The corresponding observed/expected ratio is

$$\text{OE}_{g,\kappa}^{(D2)} = \frac{O_{g,\kappa}}{E_{g,\kappa}^{(D2)}}. \quad (4.44)$$

The taxonomy-weighted variant uses weighted citation edges rather than hard subsets. The relevance weight $\omega_{\text{tax}}(i, j)$ follows Equation 4.32. In the final implementation, the weights are set to

$$(\alpha, \beta, \gamma) = (0.75, 0.50, 0.25). \quad (4.45)$$

Thus, same-topic citations receive weight 1.00, same-subfield-only citations receive weight 0.75, same-field-only citations receive weight 0.50, same-domain-only citations receive weight 0.25, and citations without a shared taxonomy level receive weight 0.

For the taxonomy-weighted topical relevance baseline, the observed and expected weighted citation counts are

$$O_{g,\text{tax}} = \sum_{(i,j) \in \mathcal{E}^*} \omega_{\text{tax}}(i, j) \mathbf{1}(c_j = g), \quad (4.46)$$

$$E_{g,\text{tax}}^{(\text{D2})} = \sum_{(i,j) \in \mathcal{E}^*} \omega_{\text{tax}}(i, j) p_{j,g}^{(\text{D2})}. \quad (4.47)$$

The taxonomy-weighted observed/expected ratio is

$$\text{OE}_{g,\text{tax}}^{(\text{D2})} = \frac{O_{g,\text{tax}}}{E_{g,\text{tax}}^{(\text{D2})}}. \quad (4.48)$$

This weighted variant tests whether the citation imbalance persists when more topically central citations are emphasized. Unlike hard topical subsets, it does not discard all cross-area citations; instead, it downweights them according to their distance in the taxonomy hierarchy.

4.7 Network-Neighborhood Baselines

The network-neighborhood baselines test whether citation imbalance remains visible when each paper is compared with locally similar papers in a citation-derived network representation. Unlike the direct baselines, which define expected citations from metadata strata, the network-neighborhood baselines define expected paper-level citation counts using bibliographic-coupling and co-citation neighborhoods.

For a known-gender Computer Science paper j , let Y_j denote the observed number of incoming citations received by j in the selected final filtered citation slice:

$$Y_j = \sum_{(i,j) \in \mathcal{E}^*} 1. \quad (4.49)$$

Here, $\mathcal{E}^*$ is restricted to citations whose target is a known-gender CS paper.

For a network representation $\nu \in \{\text{bc}, \text{co}\}$, let $\mathcal{N}_\nu(j)$ denote the retained neighborhood of paper j , and let $w_\nu(j, q)$ denote the network edge weight between j and neighbor q . The valid known-gender CS neighborhood is denoted by $\mathcal{N}_\nu^K(j)$. The expected citation count for paper j is the weighted average of the observed citation counts of its valid neighbors:

$$\hat{Y}_j^{(\nu)} = \frac{\sum_{q \in \mathcal{N}_\nu^K(j)} w_\nu(j, q) Y_q}{\sum_{q \in \mathcal{N}_\nu^K(j)} w_\nu(j, q)}. \quad (4.50)$$

Papers without any valid weighted neighbors do not receive a network-neighborhood expected value and are excluded from that baseline’s aggregation.

Because the sum of neighbor-smoothed expected counts may not exactly equal the total observed citation count in the covered paper set, expected values are calibrated by a single multiplicative

factor. Let $\mathcal{P}_\nu^K$ denote the set of known-gender CS papers with valid expected values under representation ν . The calibration factor is

$$s_\nu = \frac{\sum_{j \in \mathcal{P}_\nu^K} Y_j}{\sum_{j \in \mathcal{P}_\nu^K} \hat{Y}_j^{(\nu)}}. \quad (4.51)$$

The calibrated expected citation count is

$$\tilde{Y}_j^{(\nu)} = s_\nu \hat{Y}_j^{(\nu)}. \quad (4.52)$$

Observed and expected counts are then aggregated by gender category:

$$O_g^{(\nu)} = \sum_{j \in \mathcal{P}_\nu^K} Y_j \mathbf{1}(c_j = g), \quad (4.53)$$

$$E_g^{(\nu)} = \sum_{j \in \mathcal{P}_\nu^K} \tilde{Y}_j^{(\nu)} \mathbf{1}(c_j = g). \quad (4.54)$$

The observed/expected ratio is

$$\text{OE}_g^{(\nu)} = \frac{O_g^{(\nu)}}{E_g^{(\nu)}}. \quad (4.55)$$

N1 uses the bibliographic-coupling network. Its valid neighborhood $\mathcal{N}_{\text{bc}}^K(j)$ consists of known-gender CS neighbors in the retained bibliographic-coupling graph, and the edge weight $w_{\text{bc}}(j, q)$ is the shared-reference count. The N1 expected citation count is therefore

$$\hat{Y}_j^{(\text{bc})} = \frac{\sum_{q \in \mathcal{N}_{\text{bc}}^K(j)} w_{\text{bc}}(j, q) Y_q}{\sum_{q \in \mathcal{N}_{\text{bc}}^K(j)} w_{\text{bc}}(j, q)}. \quad (4.56)$$

N1 compares papers with similar reference foundations and is backward-looking because bibliographic coupling is based on reference lists fixed at publication time.

N2 uses the co-citation network. Its valid neighborhood $\mathcal{N}_{\text{co}}^K(j)$ consists of known-gender CS neighbors in the retained co-citation graph, and the edge weight $w_{\text{co}}(j, q)$ is the shared-citing-paper count. The N2 expected citation count is

$$\hat{Y}_j^{(\text{co})} = \frac{\sum_{q \in \mathcal{N}_{\text{co}}^K(j)} w_{\text{co}}(j, q) Y_q}{\sum_{q \in \mathcal{N}_{\text{co}}^K(j)} w_{\text{co}}(j, q)}. \quad (4.57)$$

N2 compares papers that later literature cites together. It is therefore future-informed and is interpreted as a retrospective robustness check rather than as a prospective model of citation opportunity.

This construction differs from edge-level randomization. The network-neighborhood baselines do not reassign individual citation edges to alternative targets. Instead, they estimate each paper's expected citation reception from the citation reception of its local network neighbors.

4.8 GLM Expected-Citation Models

The GLM expected-citation models provide a model-based alternative to the direct, topical, and network-neighborhood baselines. Instead of defining expected citation counts only from discrete strata or local network neighborhoods, these models estimate the expected number of citations received by each paper as a function of structural and network-derived covariates. Gender is deliberately excluded from all GLM covariates. The fitted citation counts are then aggregated by gender category using the same observed/expected framework as the other baselines.

For a paper $j \in \mathcal{P}_{CS}^K$, let Y_j denote the observed number of incoming citations received by paper j within the selected citation slice:

$$Y_j = \sum_{(i,j) \in \mathcal{E}^*} 1. \quad (4.58)$$

The GLM models use a Poisson specification for citation counts:

$$Y_j \sim \text{Poisson}(\mu_j), \quad (4.59)$$

with log link

$$\log(\mu_j) = \eta_j. \quad (4.60)$$

For model m , the linear predictor is

$$\eta_j^{(m)} = \beta_0^{(m)} + \mathbf{x}_j^{(m)\top} \boldsymbol{\beta}^{(m)}. \quad (4.61)$$

Categorical variables are one-hot encoded. Continuous count-valued structural and network features are transformed using $\log(1 + x)$ where specified below and are subsequently standardized before model fitting. The implementation uses an L2-regularized Poisson GLM with regularization parameter $\alpha = 1.0$. L2 regularization is used to stabilize estimation in the presence of correlated structural and network-derived predictors, as motivated by the feature diagnostics in Section 5.6.3. The regularization strength $\alpha = 1.0$ is treated as a fixed moderate regularization setting rather than as a tuned inferential parameter. The GLM models are used primarily to generate expected-citation predictions rather than to interpret individual coefficients.

The fitted value for paper j under model m is

$$\hat{\mu}_j^{(m)} = \exp(\hat{\eta}_j^{(m)}). \quad (4.62)$$

Because fitted totals may not exactly equal the observed total number of citations in the slice, predictions are calibrated by a single multiplicative factor:

$$s^{(m)} = \frac{\sum_{j \in \mathcal{P}_{CS}^K} Y_j}{\sum_{j \in \mathcal{P}_{CS}^K} \hat{\mu}_j^{(m)}}. \quad (4.63)$$

The calibrated expected citation count is

$$\tilde{\mu}_j^{(m)} = s^{(m)} \hat{\mu}_j^{(m)}. \quad (4.64)$$

Because the same scalar $s^{(m)}$ is applied to every paper within a model and citation slice, this calibration preserves the relative fitted expectations between papers. It changes only the common scale of the predictions so that their aggregate total matches the observed citation total; it does not alter their ranking or the relative expected citation shares of the gender categories.

The structural feature block used in M1 is

$$\mathbf{x}_j^{(M1)} = [d(j), z(j), v(j), \log(1 + \rho(j))], \quad (4.65)$$

where $d(j)$ denotes publication decade, $z(j)$ denotes the subfield or topic stratum, $v(j)$ denotes the venue-rank stratum, and $\rho(j)$ denotes the recorded reference-list size of paper j . Thus, M1 captures temporal, topical, venue-based, and reference-list structure without using any citation-network position variables.

The outgoing direct-citation feature used in M2 is

$$h_{j,\text{dir}} = \log(1 + d_{\text{out}}^{\text{CS}}(j)), \quad (4.66)$$

where $d_{\text{out}}^{\text{CS}}(j)$ denotes the number of unique Computer Science papers cited by paper j in the final CS→CS citation slice. Since the final citation edge list contains at most one edge for each ordered citing–cited paper pair, this quantity is also equal to the number of outgoing edges from j in that slice.

For bibliographic coupling and co-citation, the same paper-level feature template is used. For $\nu \in \{\text{bc}, \text{co}\}$, the retained unweighted degree, weighted strength, mean incident edge weight, and maximum incident edge weight are defined as

$$k_j^{(\nu)} = |\mathcal{N}_\nu(j)|, \quad (4.67)$$

$$s_j^{(\nu)} = \sum_{q \in \mathcal{N}_\nu(j)} w_\nu(j, q), \quad (4.68)$$

$$\bar{w}_j^{(\nu)} = \frac{s_j^{(\nu)}}{k_j^{(\nu)}}, \quad (4.69)$$

$$w_{j,\text{max}}^{(\nu)} = \max_{q \in \mathcal{N}_\nu(j)} w_\nu(j, q). \quad (4.70)$$

For papers with no retained neighbors, these quantities are set to zero. The bibliographic-coupling feature block is

$$\mathbf{h}_{j,\text{bc}} = [\log(1 + k_j^{(\text{bc})}), \log(1 + s_j^{(\text{bc})}), \log(1 + \bar{w}_j^{(\text{bc})}), \log(1 + w_{j,\text{max}}^{(\text{bc})})], \quad (4.71)$$

and the co-citation feature block is

$$\mathbf{h}_{j,\text{co}} = [\log(1 + k_j^{(\text{co})}), \log(1 + s_j^{(\text{co})}), \log(1 + \bar{w}_j^{(\text{co})}), \log(1 + w_{j,\text{max}}^{(\text{co})})]. \quad (4.72)$$

Thus, both citation-derived similarity representations enter the GLM through four explicitly defined paper-level quantities: retained degree, weighted strength, mean incident edge weight, and maximum incident edge weight. All four are $\log(1 + x)$ transformed before standardization.

The five GLM variants differ only in which feature blocks are added to the common structural specification:

Model	Feature blocks	Interpretation
M1	Structural	Decade, topical stratum, venue rank, reference-list size
M2	M1 + direct	M1 plus direct out-degree
M3	M1 + BC	M1 plus BC degree, strength, mean, and maximum weight
M4	M1 + co-citation	M1 plus co-citation degree, strength, mean, and maximum weight
M5	M1 + direct + BC + co-citation	All structural and network feature blocks

Equivalently, their covariate vectors are

$$\begin{aligned} M1 &: \mathbf{x}_j^{(M1)}, \\ M2 &: [\mathbf{x}_j^{(M1)}, \mathbf{h}_{j,\text{dir}}], \\ M3 &: [\mathbf{x}_j^{(M1)}, \mathbf{h}_{j,\text{bc}}], \\ M4 &: [\mathbf{x}_j^{(M1)}, \mathbf{h}_{j,\text{co}}], \\ M5 &: [\mathbf{x}_j^{(M1)}, \mathbf{h}_{j,\text{dir}}, \mathbf{h}_{j,\text{bc}}, \mathbf{h}_{j,\text{co}}]. \end{aligned}$$

M1 is therefore the structural reference model. M2 tests whether outgoing direct-citation behavior changes the expected-citation mapping beyond structural characteristics. M3 adds backward-looking bibliographic-coupling position, while M4 adds future-informed co-citation position. M5 combines all available structural and network-derived feature blocks.

After fitting a GLM model and obtaining calibrated expected citation counts, observed and expected citations are aggregated by gender category:

$$O_g^{(m)} = \sum_{j \in \mathcal{P}_{CS}^K} Y_j \mathbf{1}(c_j = g), \quad (4.73)$$

$$E_g^{(m)} = \sum_{j \in \mathcal{P}_{CS}^K} \tilde{\mu}_j^{(m)} \mathbf{1}(c_j = g). \quad (4.74)$$

The GLM observed/expected ratio is

$$\text{OE}_g^{(m)} = \frac{O_g^{(m)}}{E_g^{(m)}}. \quad (4.75)$$

In addition to the full-data fits, model performance is evaluated using a common 80/20 paper-level train/test split for M1-M5. Model estimation and the multiplicative calibration factor are obtained from the training subset only, and the resulting calibration factor is applied unchanged to the held-out predictions. Held-out performance is evaluated using mean Poisson deviance, Poisson D^2 , mean absolute error, median absolute error, and aggregate calibration. This validation evaluates whether the fitted expected-citation mapping generalizes to papers not used during model estimation; it should not be interpreted as prospective forecasting, particularly for models containing future-informed co-citation features.

The GLM baselines are descriptive expected-citation models. Their coefficients are not the main object of interpretation in this thesis; the focus is on gender-aggregated observed/expected ratios derived from the calibrated fitted values. M1 and M3 rely only on structural or backward-looking information, whereas M4 and M5 include future-informed co-citation features and are therefore interpreted as retrospective robustness checks. The Poisson GLM is used as a scalable approximation for citation counts, which are typically overdispersed; for this reason, the GLM results are interpreted together with the nonparametric baselines and randomized robustness checks.

4.9 Long-Memory Citation-Lag Analysis

The long-memory citation-lag analysis evaluates whether gender citation imbalance differs across citation age. The analysis is computed on final filtered citation slices, such as All→CS and CS→CS, with cited papers restricted to known-gender Computer Science papers. The aim is to compare observed citations to expected citations among papers that were eligible to be cited at the same temporal distance from the citing paper.

For a citation edge $(i, j) \in \mathcal{E}^*$, paper i is the citing paper and paper j is the cited paper. The citation lag is

$$\ell(i, j) = y_i - y_j. \quad (4.76)$$

Citation edges with missing publication years are excluded from the long-memory analysis because their lag cannot be computed. Edges with negative citation lags are also excluded.

Citation lags are grouped into fine-grained buckets:

$$B(i, j) = \begin{cases} 0-2, & 0 \leq \ell(i, j) \leq 2, \\ 3-5, & 3 \leq \ell(i, j) \leq 5, \\ 6-10, & 6 \leq \ell(i, j) \leq 10, \\ 11-20, & 11 \leq \ell(i, j) \leq 20, \\ 20+, & \ell(i, j) > 20. \end{cases} \quad (4.77)$$

These cut points distinguish early citation accumulation (0-5 years), medium-term visibility (6-10 years), and longer-term citation memory (more than 10 years), while retaining finer resolution within the early and long-lag portions of the distribution. The 10-year boundary is also consistent with the strong concentration of observed citation lags within the first decade shown in Section 5.5.1.

For each lag bucket b , the corresponding citation edge subset is

$$\mathcal{E}_b^* = \{(i, j) \in \mathcal{E}^* : B(i, j) = b\}. \quad (4.78)$$

Expected counts are computed within citing-year and lag-bucket cells. For a citing year y and lag bucket b , the eligible known-gender CS paper set is

$$\mathcal{A}_{y,b}^K = \{q \in \mathcal{P}_{CS}^K : y - y_q \in b\}. \quad (4.79)$$

Within each citing-year-lag-bucket cell, the expected gender share is

$$p_{y,b,g} = \frac{\sum_{q \in \mathcal{A}_{y,b}^K} \mathbf{1}(c_q = g)}{|\mathcal{A}_{y,b}^K|}. \quad (4.80)$$

For gender category g and lag bucket b , the observed and expected counts are

$$O_{g,b} = \sum_{(i,j) \in \mathcal{E}_b^*} \mathbf{1}(c_j = g), \quad (4.81)$$

$$E_{g,b} = \sum_{(i,j) \in \mathcal{E}_b^*} p_{y_i,b,g}. \quad (4.82)$$

The observed/expected ratio is

$$\text{OE}_{g,b} = \frac{O_{g,b}}{E_{g,b}}. \quad (4.83)$$

For summary interpretation, the fine lag buckets are also grouped into broader memory categories:

$$M(i, j) = \begin{cases} \text{short}, & 0 \leq \ell(i, j) \leq 5, \\ \text{middle}, & 6 \leq \ell(i, j) \leq 10, \\ \text{long}, & \ell(i, j) > 10. \end{cases} \quad (4.84)$$

The same observed/expected calculation is applied to these aggregated memory groups. The short-memory category captures early citation accumulation, the middle category captures medium-term visibility, and the long-memory category captures citations to work that remains visible more than ten years after publication.

4.10 Randomized Robustness Checks

The randomized robustness checks provide additional null models for evaluating whether the observed gender citation imbalance is stable under alternative assumptions about how citation targets could have been selected. Unlike the direct baselines, which compute expected citation counts from metadata strata, the randomized checks define explicit replacement mechanisms for citation edges.

4.10.1 Randomized Reference Drawing

The randomized reference drawing baseline defines expected citation targets from temporally eligible known-gender Computer Science papers. For an observed citation edge $(i, j) \in \mathcal{E}^*$, let y_i denote the publication year of the citing paper. The baseline constructs an eligibility set $\mathcal{A}_i^K$ of known-gender CS papers that could have been cited by paper i under a specified temporal rule.

The all-prior eligibility set is

$$\mathcal{A}_{i, \text{all prior}}^K = \{q \in \mathcal{P}_{CS}^K : y_q < y_i\}. \quad (4.85)$$

The age-restricted eligibility set is

$$\mathcal{A}_{i, \text{age } 0-10}^K = \{q \in \mathcal{P}_{CS}^K : 0 \leq y_i - y_q \leq 10\}. \quad (4.86)$$

The first rule compares each citation with the full pool of earlier known-gender CS papers; the second limits the pool to relatively recent papers. The 10-year window is motivated by the empirical citation-lag distribution reported in Section 5.5.1, where approximately 73% of All→CS citations occur within ten years of publication. It therefore provides a recent-work opportunity set alongside the less restrictive all-prior specification.

For each eligibility rule, two drawing schemes are used. Under uniform drawing, every eligible paper receives equal probability:

$$\Pr_{\text{uniform}}(q | i) = \frac{1}{|\mathcal{A}_i^K|}. \quad (4.87)$$

The expected probability of drawing gender category g is

$$p_{i,g}^{\text{uniform}} = \sum_{q \in \mathcal{A}_i^K} \Pr_{\text{uniform}}(q | i) \mathbf{1}(c_q = g). \quad (4.88)$$

Under prior-citation-weighted drawing, papers that had already accumulated more citations before the citing paper was published receive more weight. Let $H(q, y_i)$ denote the number of citations received by paper q before year y_i . The prior-citation weight is

$$\omega(q, y_i) = H(q, y_i) + 1. \quad (4.89)$$

The additive constant ensures that eligible papers with zero prior citations retain a non-zero probability of being selected, while preserving the monotonic weighting of candidates by prior citation visibility.

The weighted drawing probability is

$$\Pr_{\text{prior}}(q | i) = \frac{\omega(q, y_i)}{\sum_{u \in \mathcal{A}_i^K} \omega(u, y_i)}. \quad (4.90)$$

The expected probability of drawing gender category g is

$$p_{i,g}^{\text{prior}} = \sum_{q \in \mathcal{A}_i^K} \Pr_{\text{prior}}(q | i) \mathbf{1}(c_q = g). \quad (4.91)$$

For a given eligibility rule and drawing scheme, the observed and expected counts are

$$O_g^{\text{draw}} = \sum_{(i,j) \in \mathcal{E}^*} \mathbf{1}(c_j = g), \quad (4.92)$$

$$E_g^{\text{draw}} = \sum_{(i,j) \in \mathcal{E}^*} p_{i,g}. \quad (4.93)$$

Here, $p_{i,g}$ refers either to $p_{i,g}^{\text{uniform}}$ or $p_{i,g}^{\text{prior}}$, depending on the specification. The observed/expected ratio is

$$\text{OE}_g^{\text{draw}} = \frac{O_g^{\text{draw}}}{E_g^{\text{draw}}}. \quad (4.94)$$

The uniform variant controls only for temporal eligibility, while the prior-citation-weighted variant additionally accounts for cumulative advantage by making already-visible papers more likely to be drawn.

4.10.2 Network-Conditioned Edge Randomization

The network-conditioned edge randomization baseline defines replacement targets from local network neighborhoods. Unlike N1 and N2, which estimate paper-level expected citation counts from neighbor citation outcomes, this robustness check operates at the citation-edge level. For each observed citation edge (i, j) , the cited paper j is replaced probabilistically by one of its valid network neighbors.

Let $\nu \in \{\text{bc}, \text{co}\}$ denote the network representation. For an observed citation edge (i, j) , the valid replacement neighborhood is

$$\mathcal{R}_\nu^K(j) = \{q \in \mathcal{N}_\nu(j) : q \in \mathcal{P}_{CS}^K, q \neq j, d(q) = d(j)\}. \quad (4.95)$$

A replacement candidate must be a known-gender CS paper, must not be the cited paper itself, and must belong to the same publication decade as the cited paper.

A citation edge is replaceable under representation ν if its cited paper has at least one valid replacement candidate:

$$(i, j) \in \mathcal{E}_{\nu, \text{rep}}^* \iff |\mathcal{R}_\nu^K(j)| > 0. \quad (4.96)$$

Only replaceable edges are included in the network-conditioned randomization.

For a replaceable edge (i, j) , the probability of replacing cited paper j with candidate q is proportional to the network edge weight:

$$\Pr_\nu(q | j) = \frac{w_\nu(j, q)}{\sum_{u \in \mathcal{R}_\nu^K(j)} w_\nu(j, u)}. \quad (4.97)$$

The expected probability that a randomized replacement target belongs to gender category g is

$$p_{j,g}^{(\nu)} = \sum_{q \in \mathcal{R}_\nu^K(j)} \Pr_\nu(q | j) \mathbf{1}(c_q = g). \quad (4.98)$$

The observed and expected counts are

$$O_g^{(\nu, \text{rep})} = \sum_{(i,j) \in \mathcal{E}_{\nu, \text{rep}}^*} \mathbf{1}(c_j = g), \quad (4.99)$$

$$E_g^{(\nu, \text{rep})} = \sum_{(i,j) \in \mathcal{E}_{\nu, \text{rep}}^*} p_{j,g}^{(\nu)}. \quad (4.100)$$

The observed/expected ratio is

$$\text{OE}_g^{(\nu, \text{rep})} = \frac{O_g^{(\nu, \text{rep})}}{E_g^{(\nu, \text{rep})}}. \quad (4.101)$$

Prior-citation-weighted bibliographic-coupling sensitivity. As an additional sensitivity specification, the bibliographic-coupling randomization is repeated after incorporating the prior citation visibility of candidate replacement papers. The valid replacement set is unchanged: candidates must still be known-gender Computer Science papers, belong to the same publication decade as the observed cited paper, and occur in its retained bibliographic-coupling neighborhood.

For a replaceable citation edge (i, j) and candidate paper $q \in \mathcal{R}_{bc}^K(j)$, let $H(q, y_i)$ denote the number of citations received by q before the publication year y_i of the citing paper, as defined for the prior-citation-weighted reference-drawing specification. The modified replacement weight is

$$\tilde{w}_{bc}(i, j, q) = w_{bc}(j, q) [H(q, y_i) + 1]. \quad (4.102)$$

The corresponding replacement probability is

$$\Pr_{bc, \text{prior}}(q \mid i, j) = \frac{\tilde{w}_{bc}(i, j, q)}{\sum_{u \in \mathcal{R}_{bc}^K(j)} \tilde{w}_{bc}(i, j, u)}. \quad (4.103)$$

The expected probability of a replacement target belonging to gender category g is therefore

$$p_{i, j, g}^{(bc, \text{prior})} = \sum_{q \in \mathcal{R}_{bc}^K(j)} \Pr_{bc, \text{prior}}(q \mid i, j) \mathbf{1}(c_q = g). \quad (4.104)$$

Expected citation counts are then obtained over the same set of bibliographic-coupling-replaceable edges:

$$E_g^{(bc, \text{prior})} = \sum_{(i, j) \in \mathcal{E}_{bc, \text{rep}}^*} p_{i, j, g}^{(bc, \text{prior})}, \quad (4.105)$$

while the observed counts are unchanged from the standard bibliographic-coupling randomization. The resulting observed/expected ratio is

$$\text{OE}_g^{(bc, \text{prior})} = \frac{O_g^{(bc, \text{rep})}}{E_g^{(bc, \text{prior})}}. \quad (4.106)$$

This sensitivity specification separates two forms of information. Bibliographic coupling measures similarity through shared reference foundations and does not itself encode citation-based visibility. Multiplying the coupling weight by citations accumulated before the citing year therefore tests whether the bibliographic-coupling result is sensitive to prior visibility within the same similarity-conditioned replacement set.

An analogous prior-citation-weighted sensitivity specification is not applied to the co-citation randomization. Co-citation weights are themselves derived from citation activity and reflect later community association between papers. They are therefore future-informed relative to the focal citation event. Combining these weights with an additional time-local prior-citation term would mix two citation-derived visibility signals and would have a less direct interpretation as a separate null specification.

This randomization asks whether actual cited targets differ in gender composition from what would be expected if each citation were redirected to a same-decade, known-gender CS neighbor of the cited paper in the selected network representation.

4.10.3 Coverage of Replaceable Edges

The network-conditioned edge randomization is defined only for citation edges whose cited target has at least one valid replacement neighbor. Therefore, coverage must be reported alongside the observed/expected ratios.

For representation ν , edge coverage is the share of citation edges in the selected slice that are replaceable:

$$\text{Cov}_{\text{edge}}^{(\nu)} = \frac{|\mathcal{E}_{\nu, \text{rep}}^*|}{|\mathcal{E}^*|}. \quad (4.107)$$

Target coverage is the share of distinct cited papers that have at least one valid replacement neighborhood:

$$\text{Cov}_{\text{target}}^{(\nu)} = \frac{|\{j : \exists i, (i, j) \in \mathcal{E}_{\nu, \text{rep}}^*\}|}{|\{j : \exists i, (i, j) \in \mathcal{E}^*\}|}. \quad (4.108)$$

Coverage is especially important for interpreting co-citation-based randomization, because co-citation neighborhoods require papers to have sufficient later citing papers.

4.11 Interpreting Robustness Across Baselines

The baselines in this thesis are interpreted as complementary specifications of expected citation behavior. Each baseline defines a different comparison model: direct citation baselines use metadata-defined paper availability, topical relevance baselines restrict or weight citation edges by taxonomy-based proximity, network-neighborhood baselines compare papers with locally similar papers, GLM models estimate paper-level expected citation counts from structural and network-derived covariates, and randomized robustness checks define alternative citation targets under temporal or network-conditioned replacement rules.

Robustness is interpreted primarily in terms of direction rather than exact equality of magnitude. A result is considered directionally robust when the same gender category remains consistently above or below expected across substantively different baseline specifications. The exact observed/expected ratio is expected to vary because each baseline conditions on a different comparison structure. Therefore, a change in magnitude across baselines is substantively informative, but the central robustness question is whether the direction of the imbalance remains stable.

The All→CS and CS→CS slices are interpreted separately. The All→CS slice is the main specification because it measures citation reception by Computer Science papers from the broader observed corpus. The CS→CS slice is used as a stricter specification that restricts both citing and cited papers to Computer Science. Similar results across both slices indicate that the observed pattern persists when citations from outside Computer Science are excluded.

All gender-specific observed/expected ratios are computed on the known-gender subset of Computer Science papers. Papers with unknown or unresolved gender categories are excluded from gender-specific aggregation because they cannot be assigned to one of the known first-last author categories. The resulting estimates should therefore be interpreted as estimates for the known-gender CS subset rather than for the full CS corpus.

The baselines also differ in temporal interpretation. Bibliographic coupling is backward-looking because it is based on reference lists, while co-citation is future-informed because it depends on later papers citing two works together. Co-citation-based baselines are therefore interpreted as retrospective robustness checks rather than prospective models of citation opportunity at publication time.

Finally, the observed/expected comparisons are descriptive rather than causal. They quantify whether papers in each gender category receive more or fewer citations than expected under a specified comparison model. They do not identify the causal mechanism behind the imbalance.

The purpose of the multi-baseline design is to assess whether the same directional pattern remains visible under several different assumptions about citation opportunity, topical relevance, network similarity, temporal eligibility, and prior visibility.

5. Exploratory Data Analysis

5.1 Corpus and Metadata Coverage

This section describes the empirical corpus retained after preprocessing and assesses the coverage of the metadata variables used throughout the subsequent analyses. Particular attention is given to the temporal scope of the data, the Computer Science subset that defines the principal citation target population, and the availability of taxonomy, venue, reference-list, and author-gender information.

The raw corpus contained 3,720,575 publications. After the citation filtering procedure described in Chapter 4, including the removal of author-overlap self-citations, time-aware prior-coauthor citations, and papers that became isolated after edge filtering, the final analytical corpus contained 3,026,412 publications. Unless stated otherwise, the exploratory statistics in this chapter refer to this final analytical paper set.

5.1.1 Corpus Size, Time Span, and Computer Science Scope

The final analytical corpus consists of 3,026,412 publications observed between 1950 and 2024, covering 75 publication years and eight publication decades. Within this corpus, 1,542,691 publications are assigned to Computer Science according to their primary OpenAlex field, corresponding to 50.97% of all retained papers.

This distinction between the complete analytical corpus and the Computer Science subset is central to the empirical design. The main analysis uses an All→CS citation slice, in which citing publications may originate from any field represented in the retained corpus, whereas cited publications are restricted to Computer Science. The resulting main citation slice contains 11,601,416 directed citation edges. A stricter CS→CS slice, containing 9,413,932 citation edges, is retained for robustness analysis. The complete citation graph after all citation-level filtering contains 18,904,534 directed edges.

Table 5.1 summarizes the principal corpus quantities.

Table 5.1: Overview of the final analytical corpus and citation slices.

Statistic	Value
Papers before isolated-paper removal	3,720,575
Final analytical papers	3,026,412
Computer Science papers	1,542,691
Computer Science share	50.97%
Earliest publication year	1950
Latest publication year	2024
Observed publication years	75
Observed decades	8
Final citation edges	18,904,534
All→CS citation edges	11,601,416
CS→CS citation edges	9,413,932

Publication volume is highly uneven across time. Only 1,268 retained publications originate from the 1950s and 6,788 from the 1960s, compared with 644,650 publications from the 2000s and 1,318,613 from the 2010s. The 2010s alone account for 43.57% of the analytical corpus. A further 772,077 papers (25.51%) were published between 2020 and 2024. Consequently, more than two thirds of all retained publications were published from 2010 onward.

Figure 5.1 shows the annual publication counts. Publication volume increases strongly over the observation period, particularly from the 2000s onward. The apparent decline in 2024 should not

be interpreted as a substantive contraction in publication activity, because the publication data is available up to this year.

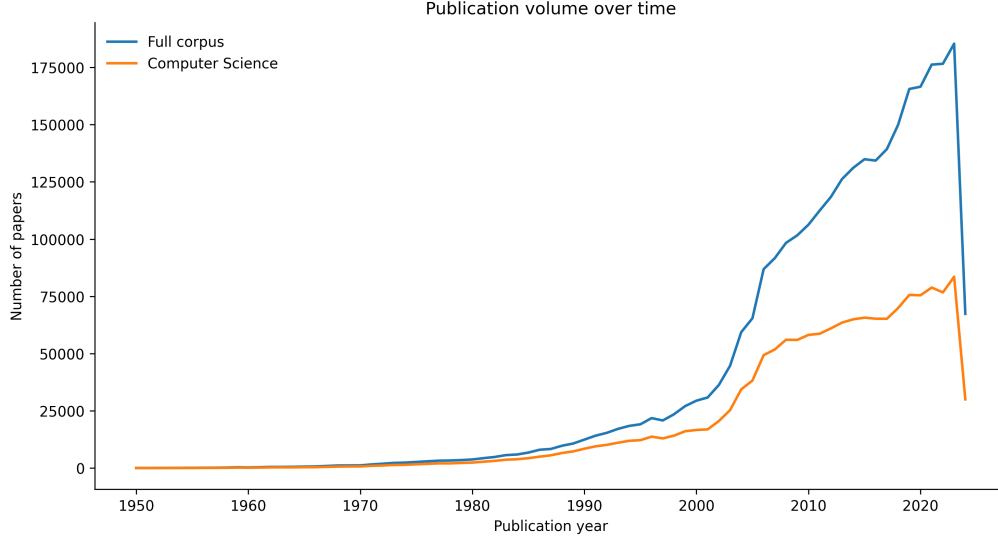


Figure 5.1: Annual publication volume in the final analytical corpus, 1950-2024.

The strong temporal imbalance is relevant for the analyses that follow. Publications from earlier periods have had a longer opportunity to accumulate citations, while both the size and composition of the available publication pool change substantially over time. For this reason, the descriptive patterns reported in this chapter are interpreted together with publication period, and the expected-citation baselines introduced in Chapter 4 explicitly incorporate temporal conditioning where appropriate.

5.1.2 Metadata Missingness and Gender Coverage

Coverage of the core bibliographic metadata is high in the final analytical corpus. Publication year, decade, taxonomy information, venue name, venue type, and the reference-list field are available for all 3,026,412 retained papers. The same complete coverage is observed within the Computer Science subset. Venue rank is the principal exception: a recognised venue-rank category is available for 1,614,963 publications, corresponding to 53.36% of the complete analytical corpus. Coverage is somewhat higher among Computer Science papers, where 879,692 of 1,542,691 publications, or 57.02%, have an available venue rank.

Gender metadata are less complete because gender classification requires a usable author name and sufficiently confident name-based inference. First-author gender is known for 2,428,923 papers (80.26%), while last-author gender is known for 2,485,836 papers (82.14%). Both author positions can be classified simultaneously for 2,110,021 papers, giving a complete first-last-author gender coverage of 69.72%. Within Computer Science the corresponding complete coverage is slightly higher at 70.10%, representing 1,081,363 papers.

Table 5.2 reports metadata availability for both the full analytical corpus and the Computer Science subset.

Table 5.2: Coverage of the principal metadata variables in the final analytical corpus and Computer Science subset.

Variable	Full corpus		Computer Science	
	Available N	Coverage	Available N	Coverage
Publication year	3,026,412	100.00%	1,542,691	100.00%
Decade	3,026,412	100.00%	1,542,691	100.00%
Taxonomy label	3,026,412	100.00%	1,542,691	100.00%
Venue type	3,026,412	100.00%	1,542,691	100.00%
Venue rank	1,614,963	53.36%	879,692	57.02%
Reference-list field	3,026,412	100.00%	1,542,691	100.00%
First-author gender	2,428,923	80.26%	1,241,641	80.49%
Last-author gender	2,485,836	82.14%	1,270,822	82.38%
Complete first-last gender	2,110,021	69.72%	1,081,363	70.10%

Gender-data coverage is not constant across publication periods. Figure 5.2 shows first-author, last-author, and complete first-last-author gender coverage by decade within the Computer Science subset. Complete first-last coverage is 77.09% in the 1950s and reaches its highest observed level of 80.82% in the 1980s. It subsequently declines to 74.42% in the 2000s, 69.32% in the 2010s, and 61.88% for papers published between 2020 and 2024.

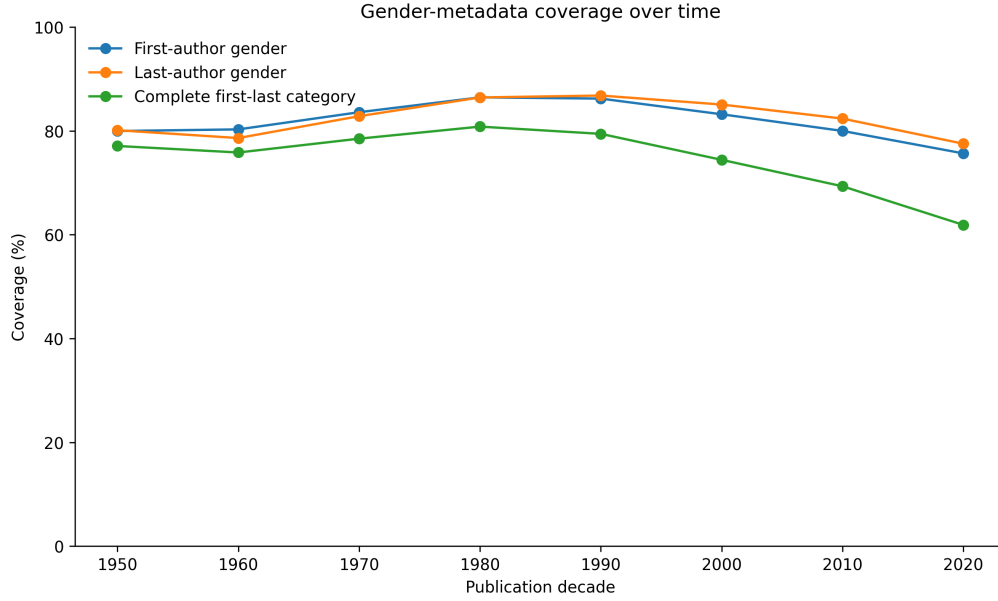


Figure 5.2: Coverage of first-author gender, last-author gender, and complete first-last-author gender classification by publication decade within the Computer Science subset.

The decline in complete gender coverage for the most recent publications is important because the principal gender-specific analyses require both the first and last author to be classified. Papers without complete first-last gender information remain part of the broader bibliographic and network representation where appropriate, but they cannot contribute as gender-classified cited papers to analyses distinguishing the MM, MW, WM, and WW categories. The similarity between overall gender coverage and Computer Science-specific coverage suggests that this limitation is not uniquely driven by the disciplinary restriction to Computer Science. Nevertheless, the temporal variation in classification coverage should be kept in mind when interpreting changes in gender composition over time.

5.2 Authorship, Gender, and Reference Characteristics

This section describes the authorship and referencing characteristics of the analytical corpus, with particular attention to the four gender categories used throughout the subsequent citation analyses. It first examines the overall composition of the MM, MW, WM, and WW groups, followed by their development across publication periods. The section then considers authorship-team size and reference-list length as additional descriptive characteristics of the publication corpus.

The gender categories used throughout the thesis are defined exclusively by the inferred gender of the first and last authors. They therefore do not describe the gender composition of the complete author team. Authorship-team size is examined only as a descriptive paper characteristic and does not alter the gender classification based on the first and last author.

5.2.1 First and Last Author Gender Composition

The analytical corpus is strongly concentrated in the MM category. Among the 2,110,021 papers for which both first-author and last-author gender can be classified, 1,506,965 papers are MM, representing 71.42% of the known-gender sample. WM papers form the second-largest category with 284,457 papers (13.48%), followed by 196,542 MW papers (9.31%) and 122,057 WW papers (5.78%).

A similar distribution is observed within Computer Science. Of the 1,081,363 Computer Science papers with complete gender information for both author positions, 790,617 are classified as MM, corresponding to 73.11% of the known-gender Computer Science sample. The remaining papers consist of 137,756 WM papers (12.74%), 97,399 MW papers (9.01%), and 55,591 WW papers (5.14%).

Table 5.3 compares the two distributions.

Table 5.3: Gender composition based on the first and last author among papers with complete gender information. Percentages are calculated within the corresponding known-gender sample.

Category	Full corpus		Computer Science	
	Papers	Share	Papers	Share
MM	1,506,965	71.42%	790,617	73.11%
MW	196,542	9.31%	97,399	9.01%
WM	284,457	13.48%	137,756	12.74%
WW	122,057	5.78%	55,591	5.14%

Figure 5.3 visualizes the same comparison. The ordering of the four categories is stable between the full corpus and Computer Science, although the Computer Science subset is somewhat more concentrated in MM papers. The MM share is 1.69 percentage points higher in Computer Science than in the full corpus. The MW, WM, and WW shares are all slightly lower.

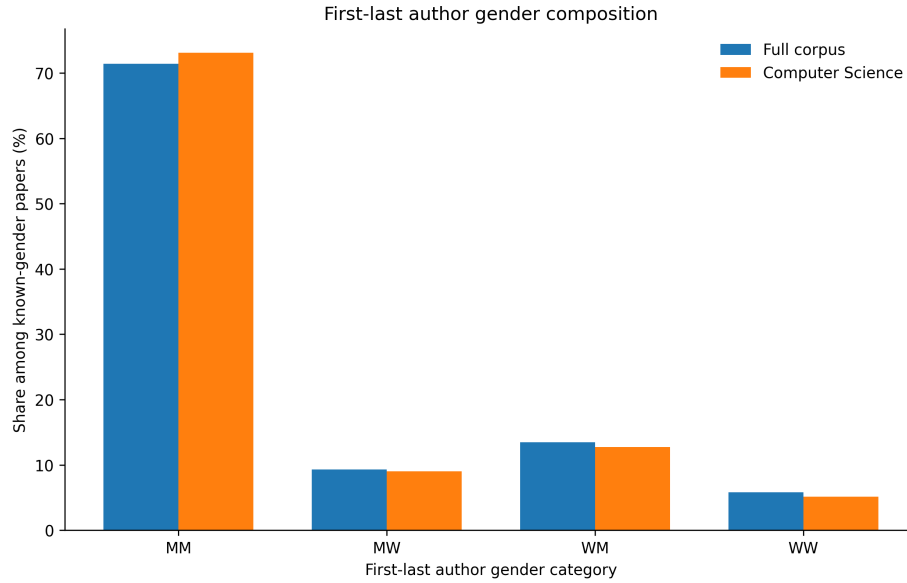


Figure 5.3: Gender composition based on the first and last author in the full analytical corpus and the Computer Science subset. Percentages are calculated among papers with complete gender classification for both author positions.

These descriptive differences are important for the later citation analysis because the available population of potentially cited Computer Science papers is itself highly gender-imbalanced. Raw citation shares therefore cannot be interpreted independently of the composition of the underlying publication pool.

5.2.2 Gender Composition over Time

The aggregate gender distribution conceals substantial temporal change. In the Computer Science subset, MM papers account for 96.65% of known-gender publications in the 1950s and 95.76% in the 1960s. Their share subsequently declines to 93.00% in the 1970s, 87.42% in the 1980s, 82.71% in the 1990s, 76.51% in the 2000s, and 71.13% in the 2010s. For papers published between 2020 and 2024, the MM share falls further to 64.74%.

The decline in the MM share is accompanied by increasing representation of all three categories containing at least one woman. The WM category exhibits the largest absolute increase. Its share rises from 0.00% in the 1950s and 0.63% in the 1960s to 3.92% in the 1980s, 6.77% in the 1990s, 10.78% in the 2000s, 14.00% in the 2010s, and 17.56% between 2020 and 2024.

The MW category follows a similar pattern, increasing from 1.38% in the 1950s to 11.27% in the most recent period. WW papers remain the least frequent category throughout the observation window, but their share also increases from 1.97% in the 1950s to 6.42% between 2020 and 2024.

Table 5.4: Gender composition of Computer Science papers by publication period. Values are percentages among papers with complete gender classification within each period.

Period	MM	MW	WM	WW
1950s	96.65%	1.38%	0.00%	1.97%
1960s	95.76%	1.03%	0.63%	2.57%
1970s	93.00%	1.84%	1.66%	3.49%
1980s	87.42%	3.86%	3.92%	4.80%
1990s	82.71%	5.81%	6.77%	4.71%
2000s	76.51%	8.21%	10.78%	4.50%
2010s	71.13%	9.76%	14.00%	5.10%
2020 to 2024	64.74%	11.27%	17.56%	6.42%

Figure 5.4 illustrates these changes as the composition of the known-gender Computer Science publication pool within each period. The declining MM share is accompanied by sustained growth in the MW, WM, and WW categories, with the change becoming particularly pronounced from the 1990s onward. Taken together, the three categories containing at least one woman increase from 3.35% of known-gender Computer Science papers in the 1950s to 35.26% between 2020 and 2024.

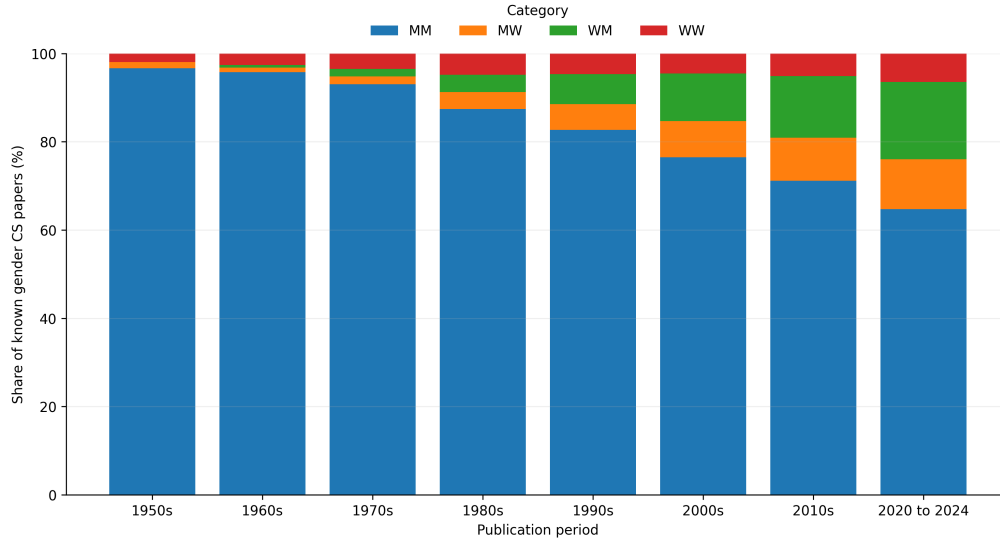


Figure 5.4: Gender composition of Computer Science papers across publication periods. Each bar shows the distribution of MM, MW, WM, and WW papers among publications with complete gender classification for both the first and last author.

The composition of the Computer Science publication pool is therefore not temporally stationary. A paper published in an earlier period faced a markedly different gender composition of potentially citable literature than a paper published in a recent period. This temporal variation provides direct descriptive motivation for the expected-citation baselines that condition on publication period rather than comparing all citations against a single global gender distribution.

5.2.3 Authorship Size and Reference-List Length

Authorship-team size varies considerably across the analytical corpus. Papers contain an average of 3.26 authors overall and 3.15 authors within Computer Science. The median is three authors in both samples. Within Computer Science, 75% of papers contain at most four authors, 90%

contain at most five, and 95% contain at most six. The 99th percentile is eight authors. The largest observed paper contains 127 authors, indicating the presence of a small upper tail of unusually large collaborations.

Figure 5.5 presents the distribution of team sizes in Computer Science. To prevent the small number of very large collaborations from determining the horizontal scale, the visualization is restricted to the 99th percentile.

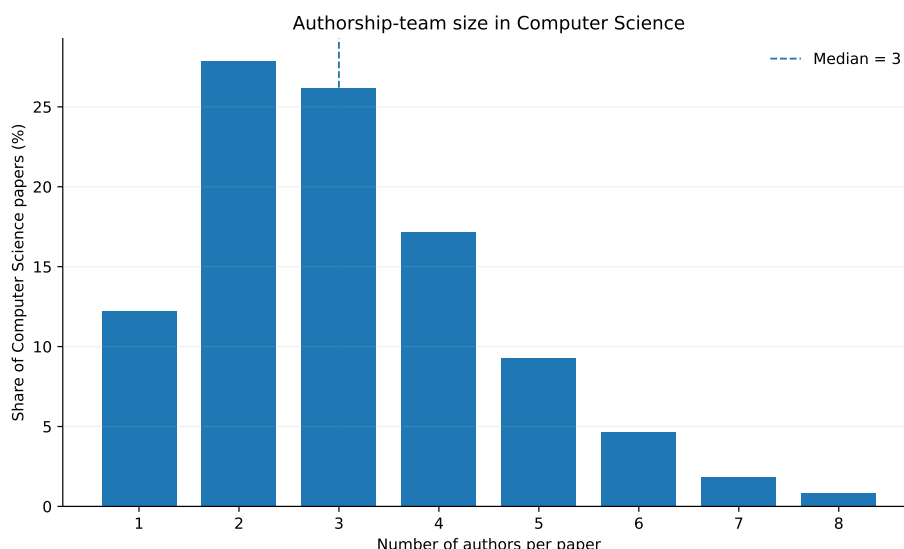


Figure 5.5: Distribution of the number of authors per Computer Science paper. The figure is restricted to the 99th percentile of the authorship-size distribution for readability.

Authorship size also differs across the four gender categories. Within Computer Science, MM papers contain an average of 2.93 authors, compared with 3.39 for MW papers and 3.43 for WM papers. WW papers have the smallest average team size at 2.48 authors. The median is three authors for MM, MW, and WM papers and two authors for WW papers. These differences are descriptive only. In particular, they do not imply that the full author team has the same gender composition as the category defined by its first and last author.

Reference-list length displays similarly substantial variation. Across the full analytical corpus, papers contain an average of 7.53 recorded references and a median of six. Computer Science papers have somewhat longer reference lists, with an average of 8.59 references and a median of seven. Within Computer Science, the 75th percentile is 12 references, the 90th percentile is 18, the 95th percentile is 22, and the 99th percentile is 34. Approximately 6.30% of Computer Science papers have no recorded references.

Figure 5.6 shows the cross-sectional reference-list distribution in Computer Science. The distribution is strongly right-skewed. The largest recorded value is 1,155 references, compared with 34 references at the 99th percentile. The visualization is therefore restricted to the 99th percentile.

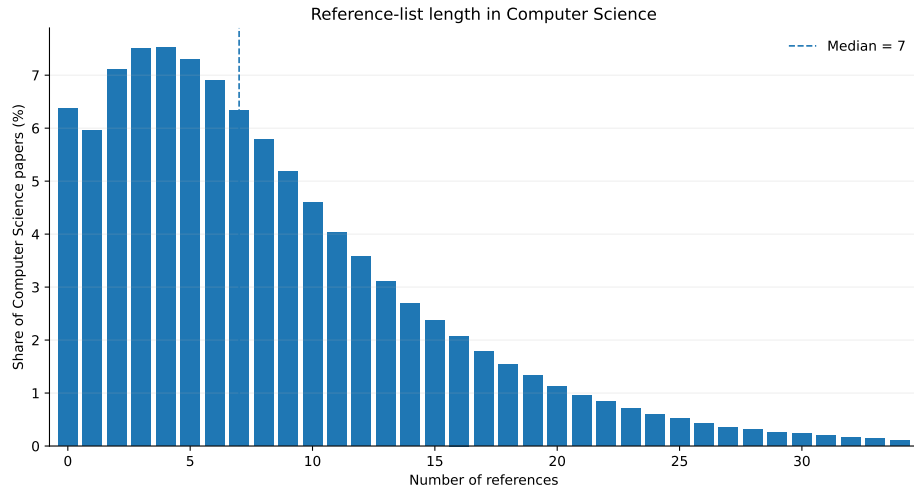


Figure 5.6: Distribution of reference-list length among Computer Science papers. The figure is restricted to the 99th percentile of the reference-list distribution for readability.

Reference-list characteristics also vary across gender categories. Among Computer Science papers, WM papers have the longest reference lists on average at 8.94 references, followed by MW papers at 8.68 and MM papers at 8.16. WW papers have the shortest average reference lists at 7.53. The median is seven references for MW and WM papers and six references for MM and WW papers.

The proportion of papers with no recorded references also differs between the groups. It is highest among WW papers at 8.48%, followed by MM papers at 7.49%, MW papers at 5.42%, and WM papers at 4.92%. Table 5.5 summarizes the principal authorship and reference-list characteristics by gender category.

Table 5.5: Authorship-team size and reference-list characteristics by gender category within Computer Science.

Category	Mean authors	Median authors	Mean refs.	Median refs.	Zero refs.
MM	2.93	3	8.16	6	7.49%
MW	3.39	3	8.68	7	5.42%
WM	3.43	3	8.94	7	4.92%
WW	2.48	2	7.53	6	8.48%

Reference-list length is also strongly associated with publication period. The temporal analysis in Figure 5.7 uses all 1,542,691 Computer Science papers in the final analytical corpus and does not apply a gender-availability restriction. In the 1950s, the median number of recorded references is 0, the 90th percentile is 2, and the 99th percentile is 4.42. These values increase to 1, 5, and 11.50 in the 1960s, respectively, and to 2, 8, and 18 in the 1970s.

The same upward pattern continues in later periods. In the 1980s, the median, 90th percentile, and 99th percentile are 3, 11, and 23 references, respectively. The corresponding values are 4, 14, and 28 in the 1990s and 5, 14, and 28 in the 2000s. A further increase is visible in the 2010s, where the median reaches 7 references, the 90th percentile 19, and the 99th percentile 35. Between 2020 and 2024, these values rise to 9, 21, and 40 references, respectively.

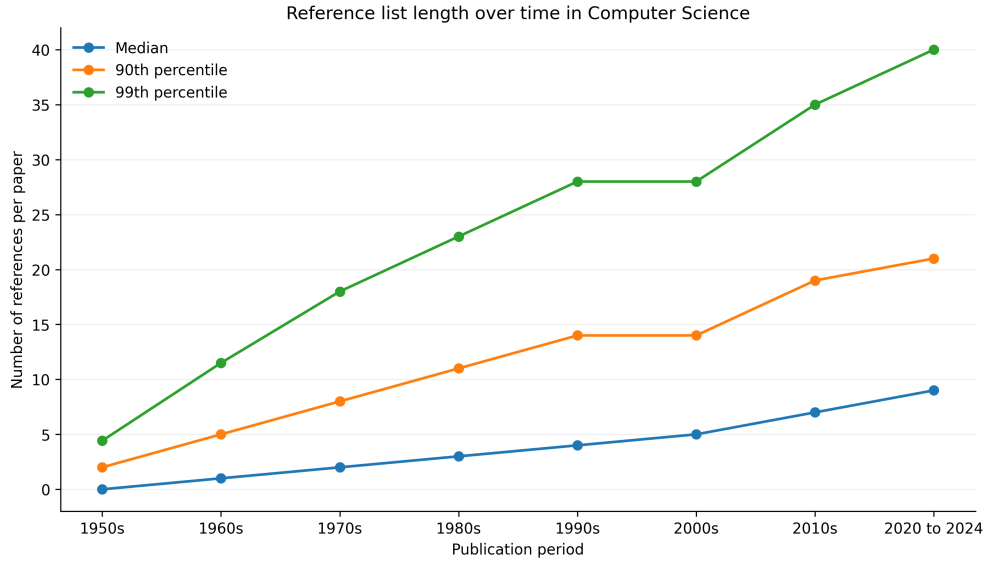


Figure 5.7: Reference-list length of Computer Science papers across publication periods, summarized by the median, 90th percentile, and 99th percentile.

The temporal increase in reference-list length indicates that more recent Computer Science papers typically generate substantially more outgoing citation opportunities than papers from earlier periods. Differences in raw citation volume across publication periods may therefore reflect changes in referencing practices in addition to changes in corpus size and publication composition. The comparatively stable upper percentiles between the 1990s and 2000s also show that the increase is not perfectly monotonic across every part of the distribution, although the longer-term trend is clearly upward.

Taken together, these authorship and referencing distributions provide structural context for the citation analyses that follow. The differences reported here are descriptive and should not be interpreted as causal gender effects because they may also reflect publication period, research area, venue characteristics, and other features of the literature. Reference-list length is therefore included as a structural covariate in the regression-based expected-citation models described in Chapter 4.

5.3 Venue and Taxonomy Composition

Venue and topical structure are important sources of heterogeneity in scholarly citation patterns. Papers published in different venues may differ in visibility, audience, and citation opportunity, while research areas differ in publication volume, referencing conventions, and gender composition. This section therefore examines the venue and taxonomy structure of the Computer Science corpus before these variables are used in the expected-citation baselines.

Importantly, venue characteristics are not used as inclusion criteria when constructing the final analytical corpus. Papers are not removed because their venue is unranked or because they belong to a particular venue type. This choice preserves the broad Computer Science publication population and avoids making external venue-rank coverage a prerequisite for inclusion. Venue information is instead used descriptively and, where appropriate, as structural conditioning information in later expected-citation models.

5.3.1 Venue Rank and Venue-Type Coverage

Venue name and venue type are available for all 1,542,691 Computer Science papers in the final analytical corpus. The corpus consists of 878,816 papers published in conference proceedings and 663,875 journal papers, corresponding to 56.97% and 43.03% of the Computer Science corpus, respectively. The dataset therefore contains substantial representation of both major publication formats rather than being dominated exclusively by either conference or journal publications.

Venue-rank information is considerably less complete. A recognised rank is available for 879,692 Computer Science papers, corresponding to 57.02% of the corpus, while 662,999 papers, or 42.98%, have no available rank. Table 5.6 summarizes this coverage.

Table 5.6: Coverage of venue metadata within the final Computer Science corpus.

Variable	Available	Missing	Coverage	Missing
Venue name	1,542,691	0	100.00%	0.00%
Venue type	1,542,691	0	100.00%	0.00%
Venue rank	879,692	662,999	57.02%	42.98%

The incomplete rank coverage provides an important reason not to restrict the analytical corpus to ranked venues. Such a restriction would remove more than two fifths of all Computer Science papers and would change the population under study according to the availability of an externally assigned venue ranking. Instead, papers without rank information are retained in the analytical corpus and represented by an unavailable rank category in the descriptive data.

The distribution of publication types is shown in Figure 5.8. Conference proceedings account for a moderate majority of the corpus, but journal publications still represent more than two fifths of all Computer Science papers.

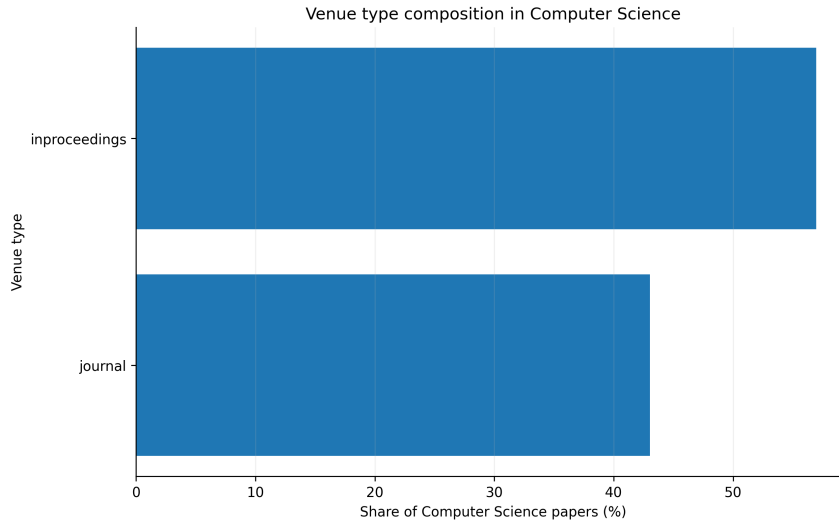


Figure 5.8: Distribution of venue types within the final Computer Science corpus.

Ranked conference and journal venues use different ranking systems. Conference venues are represented by CORE categories A^* , A, B, and C, whereas journals are represented by SCImago quartiles Q1, Q2, Q3, and Q4. These categories are therefore treated as categorical venue strata rather than as a single continuous prestige scale.

Figure 5.9 shows the rank composition of the complete Computer Science corpus, including papers without an available rank. The largest ranked group is Q1 with 270,341 papers, corresponding to

17.52% of all Computer Science papers. This is followed by B with 138,748 papers (8.99%), Q2 with 131,991 (8.56%), A* with 108,898 (7.06%), A with 87,358 (5.66%), Q3 with 71,999 (4.67%), C with 53,716 (3.48%), and Q4 with 16,641 (1.08%). The largest single category remains the unavailable-rank group at 42.98%.

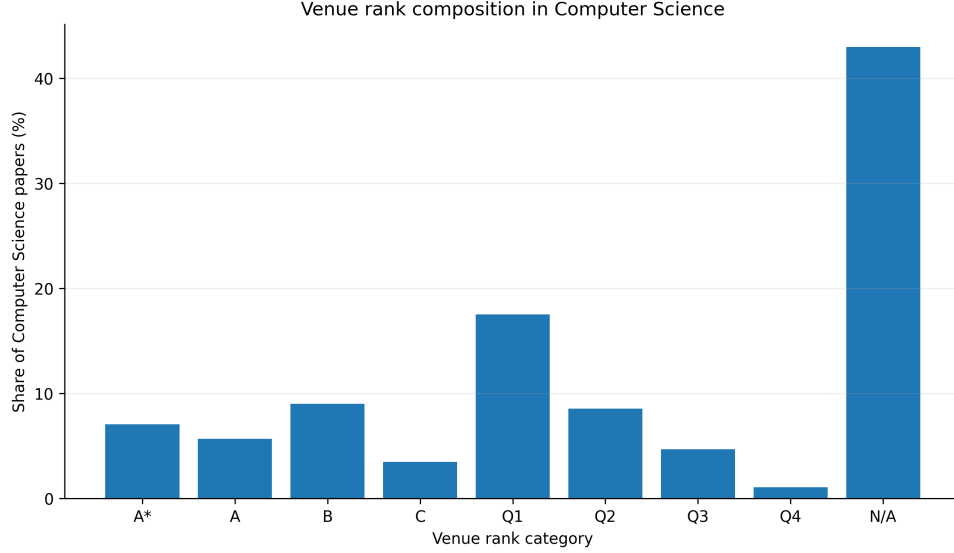


Figure 5.9: Venue-rank composition of the final Computer Science corpus. Unavailable rank information is retained as the N/A category.

Individual venue names are substantially more fragmented than these rank categories. The largest individual venue, *IEEE Access*, contains 23,125 papers and accounts for only 1.50% of the Computer Science corpus. The next largest venues are *CoRR* at 1.20% and *ICASSP* at 1.15%. Even the 20 most frequent venue names together account for only approximately 12% of Computer Science papers.

This long-tailed venue distribution makes exact venue identity unattractive as a principal matching variable for corpus-wide expected-citation baselines. Stratifying simultaneously by publication period, topical area, and exact venue would generate a very large number of small comparison cells. The later venue-conditioned baseline therefore uses the substantially coarser venue-rank strata. This preserves broad differences in venue visibility while avoiding the sparsity associated with thousands of individual venue identities.

5.3.2 Taxonomy Distribution in Computer Science

The OpenAlex taxonomy provides a hierarchical description of research area at the topic, subfield, field, and domain levels. Within the Computer Science subset, the primary taxonomy contains 11 subfields and 301 distinct primary topics. The distribution is highly concentrated at the subfield level but much more fragmented at the topic level.

Artificial Intelligence is the largest Computer Science subfield with 404,721 papers, accounting for 26.23% of the corpus. Computer Networks and Communications contains 319,111 papers (20.69%), while Computer Vision and Pattern Recognition contains 277,979 (18.02%). Together, these three subfields account for 64.94% of all Computer Science publications.

Information Systems contributes a further 154,868 papers (10.04%), followed by Computational Theory and Mathematics with 109,785 (7.12%), Signal Processing with 92,791 (6.01%), and Hardware and Architecture with 75,691 (4.91%). The remaining subfields are smaller, with each ac-

counting for less than 2.5% of the corpus.

Table 5.7: Distribution of primary Computer Science subfields.

Primary subfield	Papers	Share
Artificial Intelligence	404,721	26.23%
Computer Networks and Communications	319,111	20.69%
Computer Vision and Pattern Recognition	277,979	18.02%
Information Systems	154,868	10.04%
Computational Theory and Mathematics	109,785	7.12%
Signal Processing	92,791	6.01%
Hardware and Architecture	75,691	4.91%
Human-Computer Interaction	35,561	2.31%
Computer Science Applications	24,663	1.60%
Software	23,790	1.54%
Computer Graphics and Computer-Aided Design	23,731	1.54%

The same distribution is visualized in Figure 5.10. The concentration is substantial. The top five subfields account for 82.09% of the Computer Science corpus, and the top ten account for 98.46%.

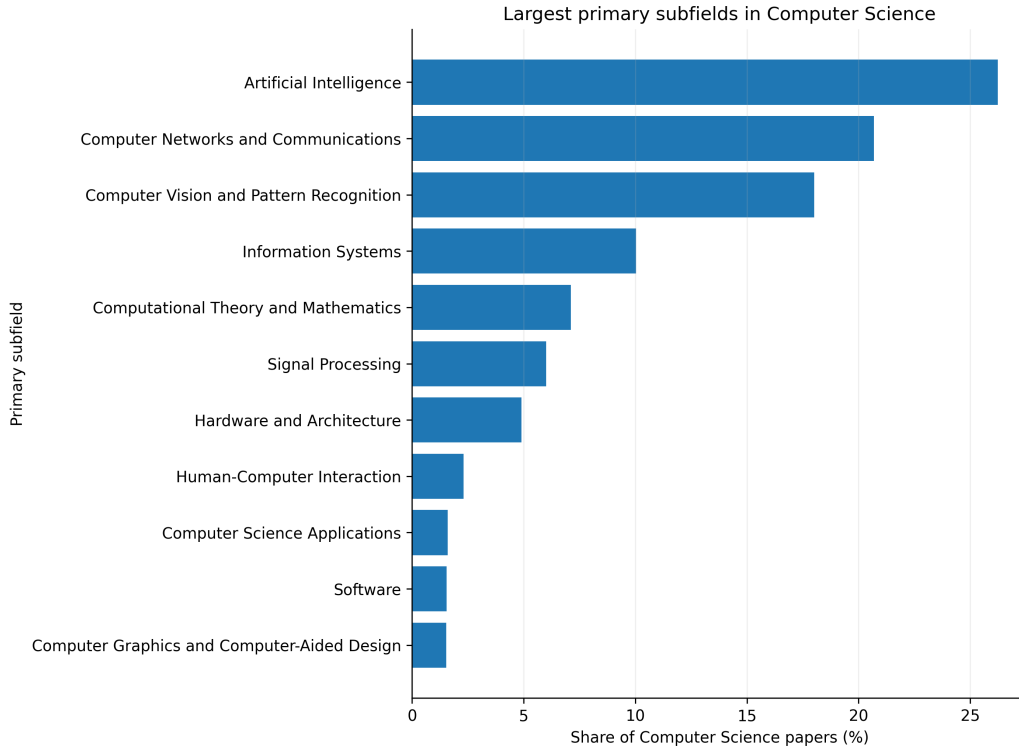


Figure 5.10: Distribution of papers across primary Computer Science subfields.

The topic level has a very different structure. There are 301 distinct primary topics, but no individual topic represents a large fraction of the corpus. The largest topic, Parallel Computing and Performance Optimization, accounts for only 2.34% of Computer Science papers. The five largest topics together account for 8.94%, the ten largest for 15.24%, and the twenty largest for 26.08%.

This difference between the two taxonomy levels is important for the later analysis. Subfields

provide relatively large and stable comparison groups, whereas exact topics partition the corpus into a much larger number of smaller cells. The subfield level is therefore well suited to the main topically-conditioned expected-citation baseline. It captures substantial variation in research area while limiting sparsity in the comparison sets.

The finer topic level nevertheless remains informative. Ninety of the 301 primary topics contain at least 5,000 Computer Science papers with complete first and last author gender information, so topic-level analysis is feasible for a substantial subset of large topics. However, the overall topic distribution remains highly fragmented. For this reason, topic information is used primarily to identify close topical relationships between citation edges and to support the same-topic and taxonomy-based relevance analyses, rather than replacing subfields as the principal corpus-wide stratification level.

This distinction also avoids constructing overly restrictive cells that combine publication period, detailed topic, and venue information. Such combinations would provide increasingly close comparisons but would substantially reduce the number of papers available within many cells. The taxonomy hierarchy therefore supports different levels of analysis: subfields provide robust structural strata, while topics provide finer-grained information about citation relevance.

5.3.3 Gender Composition across Venues and Subfields

Gender composition is not constant across either venue strata or research areas. This heterogeneity is particularly important because differences in citation outcomes at the aggregate level can partly reflect where papers from different gender categories are published rather than differences within otherwise comparable publication environments.

Variation across venue-rank categories is visible but relatively moderate. Among known-gender Computer Science papers in A^* venues, 76.66% are MM, 8.58% are MW, 10.94% are WM, and 3.82% are WW. In A venues the corresponding shares are 73.58%, 9.18%, 12.35%, and 4.90%, while in B venues they are 72.60%, 9.38%, 13.92%, and 4.10%.

C venues have the lowest MM share among the conference rank categories at 68.86%, accompanied by 10.14% MW, 15.02% WM, and 5.97% WW. Journal quartiles show somewhat smaller differences. MM shares range from 72.72% in Q2 to 74.76% in Q1, while WW shares increase from 4.23% in Q1 to 6.67% in Q4. Papers without an available venue rank have a composition of 72.40% MM, 9.20% MW, 12.80% WM, and 5.61% WW.

Table 5.8: Gender composition within venue-rank categories among Computer Science papers with complete first and last author gender information.

Venue rank	MM	MW	WM	WW
A^*	76.66%	8.58%	10.94%	3.82%
A	73.58%	9.18%	12.35%	4.90%
B	72.60%	9.38%	13.92%	4.10%
C	68.86%	10.14%	15.02%	5.97%
Q1	74.76%	8.44%	12.57%	4.23%
Q2	72.72%	8.98%	12.49%	5.81%
Q3	73.10%	8.21%	12.61%	6.08%
Q4	73.16%	8.08%	12.09%	6.67%
N/A	72.40%	9.20%	12.80%	5.61%

Figure 5.11 visualizes these differences. Although the MM category remains dominant in every rank stratum, the composition is not identical across strata. Venue-rank structure can therefore contribute to aggregate gender differences in citation reception. This provides empirical motivation

for including a venue-conditioned expected-citation baseline rather than assuming that papers from all venue strata have directly comparable citation opportunities.

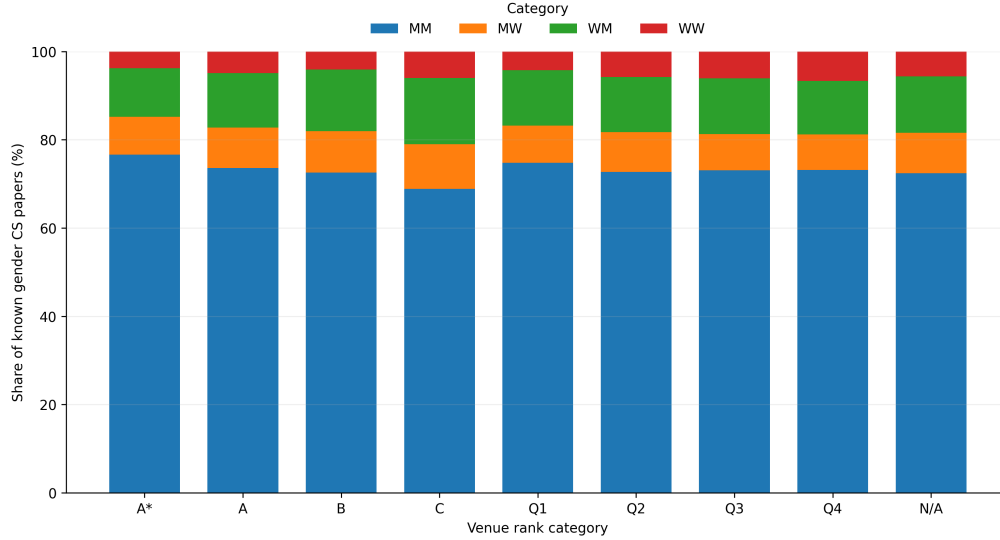


Figure 5.11: Gender composition across venue-rank categories in Computer Science. Percentages are calculated within each venue-rank stratum among papers with complete first and last author gender information.

Venue type itself shows a smaller compositional difference. Conference proceedings consist of 72.19% MM, 9.45% MW, 13.33% WM, and 5.03% WW papers, while journals consist of 74.39% MM, 8.39% MW, 11.92% WM, and 5.29% WW papers. Venue type is therefore useful descriptive information, but the variation is less pronounced than the variation observed across research subfields.

Subfield composition differs much more substantially. All 11 Computer Science subfields contain more than 5,000 papers with complete gender information, making subfield-level comparisons sufficiently well supported for descriptive analysis. Figure 5.12 shows the resulting distributions.

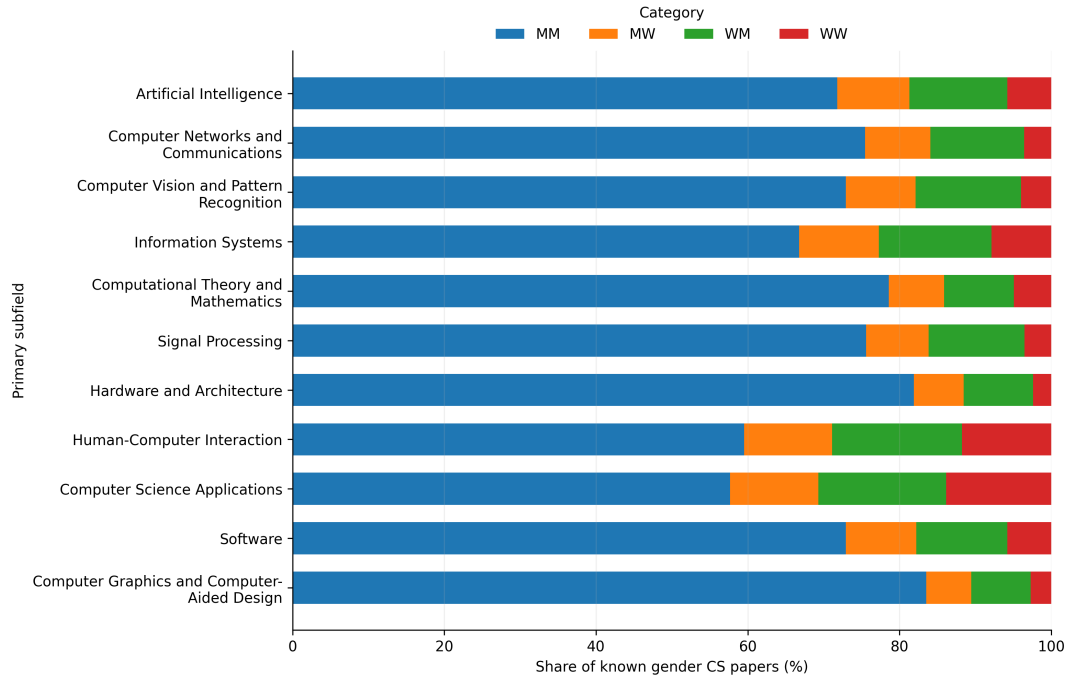


Figure 5.12: Gender composition across primary Computer Science subfields. Percentages are calculated within each subfield among papers with complete first and last author gender information.

The MM share ranges from 57.65% in Computer Science Applications and 59.51% in Human-Computer Interaction to 81.89% in Hardware and Architecture and 83.53% in Computer Graphics and Computer-Aided Design. Expressed equivalently, the combined share of MW, WM, and WW papers ranges from only 16.47% in Computer Graphics and Computer-Aided Design to 42.35% in Computer Science Applications.

Human-Computer Interaction also has a relatively high combined non-MM share of 40.49%, followed by Information Systems at 33.22%. Artificial Intelligence is closer to the overall Computer Science composition, with 71.79% MM, 9.51% MW, 12.90% WM, and 5.80% WW. Computer Networks and Communications, by contrast, contains 75.46% MM papers, while Computational Theory and Mathematics contains 78.60% and Hardware and Architecture 81.89%.

Table 5.9: Gender composition across primary Computer Science subfields. Percentages are calculated among papers with complete first and last author gender information.

Primary subfield	MM	MW	WM	WW
Artificial Intelligence	71.79%	9.51%	12.90%	5.80%
Computer Networks and Communications	75.46%	8.60%	12.38%	3.55%
Computer Vision and Pattern Recognition	72.93%	9.19%	13.90%	3.98%
Information Systems	66.78%	10.52%	14.84%	7.86%
Computational Theory and Mathematics	78.60%	7.27%	9.20%	4.93%
Signal Processing	75.62%	8.22%	12.66%	3.50%
Hardware and Architecture	81.89%	6.55%	9.19%	2.37%
Human-Computer Interaction	59.51%	11.60%	17.14%	11.74%
Computer Science Applications	57.65%	11.68%	16.81%	13.86%
Software	72.93%	9.30%	11.98%	5.78%
Computer Graphics and Computer-Aided Design	83.53%	5.94%	7.82%	2.71%

The magnitude of this variation provides a particularly strong justification for conditioning expected citations on topical structure. An aggregate comparison of gender categories would combine papers from research areas with substantially different gender compositions and potentially different citation environments. The subfield-conditioned baseline therefore compares citation outcomes against gender availability within more comparable topical contexts.

Variation is also visible at the individual venue and topic levels. Among the 30 individual venues containing at least 3,000 known-gender Computer Science papers, MM shares range from approximately 65% in *IEEE Access* to more than 85% in venues such as *Communications of the ACM* and *IEEE Transactions on Information Theory*. Similarly, many large topics show meaningful compositional differences. For example, Statistical Machine Translation and Natural Language Processing contains 64.99% MM papers, while Parallel Computing and Performance Optimization contains 82.88%.

These finer differences show that both venue identity and detailed topic can contain additional structural information. However, they also illustrate the trade-off between comparability and sample size. Exact venues are highly fragmented, and the 301-topic taxonomy produces many substantially smaller groups than the 11-subfield taxonomy. Using these variables simultaneously with publication period and other structural characteristics would create many small or empty comparison cells.

For this reason, the later expected-citation framework uses multiple complementary levels of adjustment rather than one maximally crossed stratification. Publication period controls temporal exposure, subfield conditioning controls broad topical composition, and venue-rank conditioning captures coarse venue visibility. Topic information is retained for the separate topical-relevance analyses, where citation edges are classified by whether they remain within the same topic or subfield and can additionally be weighted by hierarchical taxonomy proximity. This design allows the analysis to account for the strong structural differences revealed in the EDA without discarding large parts of the corpus or creating excessively sparse comparison groups.

5.4 Direct Citation Structure

The direct citation network provides the basic representation of citation activity used throughout the subsequent analyses. Each node represents a paper, and each directed edge runs from a citing paper to a cited paper. Consequently, out-degree captures the number of references from a paper to other papers represented in the analytical corpus, whereas in-degree captures the number of citations received from within that corpus.

Examining the structure of this network is important before defining expected citation baselines. Citation opportunities are not distributed uniformly across papers: papers differ substantially in the number of references they make, in whether they receive any citations at all, and in the extent to which citation attention is concentrated among a relatively small set of highly cited papers. The diagnostics in this section therefore establish the distributional properties that motivate the use of conditioned expected-citation comparisons rather than comparisons based solely on raw citation counts.

5.4.1 Direct Citation Network Size and Degree Distributions

The full direct citation network contains 3,026,412 papers and 18,904,534 directed citation edges. Of all papers in the network, 2,733,997 have at least one outgoing citation edge to another paper represented in the corpus, whereas 1,915,670 receive at least one incoming citation edge. Thus, outgoing citation activity is substantially more widespread than citation reception: 90.34% of papers have positive out-degree, compared with 63.30% with positive in-degree.

Because every directed edge contributes once to the total in-degree and once to the total out-

degree, the two distributions have the same mean of approximately 6.25 edges per paper. Their distributions nevertheless differ markedly. The median in-degree is only 1, whereas the median out-degree is 5. Differences are even more pronounced in the upper tail. The 90th percentiles are 12 and 14 for in-degree and out-degree, respectively, while the corresponding 99th percentiles are 78 and 28. At the 99.9th percentile, in-degree reaches 359 compared with 59 for out-degree. The maximum observed in-degree is 48,576, while the maximum out-degree is 1,090.

Table 5.10 summarizes these differences.

Table 5.10: Degree distributions in the full direct citation network.

Statistic	In-degree	Out-degree
Mean	6.25	6.25
Median	1	5
75th percentile	4	9
90th percentile	12	14
95th percentile	23	18
99th percentile	78	28
99.9th percentile	359	59
Maximum	48,576	1,090

The different tail behaviour is particularly clear in Figure 5.13, which presents the complementary cumulative degree distributions on logarithmic axes. For a degree threshold x , the CCDF gives the proportion of papers with degree at least x . Out-degree decreases relatively rapidly after the lower degree range, whereas the in-degree distribution extends over several additional orders of magnitude. Citation reception is therefore considerably more concentrated than citation production: most papers receive relatively few citations, while a small number accumulate exceptionally large numbers of incoming citations.

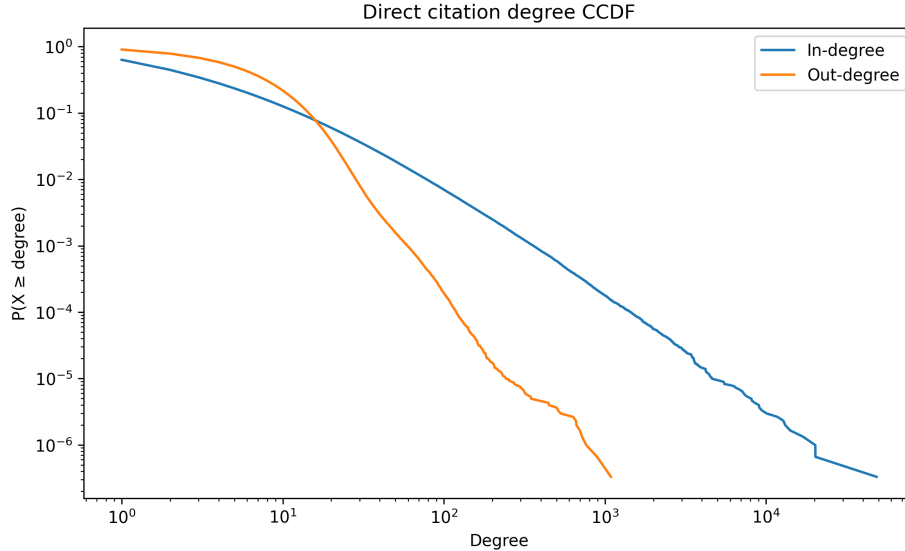


Figure 5.13: Complementary cumulative distributions of direct-citation in-degree and out-degree on logarithmic axes.

This asymmetry has two implications for the analyses that follow. First, raw citation counts cannot be treated as approximately homogeneous outcomes across papers, since citation reception is strongly right-skewed and contains an extreme upper tail. Second, the large heterogeneity in

citation reception and citation opportunities motivates the later use of expected-citation baselines that condition on paper characteristics and local citation structure rather than comparing gender categories using raw means alone. Together with the substantial mass at zero examined below, the strongly asymmetric count distribution also motivates treating citation counts as count-valued outcomes in the regression-based specifications rather than applying a Gaussian outcome model directly to the raw counts.

5.4.2 Citation Reception by Gender Category

Gender-specific citation reception is examined on the known-gender Computer Science subset. This restriction corresponds to the main All→CS analytical slice used in the subsequent gender-specific observed/expected comparisons, in which cited papers belong to the known-gender Computer Science target population while incoming citations may originate from the broader analytical citation network. The subset contains 1,081,363 Computer Science papers classified into the four first-last author gender categories MM, MW, WM, and WW. Across this target population, 8,686,237 incoming citations are observed in the All→CS slice.

Raw citation reception differs across these categories. MM papers have the highest mean citation count, at 8.781 citations per paper, followed by MW papers at 6.586, WM papers at 5.898, and WW papers at 5.210. The median is 1 citation for all four gender categories.

The differences become more visible toward the upper tail. At the 90th percentile, the number of received citations is 16 for MM papers, 13 for MW, 12 for WM, and 10 for WW. At the 99th percentile, the corresponding values are 111, 84, 73, and 66 citations. The descriptive distributions are summarized in Table 5.11.

Table 5.11: Direct citation reception among known-gender Computer Science papers by first-last author gender category in the All→CS citation slice.

Category	Papers	Mean	Median	P90	P99
MM	790,617	8.781	1	16	111
MW	97,399	6.586	1	13	84
WM	137,756	5.898	1	12	73
WW	55,591	5.210	1	10	66

The ordering is therefore consistent across the mean and the upper quantiles, with MM papers receiving more citations in raw terms and WW papers receiving fewer. At the same time, the identical median of one citation across all four categories contrasts with the larger differences in the means and upper quantiles. The group-level averages are therefore strongly influenced by the right-skewed citation distribution and the comparatively small share of papers located in its upper tail. A comparison of raw means alone would consequently combine differences in typical citation reception with differences in the frequency and magnitude of highly cited observations.

More importantly, these statistics are unconditional. The four gender categories need not have the same distribution over publication periods, subfields, topics, venue characteristics, citation opportunities, or network positions. Differences along these dimensions may correspond to systematically different citation opportunities. The descriptive ordering observed here is therefore not interpreted as evidence of citation imbalance by itself. Instead, it provides the empirical motivation for the observed/expected framework used later in the thesis, where observed citation reception is compared with expectations defined under alternative temporal, topical, venue-based, and network-neighborhood comparison structures.

5.4.3 Uncited Papers and Heavy-Tailed Citation Outcomes

The strong right-skew in citation reception is accompanied by a substantial mass at zero. In the full direct citation network, 1,110,742 of the 3,026,412 papers receive no incoming citation from another paper represented in the analytical corpus, corresponding to 36.70%. Thus, the distribution of citation outcomes combines a large group of uncited or weakly cited papers with a comparatively small group of papers receiving very large numbers of citations.

The prevalence of zero-citation outcomes also differs across the known-gender Computer Science categories in the All→CS slice. The uncited share is lowest for MM papers, at 33.41%, followed by MW papers at 36.69% and WM papers at 37.46%. For WW papers, 39.28% receive no incoming citation. Consequently, approximately one third of MM papers and almost two fifths of WW papers are uncited within the corresponding analytical citation slice.

Table 5.12: Known-gender Computer Science papers receiving zero direct citations in the All→CS citation slice, by first-last author gender category.

Category	Uncited papers	Uncited share (%)
MM	264,133	33.41
MW	35,731	36.69
WM	51,605	37.46
WW	21,837	39.28

Across the four known-gender categories, 373,306 papers receive no citation in the All→CS slice, while 708,057 are cited at least once.

The zero mass and the heavy upper tail should be considered jointly. Across gender categories, the median paper receives only one citation, yet the 99th percentiles range from 66 to 111 citations, and the full network contains papers with citation counts several orders of magnitude larger still. Mean citation reception is therefore not representative of a typical paper and is sensitive to the comparatively small number of highly cited observations.

These distributional properties inform several later methodological choices. First, the main analyses evaluate observed citations relative to an expected number of citations rather than treating differences in unconditional group means as the quantity of primary interest. Second, temporal and topical conditioning is introduced to define more comparable sets of citation opportunities. Third, citation-derived neighborhood baselines provide comparisons against papers occupying locally similar positions in bibliographic-coupling and co-citation space, supplying alternative comparison structures based on dimensions of scholarly proximity not represented by raw citation counts alone. Finally, the large mass at zero and strongly asymmetric positive tail motivate explicit evaluation of model fit and calibration when paper-level citation counts are modelled.

The EDA therefore establishes a distinction that is maintained throughout the remainder of the thesis: differences in observed citation reception are empirically present, but determining whether those differences constitute over- or under-citation requires a corresponding model of how many citations papers would be expected to receive under comparable citation opportunities.

5.5 Temporal, Topical, and Gendered Citation Flows

The direct citation network is structured not only by differences in citation volume, but also by publication time, topical proximity, and the composition of the citing and cited paper populations. These dimensions determine which papers constitute plausible citation opportunities for one another and therefore provide an important descriptive basis for the expected-citation models introduced later in the thesis.

The diagnostics in this section focus primarily on the All→CS citation slice used in the main analysis, in which cited papers belong to Computer Science while citing papers may originate from any field. The more restrictive CS→CS slice is additionally examined where useful as a within-field comparison. The objective remains descriptive: the observed temporal, topical, and gendered flows characterize the citation environment and motivate subsequent conditioning choices, but are not themselves interpreted as evidence of citation imbalance.

5.5.1 Citation Flows across Decades and Citation Lags

Citation opportunity is inherently time-dependent. A paper can only cite work that has already been published, while older papers have had more time to accumulate citations. At the same time, citing papers do not distribute their references uniformly across the available historical literature. The All→CS citation-lag distribution shows that citation attention is strongly concentrated in relatively recent prior work while retaining a substantial longer-term component.

Citation lag is defined as the difference between the publication year of the citing paper and that of the cited paper. Across the 11,601,416 temporally valid All→CS citation edges, 19.09% have a lag of 0-2 years, 27.55% a lag of 3-5 years, and 26.35% a lag of 6-10 years. Thus, approximately 73% of observed citation edges connect to work published no more than ten years earlier. A further 18.98% have lags of 11-20 years, while 8.03% extend more than twenty years into the past.

Figure 5.14 summarizes these broad citation-age groups.

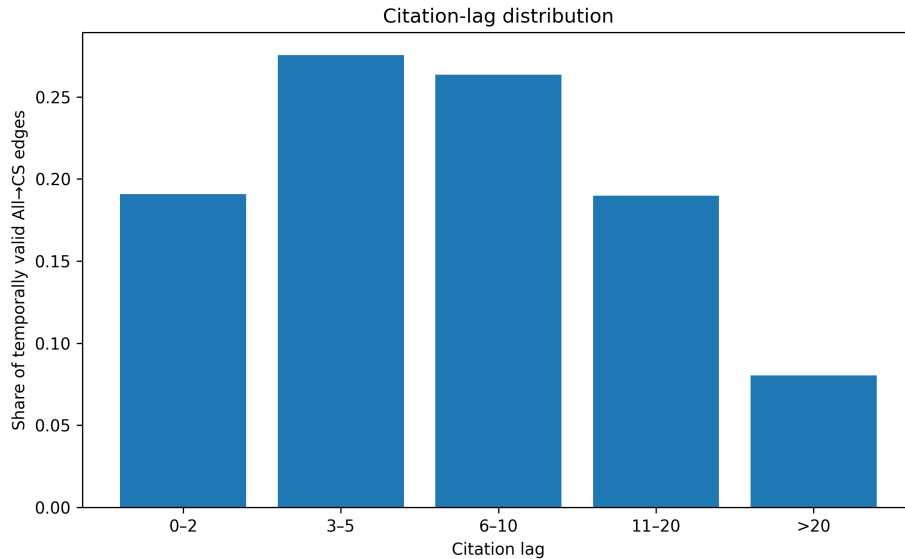


Figure 5.14: Distribution of citation lags among temporally valid All→CS citation edges. Citation lag is measured as the difference between the publication years of the citing and cited papers.

The annual distribution shown in Figure 5.15 provides a more detailed view of the same pattern. The share of citation edges rises sharply during the first years after publication, peaks at approximately two to three years of lag, and then declines progressively. Importantly, the decline is gradual rather than abrupt. Citation edges remain observable several decades after publication, producing a long right tail in the age distribution.

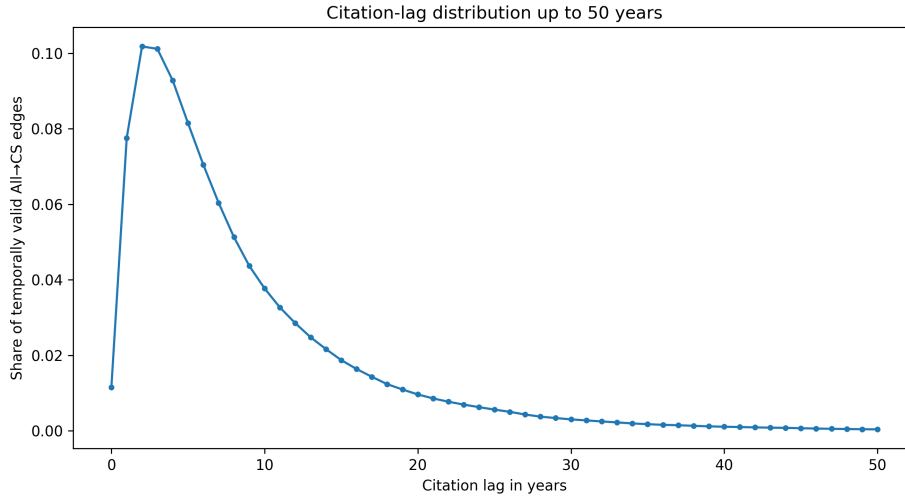


Figure 5.15: Exact citation-lag distribution for temporally valid All→CS citation edges, shown for citation lags of up to 50 years.

This temporal dependence is also visible after aggregating publication years into decades. Figure 5.16 presents a row-normalized citing-decade-cited-decade matrix. Each row therefore describes the distribution of citation targets selected by papers published in a particular decade.

The matrix exhibits a pronounced near-diagonal structure. Papers tend to cite work from their own decade and, especially in more recent periods, the immediately preceding decade. For example, papers published in the 2010s direct approximately 45% of their citations to papers from the 2000s and 39% to papers from the 2010s. For papers published in the 2020s, approximately 57% of citation edges point to work from the 2010s, compared with approximately 20% to other papers from the 2020s and 15% to papers from the 2000s. Citation flows to progressively older decades remain present but become much smaller as temporal distance increases.

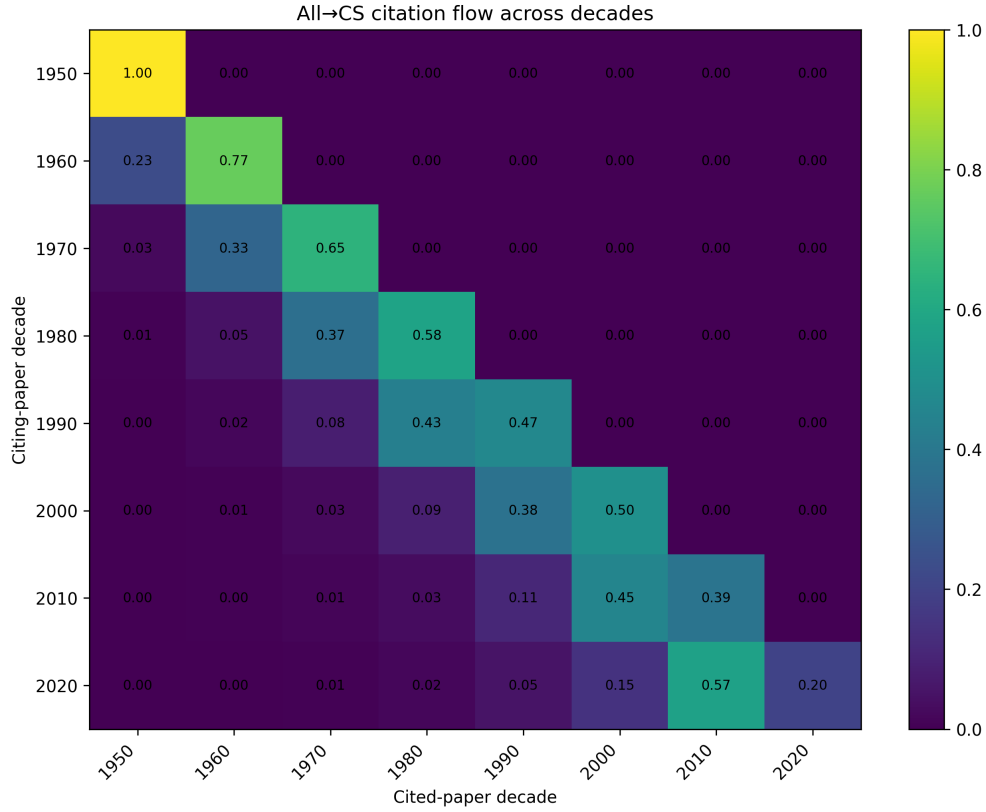


Figure 5.16: Row-normalized All→CS citation flows across publication decades. Rows represent citing-paper decades and columns represent cited-paper decades.

These temporal patterns have direct implications for the later analysis. Comparing papers against a single corpus-wide citation expectation would combine papers exposed to markedly different citation environments. This is especially important because the composition of the Computer Science literature, including its gender composition, changes over time. Publication period may therefore be associated simultaneously with the size and composition of the available cited paper pool.

The observed temporal localization consequently motivates conditioning the direct expected-citation baselines on publication period rather than assuming a temporally homogeneous opportunity set. The substantial long-lag tail provides an additional motivation for examining citation imbalance by citation age. If citation patterns differ between recent and much older literature, an aggregate estimate could conceal differences between short-term visibility and continued presence in longer-term scientific memory.

5.5.2 Citation Flows across Taxonomy Levels

Citation opportunities are also strongly structured by topical proximity. The OpenAlex taxonomy provides nested topic, subfield, and field assignments, allowing the relationship between citing and cited papers to be characterized at different levels of disciplinary resolution.

Figure 5.17 shows the proportion of All→CS citation edges for which the citing and cited papers remain within the same category at each taxonomic level. Topical concentration becomes progressively stronger as the taxonomy is broadened. A total of 43.14% of citation edges connect papers assigned to the same topic. The corresponding share rises to 58.98% at the subfield level and 81.14% at the field level.

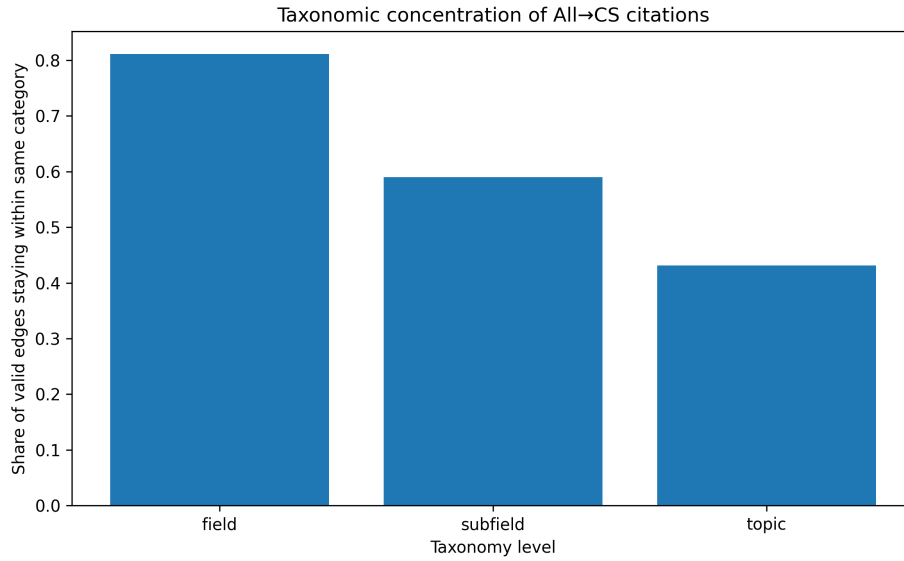


Figure 5.17: Taxonomic concentration of All→CS citation edges. Bars show the share of valid citation edges for which the citing and cited papers belong to the same field, subfield, or topic.

These cumulative shares are nested. In particular, a same-topic citation also belongs to the same subfield and field, so the three bars should not be interpreted as mutually exclusive categories. To distinguish the different forms of topical proximity explicitly, the citation edges were additionally partitioned according to their closest shared level in the taxonomy.

The resulting hierarchical distribution is shown in Figure 5.18. Of all classified All→CS citation edges, 43.14% connect papers within the same topic. A further 15.85% cross topic boundaries while remaining within the same subfield, and 22.16% cross subfield boundaries while remaining within the same field. The remaining 18.86% are cross-field citations.

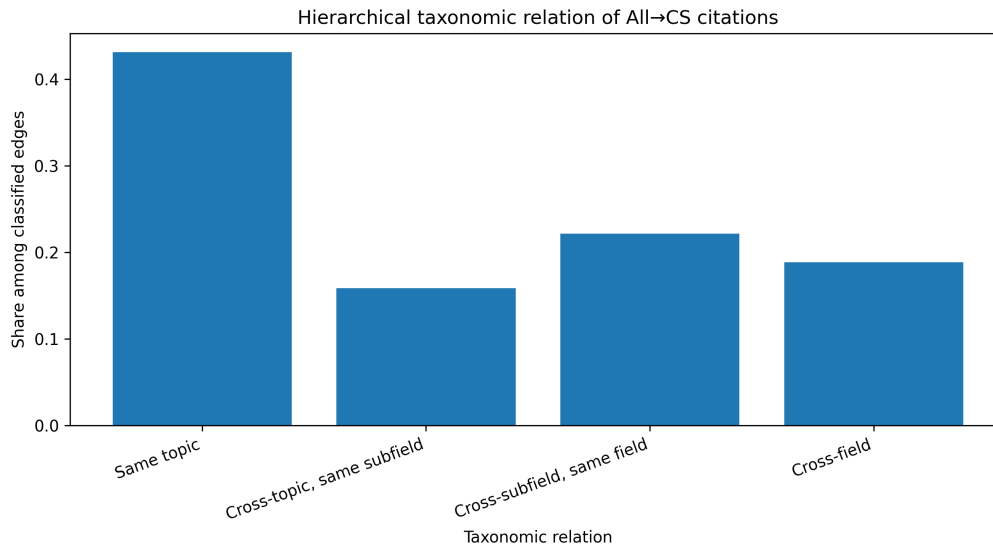


Figure 5.18: Hierarchical taxonomic relationship of All→CS citation edges. Categories are mutually exclusive and distinguish same-topic citations, cross-topic citations within a shared subfield, cross-subfield citations within a shared field, and cross-field citations.

The hierarchical distribution is particularly relevant for the subsequent definition of topical relevance. An exclusively same-topic criterion would capture less than half of the observed citation relationships. Conversely, treating all citations to Computer Science papers as equally topically related would ignore substantial differences between citations connecting papers within a narrow research area and those spanning subfields or fields. The data therefore support a graded definition of topical proximity rather than either extreme.

The importance of subfield structure is further illustrated within the CS→CS citation slice. Figure 5.19 presents row-normalized citation flows among the major Computer Science subfields. The matrix is strongly diagonal, indicating substantial within-subfield citation concentration, but the magnitude differs across research areas. Approximately 80% of citations originating from Computer Vision and Pattern Recognition remain within that subfield, while the corresponding within-subfield shares are about 74% for Artificial Intelligence, 78% for Computer Networks and Communications, and 73% for Computer Graphics and Computer-Aided Design. Lower diagonal shares occur for areas such as Computer Science Applications and Software, where approximately 54-55% of citations remain within the same subfield.

The off-diagonal structure also shows that cross-subfield citation is not random. Particular research areas have substantial citation connections to specific neighboring subfields, demonstrating that different parts of Computer Science operate within different local citation environments.

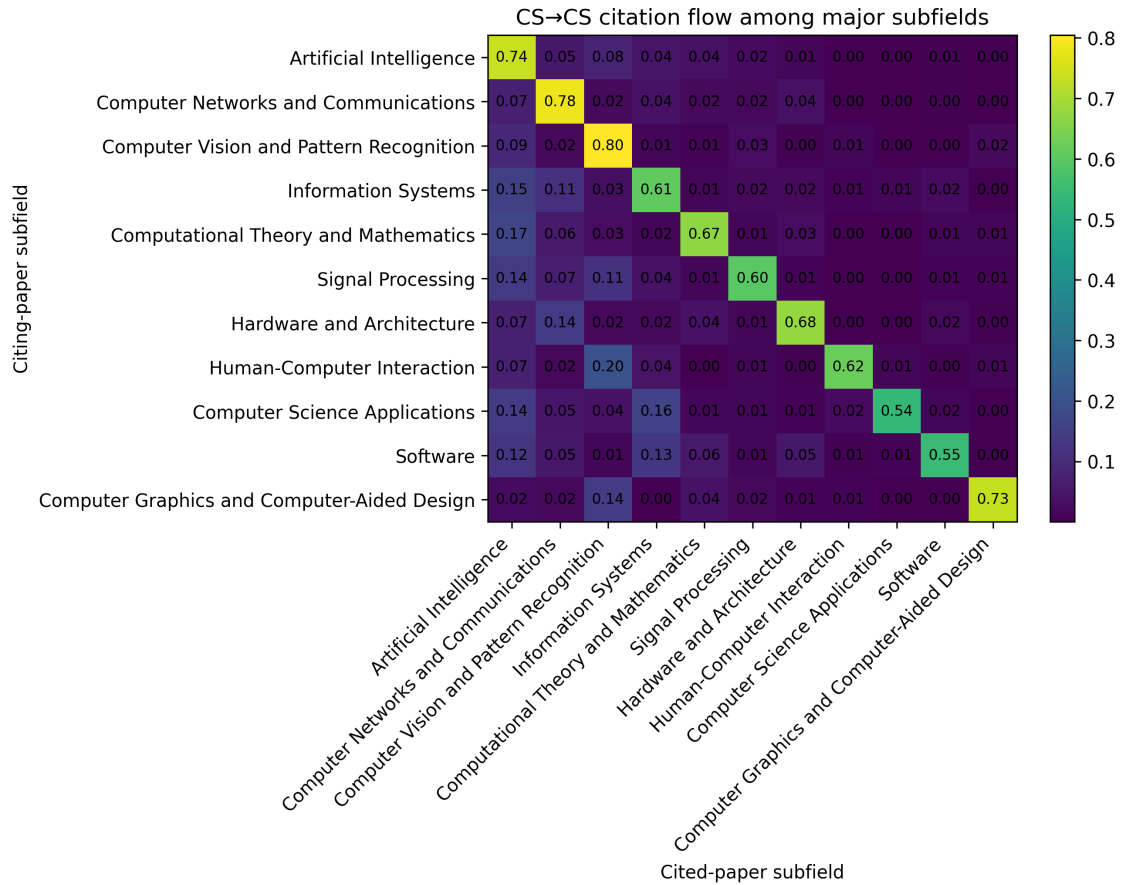


Figure 5.19: Row-normalized CS→CS citation flows among major Computer Science subfields. Rows represent citing-paper subfields and columns represent cited-paper subfields.

These diagnostics motivate the topical adjustments introduced in the subsequent expected-citation baselines. Papers should not be treated as equally plausible citation targets merely because they

belong to Computer Science. Topic and subfield membership provide increasingly localized comparison sets, while the hierarchical taxonomy additionally supplies a natural ordering of topical proximity. This motivates both categorical topical-match specifications and taxonomy-weighted variants that allow same-topic, same-subfield, same-field, and more distant citation relationships to contribute differently to expected citation calculations.

5.5.3 Gender-to-Gender Citation Flow Patterns

The final flow diagnostic considers the first-last author gender categories of both the citing and cited papers. Because the four categories differ substantially in prevalence, absolute citation-flow counts are dominated by the largest category and are difficult to compare directly across citing groups. The matrices are therefore row-normalized, so that each row describes the composition of known-gender citation targets selected by a given citing category.

Gender information is available on both ends of 5,948,956 All→CS citation edges, corresponding to 51.28% of that citation slice. For CS→CS, 4,812,158 edges have known gender on both sides, or 51.12%. The gender-flow patterns below therefore describe the known-gender portion of the citation network rather than all citation edges.

For the primary All→CS slice, Figure 5.20 shows that MM papers constitute the largest cited category for every citing group. Among known-gender citations made by MM papers, 81.50% point to MM papers, 6.92% to MW papers, 8.50% to WM papers, and 3.08% to WW papers.

The proportion directed to MM papers becomes progressively lower for the other citing categories: 77.52% for MW citing papers, 76.97% for WM papers, and 72.97% for WW papers. Conversely, the share directed to WW papers increases from 3.08% among MM citing papers to 4.08% among MW papers, 4.15% among WM papers, and 6.79% among WW papers. Similar differences are visible for the MW and WM cited categories.

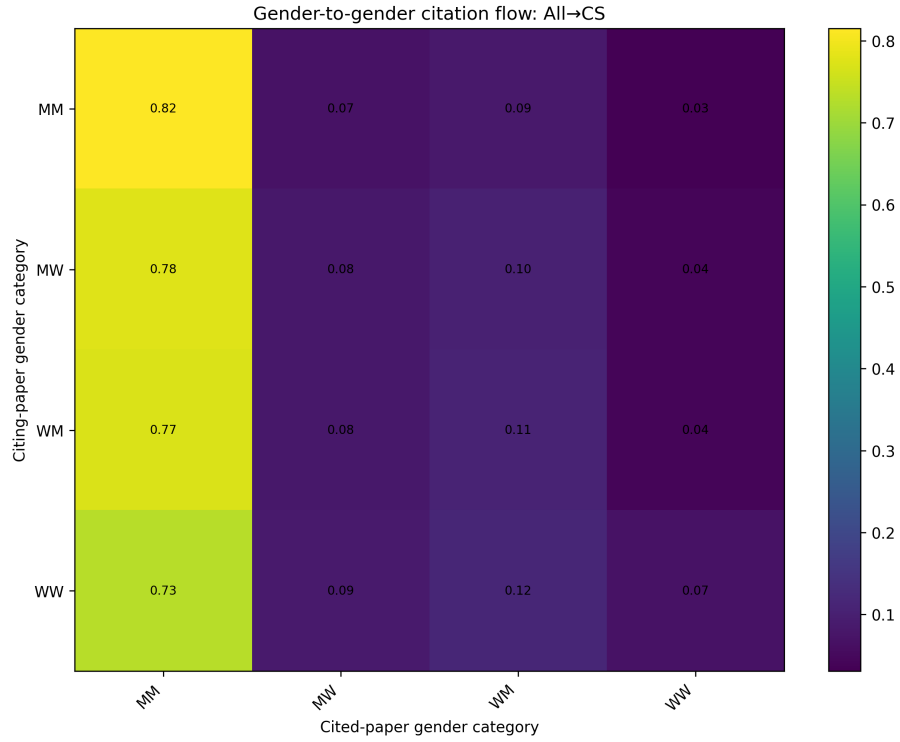


Figure 5.20: Row-normalized gender-to-gender citation flows for the All→CS citation slice. Each row shows the distribution of known-gender citation targets for a given citing-paper first-last author gender category.

The corresponding CS→CS matrix, Figure 5.21, is highly similar. For example, 81.32% of known-gender citations from MM papers point to MM papers, compared with 73.54% among WW citing papers. The WW target share ranges from 3.06% for MM citing papers to 6.28% for WW citing papers. The close agreement between the All→CS and CS→CS matrices indicates that the broad descriptive pattern persists when citations from outside Computer Science are excluded.

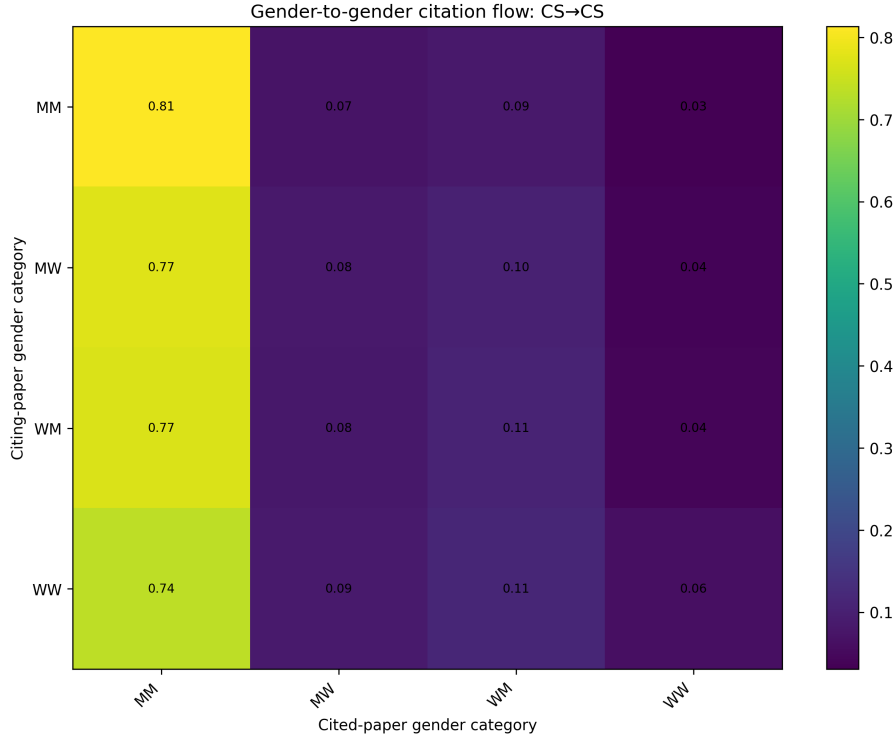


Figure 5.21: Row-normalized gender-to-gender citation flows for the CS→CS citation slice. Each row shows the distribution of known-gender citation targets for a given citing-paper first-last author gender category.

These differences should not be interpreted directly as gender-specific citation preferences. Row normalization controls for differences in the total number of citations made by each citing category, but it does not control for the composition of the papers available to be cited. The underlying pool of Computer Science papers is itself strongly imbalanced across the four gender categories, and its composition varies across publication periods, topics, subfields, and venues. A larger observed citation share toward one category may therefore arise partly because that category constitutes a larger share of the relevant citation opportunity set.

The descriptive matrices instead show that the composition of citation targets varies systematically across citing categories and that these patterns are similar under the main All→CS definition and the more restrictive CS→CS slice. This motivates replacing raw citation shares with observed-versus-expected comparisons in the subsequent analyses. Expected citation flows must account for the composition of the relevant paper pool before deviations can be interpreted as over- or under-citation.

Taken together, the temporal, taxonomic, and gender-flow diagnostics show that the citation network is highly structured. Citation relationships are strongly localized in time and topic, yet substantial long-lag and cross-topic citation flows remain. At the same time, raw gender-to-gender citation shares vary across citing categories but are inseparable from the composition of the underlying citation opportunity sets. These observations provide the empirical basis for the conditioned temporal and topical baselines, the taxonomy-sensitive relevance specifications, the citation-lag analysis, and the comparison of the primary All→CS results with the CS→CS robustness slice.

5.6 Citation-Derived Network Diagnostics

The bibliographic-coupling and co-citation representations provide alternative definitions of local scholarly similarity that are used later to construct network-neighborhood baselines and network-derived model features. Unlike the direct citation graph, however, these representations are intentionally sparse approximations: only retained similarity relationships are represented, and not every paper obtains a usable network neighborhood.

The purpose of the diagnostics in this section is therefore methodological rather than inferential. First, network coverage is examined to establish which parts of the Computer Science corpus can contribute to the bibliographic-coupling and co-citation analyses. Second, retained edge weights are examined to characterize the strength and skewness of the similarity relations encoded by the two networks. Third, paper-level network features are compared across gender categories and their distributional properties are assessed. These diagnostics determine how the later network-based results should be interpreted, which transformations are appropriate for network features, and how strongly coverage limitations constrain the corresponding estimands.

5.6.1 Coverage and Sparsity across Network Representations

The three network representations differ substantially in scale and coverage. The final direct citation graph contains 3,026,412 nodes and 18,904,534 directed edges. By comparison, the retained bibliographic-coupling graph contains 934,330 nodes and 3,864,772 undirected edges, while the retained co-citation graph contains 188,084 nodes and 232,265 undirected edges.

All three representations are sparse relative to the number of theoretically possible paper pairs. The direct graph has a density of approximately 2.06×10^{-6} , the bibliographic-coupling graph approximately 8.85×10^{-6} , and the co-citation graph approximately 1.31×10^{-5} . Although the latter two densities are numerically somewhat higher, this reflects their much smaller retained node sets rather than broad corpus coverage.

Table 5.13 summarizes the resulting network sizes and coverage rates.

Table 5.13: Size, density, and paper coverage of the final network representations. Coverage denotes the share of papers appearing as a node in the retained representation.

Representation	Nodes	Edges	CS coverage	Known-gender CS coverage
Direct citation	3,026,412	18,904,534	100.00%	100.00%
Bibliographic coupling	934,330	3,864,772	38.41%	36.00%
Co-citation	188,084	232,265	7.56%	7.84%

The difference in coverage is consequential for the network-neighborhood baselines. Bibliographic coupling provides a retained neighborhood for slightly more than one third of known-gender CS papers, whereas co-citation covers fewer than one in twelve. Consequently, N1 and especially N2 do not estimate expected citation reception for the entire known-gender CS corpus. Their results pertain to the subset of papers for which the corresponding retained network contains a usable neighborhood.

Coverage also varies across gender categories. As shown in Figure 5.22, bibliographic-coupling coverage is 36.05% for MM papers, 36.97% for MW papers, and 37.53% for WM papers, but falls to 29.80% for WW papers. Co-citation coverage is lower throughout and decreases from 8.44% for MM papers to 6.86% for MW, 6.12% for WM, and 5.31% for WW papers.

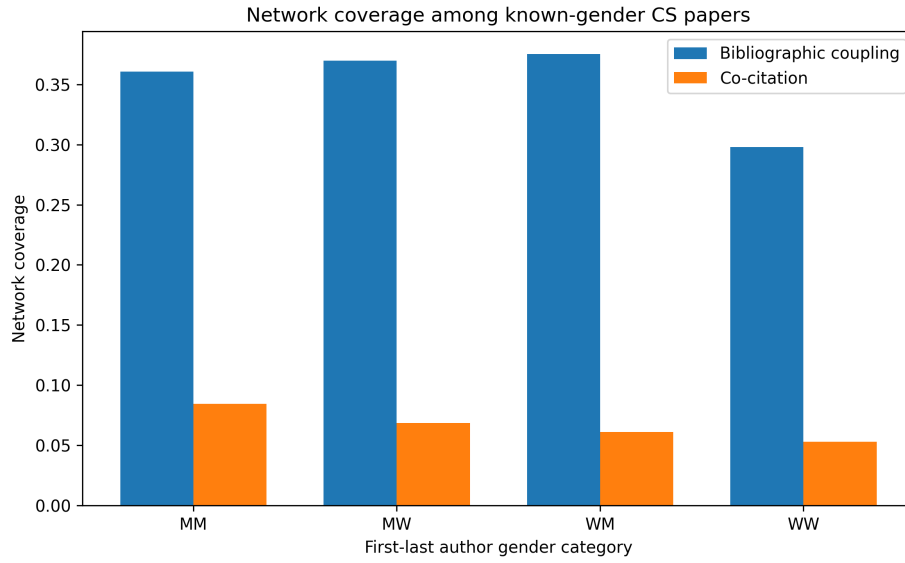


Figure 5.22: Coverage of the retained bibliographic-coupling and co-citation networks among known-gender Computer Science papers. Coverage denotes the share of papers appearing in the corresponding network representation.

The unequal coverage rates mean that comparisons based on the two citation-derived representations must not be interpreted as if the same paper population were observed in each network. In particular, the lower WW coverage indicates that the composition of the covered sample differs from the composition of the full known-gender CS corpus. Coverage is therefore reported alongside the later network-neighborhood estimates rather than treated as a purely technical preprocessing detail.

The temporal distribution of coverage provides an additional distinction between the two representations. Bibliographic-coupling coverage generally increases across publication cohorts, from 13.27% among papers from the 1970s to 29.90% in the 1990s, 34.07% in the 2000s, and 40.04% in the 2010s, before declining moderately to 36.75% in the 2020s.

Co-citation follows a markedly different pattern. Coverage is 18.31% among known-gender CS papers from the 1980s, but declines to 15.26% in the 1990s, 11.22% in the 2000s, 6.13% in the 2010s, and only 1.40% in the 2020s. Figure 5.23 shows these contrasting temporal profiles.

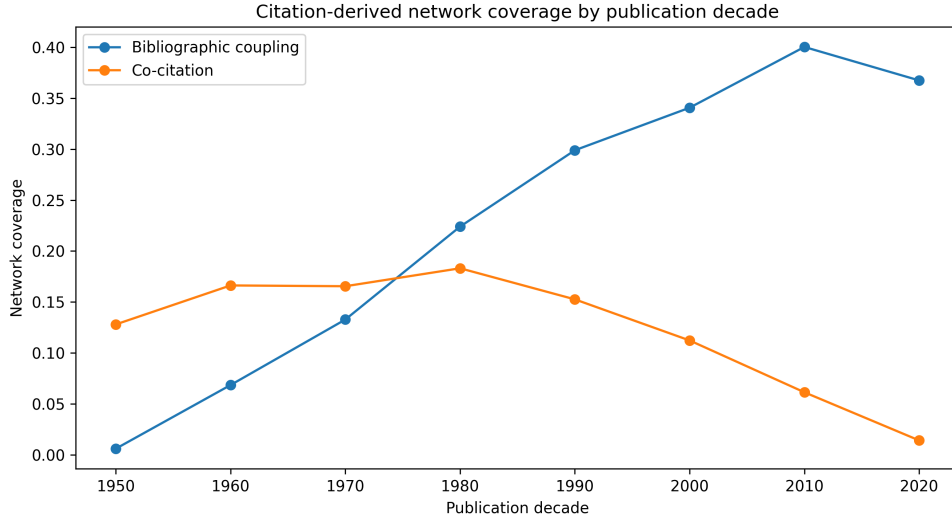


Figure 5.23: Coverage of the retained bibliographic-coupling and co-citation networks among known-gender Computer Science papers by publication decade.

The sharp decline in co-citation coverage for recent publication cohorts is consistent with the future-informed nature of this representation. Two papers can only become co-cited after subsequent papers cite them together, so recently published papers have had fewer opportunities to acquire co-citation relationships. Bibliographic coupling does not depend on subsequent recognition in the same way, because it is defined from the papers’ reference lists at publication. This distinction motivates particular caution when interpreting N2: co-citation neighborhoods represent later community association and are available for a temporally selective subset of the corpus.

5.6.2 Co-Citation and Bibliographic-Coupling Weight Distributions

The two citation-derived networks encode different notions of edge strength. In the bibliographic-coupling graph, the primary edge weight is the number of references shared by two papers. In the co-citation graph, the primary weight is the number of distinct citing papers that cite the pair together. Jaccard similarity is additionally available in the retained network artifacts, but the raw overlap counts provide the principal edge-strength measure examined here.

The retained bibliographic-coupling graph is dominated by relatively low shared-reference counts. The minimum retained edge weight is 3, the median is 4, and 75% of retained edges have weight at most 5. The 90th and 99th percentiles are 6 and 11 shared references, respectively. The distribution nevertheless contains a considerably longer upper tail, with a 99.9th percentile of 18 and a maximum of 195 shared references.

The co-citation distribution is substantially more dispersed. The minimum retained edge weight is 5 shared citing papers and the median is 8. The 90th percentile reaches 23, the 99th percentile 94, and the 99.9th percentile 331. The maximum co-citation weight is 7,178, illustrating the presence of a small number of paper pairs that are repeatedly associated by later citing work.

Table 5.14: Selected quantiles of retained edge weights in the bibliographic-coupling and co-citation networks.

Statistic	Bibliographic coupling	Co-citation
Minimum	3	5
Median	4	8
75th percentile	5	12
90th percentile	6	23
95th percentile	8	36
99th percentile	11	94
99.9th percentile	18	331
Maximum	195	7,178

Figures 5.24 and 5.25 visualize the retained weight distributions after applying a $\log(1 + x)$ transformation to the horizontal axis. This transformation is used only to make the strong right-skew visible over a manageable plotting range.

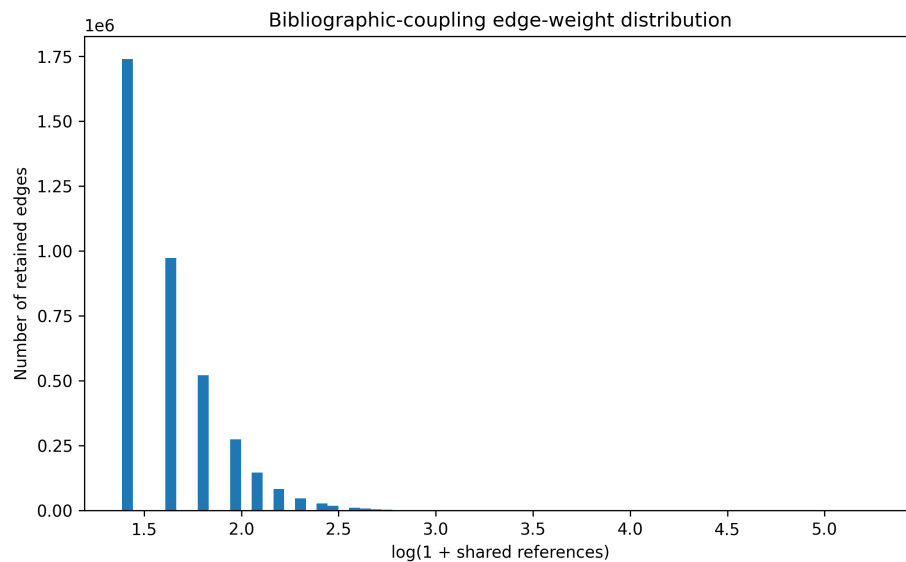


Figure 5.24: Distribution of retained bibliographic-coupling edge weights, expressed as $\log(1 + \text{shared references})$.

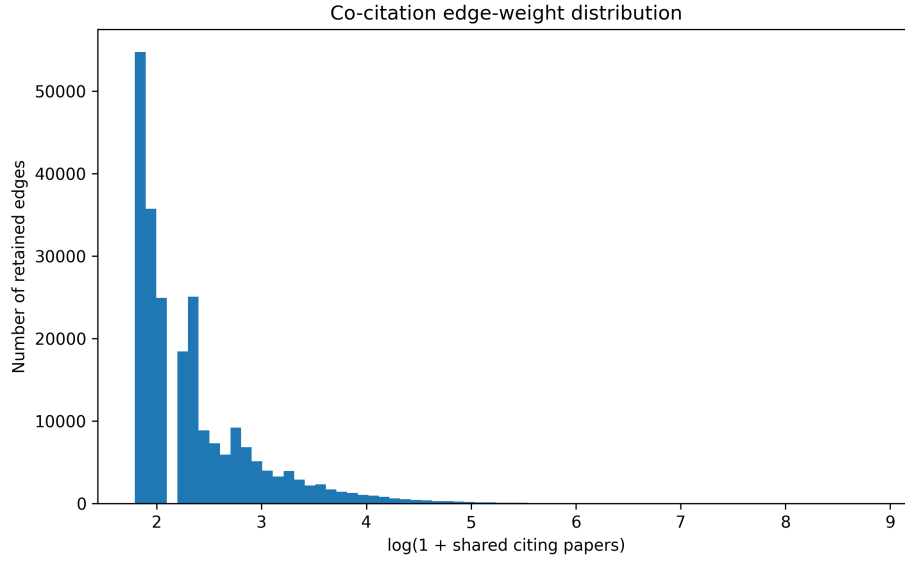


Figure 5.25: Distribution of retained co-citation edge weights, expressed as $\log(1 + \text{shared citing papers})$.

The mean retained weights further illustrate the contrast between the representations. Bibliographic-coupling edges have a mean weight of 4.26 and a median of 4, whereas co-citation edges have a mean of 13.29 and a median of 8. In both cases the mean exceeds the median, but the discrepancy is particularly large for co-citation because of its much heavier upper tail.

The raw overlap count and Jaccard similarity should also not be treated as interchangeable measures. Their Spearman correlation is only 0.35 in the bibliographic-coupling network and 0.07 in the co-citation network. A large shared-count edge can therefore represent a different notion of similarity from a high proportional overlap, particularly when the underlying reference or citer sets differ substantially in size.

For the later analyses, the most important implication is that network relations differ considerably in strength and that their weight distributions are strongly asymmetric. Unweighted neighborhood averages would discard this variation, whereas the weighted neighborhood baselines allow stronger retained relationships to contribute more to the local expectation. The skewness of these edge weights also anticipates the need to transform network-derived features before including them in the model-based specifications.

5.6.3 Network Degree and Strength by Gender Category

The retained citation-derived networks additionally generate the paper-level features used in the GLM specifications. For each paper and each network representation, four quantities are constructed: retained degree, weighted strength, mean incident edge weight, and maximum incident edge weight. Degree counts the number of retained neighbors, whereas strength sums the weights of all incident edges. Mean and maximum edge weights describe the typical and strongest retained relationship of a paper. These are the four representation-specific quantities that later enter M3 for bibliographic coupling and M4 for co-citation.

These quantities are subsequently used as representation-specific predictors in the model-based analysis after applying a $\log(1 + x)$ transformation and standardization. The EDA therefore examines both their distribution across gender categories and their overall skewness. As with the earlier citation-reception comparisons, differences reported here are descriptive and should not be interpreted as causal or as direct evidence of gender imbalance. Moreover, they are conditional on

network coverage: papers with no retained node representation do not enter these summaries.

In the bibliographic-coupling network, median degree is 4 for MM, MW, and WM papers and 3 for WW papers. Median strength is 15 for MM, MW, and WM papers and 13 for WW papers. Differences are more visible in the upper tail. The 99th percentile of degree ranges from 49 for WW papers to 55 for MW papers, while the 99th percentile of strength ranges from approximately 238 for WW papers to 275 for MW papers.

Table 5.15: Selected degree and strength summaries by gender category among known-gender CS papers covered by the retained citation-derived networks.

Representation	Gender	Median degree	P99 degree	Median strength	P99 strength
Bibliographic coupling	MM	4	53.00	15	257.00
	MW	4	55.00	15	275.00
	WM	4	54.00	15	266.00
	WW	3	49.00	13	238.33
Co-citation	MM	2	12.00	13	371.00
	MW	2	12.22	13	394.54
	WM	2	12.00	12	323.00
	WW	2	10.00	12	316.76

The co-citation network is considerably smaller and has lower retained degree. All four gender categories have a median degree of 2. Median strength is 13 for MM and MW papers and 12 for WM and WW papers. Again, the larger differences occur in the upper tail rather than at the median. The 99th percentile of co-citation strength is 371 for MM, approximately 395 for MW, 323 for WM, and approximately 317 for WW papers.

Figures 5.26 and 5.27 visualize the strength distributions after a $\log(1 + x)$ transformation. The plots show substantial within-category variation and considerable overlap between categories, reinforcing that these network measures should be treated as continuous structural covariates rather than interpreted from group averages alone.

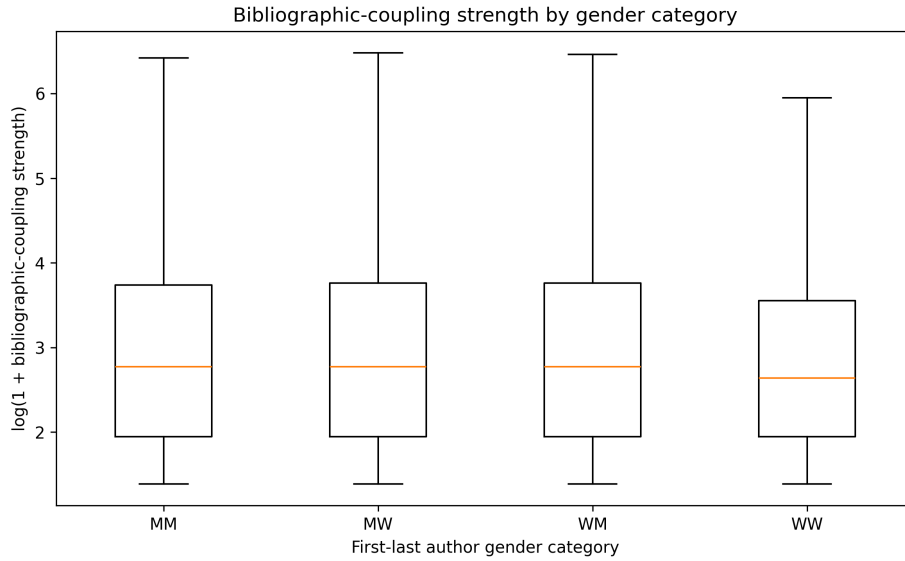


Figure 5.26: Distribution of $\log(1 + \text{strength})$ in the retained bibliographic-coupling network by first-last author gender category.

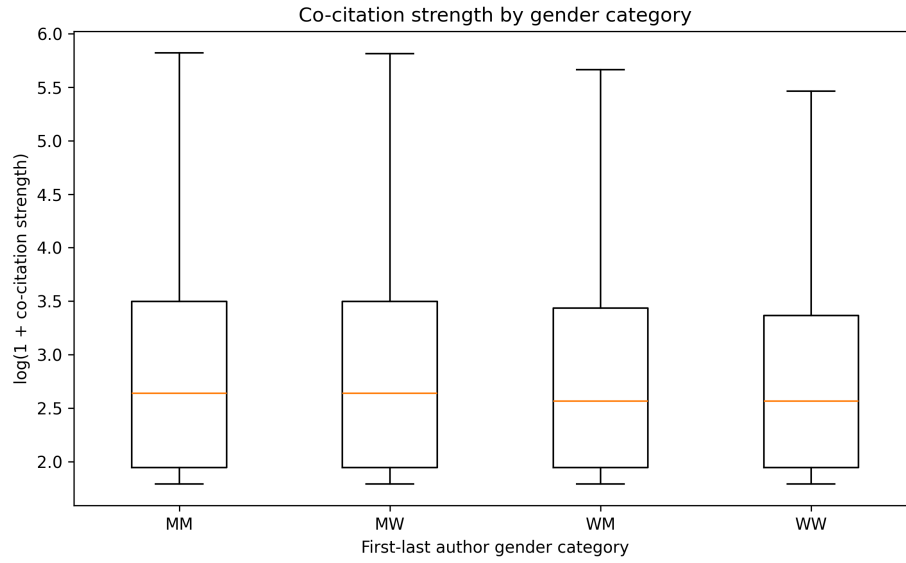


Figure 5.27: Distribution of $\log(1 + \text{strength})$ in the retained co-citation network by first-last author gender category.

The raw feature distributions are strongly right-skewed. For bibliographic coupling, skewness is 4.49 for degree and 5.66 for strength, while mean and maximum edge weight have skewness values of 11.86 and 4.50. In the co-citation representation the asymmetry is even more pronounced: degree has skewness 2.79, strength 33.27, mean edge weight 97.22, and maximum edge weight 55.55.

Applying $\log(1 + x)$ substantially reduces this asymmetry in every case. Bibliographic-coupling degree and strength skewness fall to 0.65 and 0.45, respectively, while co-citation degree and strength fall to 1.09 and 1.04. The transformation also strongly reduces the skewness of mean and maximum edge weights, although these distributions remain more asymmetric.

Table 5.16: Raw and $\log(1 + x)$ skewness of paper-level network features among covered known-gender CS papers.

Representation	Feature	Raw skewness	$\log(1+x)$ skewness
Bibliographic coupling	Degree	4.49	0.65
	Strength	5.66	0.45
	Mean weight	11.86	1.73
	Maximum weight	4.50	0.89
Co-citation	Degree	2.79	1.09
	Strength	33.27	1.04
	Mean weight	97.22	2.16
	Maximum weight	55.55	1.76

These diagnostics provide a direct empirical justification for applying $\log(1 + x)$ transformations to the citation-derived network features in the later GLM specifications. Specifically, the transformed degree, strength, mean-weight, and maximum-weight features form the bibliographic-coupling and co-citation feature blocks used in M3, M4, and M5. Without transformation, a comparatively small number of highly connected or very strongly associated papers would dominate the numerical scale of the corresponding predictors.

The network features are also not independent of one another. In the bibliographic-coupling

representation, degree and strength have a Spearman correlation of 0.96, while mean and maximum edge weight correlate at 0.89. In the co-citation representation, degree and strength correlate at 0.85, and mean and maximum edge weight at 0.97. Co-citation strength is also strongly associated with maximum edge weight ($\rho = 0.89$).

The high correlations indicate that several network features encode overlapping dimensions of local network position. This does not make them descriptively redundant, but it does mean that coefficients attached to individual network features in multivariable models should not be interpreted as fully independent structural effects. The EDA therefore motivates both transformation of these features and an explicit redundancy check before interpreting the representation-specific GLM specifications.

Overall, the citation-derived network diagnostics establish three constraints that carry forward into the later analysis. First, N1 and especially N2 apply to covered subsets rather than the full CS corpus, making coverage an essential part of their interpretation. Second, bibliographic-coupling and co-citation relationships vary substantially in strength, supporting weighted local expectations rather than purely binary neighborhoods. Third, paper-level network features are strongly right-skewed and correlated, motivating transformed predictors and careful interpretation of model coefficients. The particularly strong temporal selectivity of co-citation coverage additionally reinforces the distinction between bibliographic coupling as a reference-based similarity representation and co-citation as a later, community-generated association.

5.7 Exploratory Summary

The exploratory analysis reveals that citation behaviour in the analytical corpus is strongly structured along temporal, topical, and network dimensions. These patterns make unconditional comparisons of raw citation counts insufficient for assessing gender-related citation imbalance and motivate the progressively conditioned analyses developed in the following chapters.

First, direct citation reception is highly heterogeneous. Most papers receive relatively few citations, a substantial fraction remain uncited within the analytical network, and a small upper tail accumulates very large citation counts. Raw citation reception also differs across first-last author gender categories, but these differences occur within a strongly skewed distribution and cannot be separated descriptively from differences in paper composition and citation opportunity. The subsequent analysis therefore focuses on observed-versus-expected citation outcomes rather than treating differences in unconditional group means as evidence of over- or under-citation.

Second, citation opportunities are strongly dependent on publication time. All→CS citations are concentrated on relatively recent work, while a meaningful long-lag tail extends citation attention over several decades. Citation flows likewise concentrate within the citing paper’s own and preceding publication cohorts. Because publication cohorts differ both in their available citation pools and in the composition of the CS literature, temporal exposure must be incorporated into the expected-citation framework. This motivates the decade-conditioned baselines and the separate citation-lag analysis used to examine whether citation patterns vary across shorter- and longer-term scholarly attention.

Third, citations are strongly but not exclusively localized by research area. A large share of All→CS citation edges connects papers within the same topic, subfield, or field, while substantial flows also cross these boundaries. The CS→CS subfield matrix further shows that the extent of within-subfield concentration differs across areas of Computer Science. Consequently, neither a corpus-wide opportunity set nor an exclusively same-topic definition provides a complete representation of topical relevance. These observations motivate the topic- and subfield-conditioned direct baselines as well as the hierarchical and taxonomy-weighted relevance specifications introduced later.

Fourth, the raw composition of citation targets varies across citing gender categories in both the All→CS and CS→CS slices. However, these descriptive flows are inseparable from the strongly unequal composition of the papers available to be cited and from temporal and topical differences between citing groups. The gender-flow matrices are therefore treated as descriptive evidence that citation behaviour is compositionally structured, rather than as direct measures of citation preference. The observed-versus-expected framework is required to determine whether observed gender-specific citation shares differ from those implied by an appropriate comparison set.

Finally, the bibliographic-coupling and co-citation representations provide additional but substantially more selective views of scholarly proximity. Their incomplete and unequal coverage implies that the corresponding neighborhood baselines estimate citation expectations for covered subsets of the CS literature rather than for the full corpus. Co-citation coverage is particularly low for recent papers, reflecting the fact that co-citation relationships depend on subsequent citing activity and therefore incorporate future information relative to the cited papers. Bibliographic coupling and co-citation are consequently interpreted as complementary representations of local scholarly similarity rather than interchangeable alternatives.

The citation-derived network measures are also strongly right-skewed, with a small number of papers occupying exceptionally high-degree or high-strength positions. This motivates the use of weighted neighborhoods in the network baselines and the $\log(1 + x)$ transformation of network-derived predictors before their inclusion in the regression models. Strong correlations among several network features further motivate caution when interpreting individual coefficients in multivariable specifications.

Taken together, the EDA establishes the central methodological premise of the thesis: citation opportunities are heterogeneous, and much of this heterogeneity is systematically related to publication period, research area, and network position. The subsequent analyses therefore introduce increasingly structured expectations rather than relying on a single global benchmark. Direct baselines progressively condition on temporal, topical, and venue characteristics; network-neighborhood baselines define expectations from locally similar papers in bibliographic-coupling and co-citation space; and the regression models evaluate these dimensions jointly at the paper level. The primary All→CS analysis is complemented by the more restrictive CS→CS slice to assess the robustness of the resulting patterns to the definition of the citing population.

6. Results

6.1 Direct Citation Imbalance

This section reports the direct citation results for the four known first-last author gender categories. The primary analysis considers citations from papers in all fields to known-gender Computer Science papers (All→CS), while the CS→CS slice is used to assess whether the same patterns are present when both the citing and cited papers belong to Computer Science. Results are first described in terms of the observed citation distribution and are then compared with the expectations generated by the D0-D3 baselines defined in Chapter 4.

6.1.1 Observed Citation Outcomes by Gender Category

The final dataset contained 1,542,691 Computer Science papers, of which 1,081,363 (70.10%) could be assigned to one of the four known first-last author gender categories. These papers received 8,686,237 citations in the primary All→CS slice, originating from 1,966,202 unique citing papers. In total, 708,057 known-gender CS papers received at least one citation in this slice.

The distribution of papers across gender categories was highly uneven. MM papers accounted for 73.11% of known-gender CS papers, followed by WM papers at 12.74%, MW papers at 9.01%, and WW papers at 5.14%. The observed citation distribution was more concentrated towards MM papers than the underlying paper distribution. MM papers received 6,942,653 citations, corresponding to 79.93% of all citations in the primary slice. By comparison, MW papers received 641,435 citations (7.38%), WM papers 812,529 citations (9.35%), and WW papers 289,620 citations (3.33%).

Table 6.1: Observed citation outcomes for known-gender Computer Science papers in the All→CS citation slice.

Category	Papers	Paper share	Citations	Citation share	Mean citations	Uncited share
MM	790,617	73.11%	6,942,653	79.93%	8.781	33.41%
MW	97,399	9.01%	641,435	7.38%	6.586	36.69%
WM	137,756	12.74%	812,529	9.35%	5.898	37.46%
WW	55,591	5.14%	289,620	3.33%	5.210	39.28%

The paper-level citation distributions show the same ordering. Average citation counts decreased from 8.781 for MM papers to 6.586 for MW, 5.898 for WM, and 5.210 for WW papers. The median was one citation in all four categories, indicating that the difference in means was concentrated primarily in the upper part of the citation distribution. At the 90th percentile, citation counts were 16 for MM, 13 for MW, 12 for WM, and 10 for WW papers. The share of papers with zero citations also increased across these categories, from 33.41% for MM papers to 39.28% for WW papers.

These descriptive differences do not by themselves account for the different representation of the four categories in the available paper population. The D0-D3 baselines therefore compare these observed citation counts with increasingly structured expectations.

6.1.2 Observed/Expected Results under D0-D3

Table 6.2 reports the observed-to-expected (O/E) citation ratios for the primary All→CS slice. An O/E ratio of one denotes equality between the observed and expected citation counts, while values below and above one indicate fewer and more observed citations than expected, respectively.

Table 6.2: Observed-to-expected citation ratios under the D0-D3 baselines for the All→CS citation slice.

Baseline	MM	MW	WM	WW
D0: global paper share	1.093	0.820	0.734	0.649
D1: decade	1.041	0.938	0.884	0.695
D2: decade $\times$ subfield	1.041	0.935	0.877	0.712
D3: decade $\times$ venue rank	1.031	0.949	0.907	0.762

Under the unconditioned D0 baseline, the differences between observed and expected citations were substantial. MM papers received 9.32% more citations than expected from their 73.11% share of the known-gender CS paper population, corresponding to an O/E ratio of 1.093. In contrast, MW papers received 18.01% fewer citations than expected ($O/E = 0.820$), WM papers 26.57% fewer ($O/E = 0.734$), and WW papers 35.14% fewer ($O/E = 0.649$). In absolute terms, D0 expected approximately 6.35 million citations to MM papers compared with 6.94 million observed, whereas the three remaining categories each received fewer citations than their paper-share expectations.

Under the decade-conditioned D1 baseline, the magnitude of these differences was substantially smaller for MM, MW, and WM papers. The MM ratio fell from 1.093 to 1.041, while the MW and WM ratios increased from 0.820 to 0.938 and from 0.734 to 0.884, respectively. The WW ratio also increased, but remained markedly further below one at 0.695. Under the decade-conditioned D1 baseline, WW papers received 30.50% fewer citations than expected.

Adding subfield conditioning in D2 produced comparatively small changes relative to D1. The O/E ratio was 1.041 for MM, 0.935 for MW, 0.877 for WM, and 0.712 for WW papers. The corresponding deviations from expectation were +4.14%, -6.46% , -12.35% , and -28.77% , respectively. The D2 specification therefore produces only a limited additional change relative to the decade-conditioned D1 baseline.

The D3 baseline, which conditions on publication decade and venue-rank stratum rather than on subfield, produced the ratios closest to one among the four direct baselines. MM papers remained above expectation at $O/E = 1.031$, corresponding to 3.09% more citations than expected. MW and WM papers remained below expectation at 0.949 and 0.907, respectively, corresponding to deficits of 5.13% and 9.29%. The largest remaining difference was again observed for WW papers, whose O/E ratio increased to 0.762 but still represented 23.81% fewer citations than expected under D3.

Across all four specifications, the direction and ordering of the results were therefore stable. MM papers were observed above their baseline expectation in every specification, whereas MW, WM, and WW papers were observed below expectation. Across the decade- and structure-conditioned specifications, the deviations were smaller, particularly for MW and WM papers, but remained present. WW papers consistently exhibited the lowest O/E ratio, ranging from 0.649 under D0 to 0.762 under D3.

6.1.3 All-to-CS versus CS-to-CS Direct Citation Results

The primary All→CS results were closely reproduced when the analysis was restricted to citations originating from CS papers. The CS→CS slice contained 7,023,500 citations to known-gender CS papers, corresponding to 80.86% of the citations in the primary All→CS slice. These citations originated from 1,331,175 unique CS papers and reached 645,672 unique known-gender CS papers.

At the descriptive level, the citation shares in the restricted slice were almost identical to those in the primary analysis. MM papers received 79.76% of CS→CS citations, compared with 79.93% in All→CS. The corresponding shares were 7.53% versus 7.38% for MW, 9.43% versus 9.35% for WM, and 3.28% versus 3.33% for WW papers.

Table 6.3: Comparison of observed-to-expected citation ratios between the All→CS and CS→CS citation slices.

Baseline	Citation slice	MM	MW	WM	WW
D0	All→CS	1.093	0.820	0.734	0.649
	CS→CS	1.091	0.836	0.740	0.638
D1	All→CS	1.041	0.938	0.884	0.695
	CS→CS	1.039	0.958	0.893	0.684
D2	All→CS	1.041	0.935	0.877	0.712
	CS→CS	1.039	0.955	0.886	0.702
D3	All→CS	1.031	0.949	0.907	0.762
	CS→CS	1.028	0.968	0.916	0.751

The similarity between the two citation slices persisted across all four baselines. Under D0, for example, the O/E ratios changed only from 1.093 to 1.091 for MM, from 0.820 to 0.836 for MW, from 0.734 to 0.740 for WM, and from 0.649 to 0.638 for WW papers. The same pattern remained under D1-D3. Restricting citing papers to Computer Science slightly increased the MW and WM ratios, while producing slightly lower ratios for MM and WW papers, but none of these changes altered the direction or relative ordering of the results.

Under the most extensively conditioned direct specification, D3, the CS→CS O/E ratios were 1.028 for MM, 0.968 for MW, 0.916 for WM, and 0.751 for WW papers, compared with 1.031, 0.949, 0.907, and 0.762 in the All→CS slice. The direct citation imbalance observed in the primary analysis therefore persists when citations originating outside Computer Science are excluded. The same qualitative pattern is present within the CS citation network itself, with WW papers remaining the category furthest below their baseline expectation in every specification.

6.2 Citation Flow and Citing-Side Patterns

The direct citation results in Section 6.1 describe how citations were distributed across the four cited-paper gender categories. This section instead examines the citing side of the citation network. In particular, it considers citation flows between the gender categories of citing and cited papers and the composition of outgoing references within each citing category.

The analysis is descriptive rather than an alternative expected-citation baseline. Consequently, differences in the observed flow matrix should not be interpreted as adjusted citation imbalances, since they also reflect differences in the availability, publication period, topic, venue, and network position of papers belonging to the four gender categories.

6.2.1 Observed Gender-to-Gender Citation Flows

In the primary All→CS citation slice, 5,948,956 citation edges connected papers for which the first-last author gender category was known on both the citing and cited side, corresponding to 51.28% of that slice. For the more restrictive CS→CS slice, 4,812,158 edges had known gender categories on both sides, corresponding to 51.12%. The citation-flow results therefore describe the known-gender portion of the two analytical citation slices rather than all citation edges.

In both slices, MM papers constituted the largest cited category for every citing group. In the primary All→CS slice, 81.50% of known-gender citations made by MM papers were directed towards MM papers. The corresponding MM target share was lower among the other citing categories, at 77.52% for MW papers, 76.97% for WM papers, and 72.97% for WW papers.

The opposite gradient was visible for citations directed towards WW papers. WW papers accounted for 3.08% of known-gender citations from MM papers, 4.08% from MW papers, 4.15% from WM papers, and 6.79% from WW papers. Differences were also present for citations directed towards the MW and WM categories.

The CS→CS restriction produced a highly similar descriptive structure. MM papers again constituted the largest target category for every citing group, while the MM target share declined from 81.32% among MM citing papers to 73.54% among WW citing papers. Conversely, the WW target share increased from 3.06% among MM citing papers to 6.28% among WW citing papers.

The similarity between the All→CS and CS→CS citation-flow structures shows that the broad descriptive pattern persists when citations originating outside Computer Science are excluded. However, the relative sizes of the four citing-paper groups differ substantially, so absolute citation-flow counts cannot be compared directly as measures of citing preference. The within-citing-category composition of references provides a more informative descriptive comparison.

6.2.2 Row-Normalized Citing-Side Reference Composition

To compare citation behaviour across citing categories, the citation-flow matrix was normalized within each citing category. Each row therefore sums to 100% and represents the distribution of known-gender references made by papers belonging to that citing category.

Table 6.4 summarizes the principal differences in the primary All→CS slice. Because the largest descriptive gradient occurs between citations to MM and WW papers, these two target categories are reported directly.

Table 6.4: Selected within-citing-category citation shares in the All→CS citation slice. Values denote the percentage of known-gender references directed towards MM and WW papers.

Citing category	MM target	WW target
MM	81.50%	3.08%
MW	77.52%	4.08%
WM	76.97%	4.15%
WW	72.97%	6.79%

MM papers displayed the most MM-concentrated reference composition: 81.50% of their known-gender citations were directed towards MM papers. The corresponding share decreased to 77.52% for MW, 76.97% for WM, and 72.97% for WW citing papers. Equivalently, the combined share directed towards MW, WM, or WW papers increased from 18.50% among MM citers to 22.48% among MW, 23.03% among WM, and 27.03% among WW citers.

The clearest individual-category difference concerned WW citation targets. WW papers represented 3.08% of known-gender references made by MM papers, compared with 4.08% for MW citing papers, 4.15% for WM citing papers, and 6.79% for WW citing papers. For MM citing papers specifically, the remaining known-gender references were distributed as 6.92% to MW papers and 8.50% to WM papers.

The same qualitative pattern was observed after restricting the analysis to CS→CS citations. Table 6.5 reports the corresponding row-normalized citation shares.

Table 6.5: Within-citing-category composition of cited gender categories for the CS→CS citation network. Each row sums to 100%.

Citing category	MM	MW	WM	WW
MM	81.32%	7.06%	8.56%	3.06%
MW	77.41%	8.30%	10.33%	3.97%
WM	76.90%	8.23%	10.81%	4.06%
WW	73.54%	8.69%	11.50%	6.28%

Within Computer Science, MM papers again directed the largest proportion of their references towards MM papers, at 81.32%, whereas this share declined to 77.41% for MW, 76.90% for WM, and 73.54% for WW citing papers. Conversely, the combined share directed towards MW, WM, and WW papers increased from 18.68% among MM citers to 22.59% among MW, 23.10% among WM, and 26.46% among WW citers. WW-to-WW citations constituted 6.28% of the references from WW citing papers, approximately twice the 3.06% share observed among MM citing papers.

Differences in reference-list length were comparatively moderate across the four citing categories. Among known-gender CS papers, the mean number of references was 8.16 for MM papers, 8.68 for MW papers, 8.94 for WM papers, and 7.53 for WW papers, with medians of six, seven, seven, and six references, respectively. The corresponding mean direct out-degrees were 6.69, 7.25, 7.47, and 6.41. Row normalization therefore allows the citation composition to be compared independently of these differences in the absolute number of outgoing references.

Overall, the descriptive citing-side results show a consistent gradient in reference composition. MM citing papers had the most MM-dominated reference lists, while citing papers involving female first or last authors allocated progressively larger shares of their known-gender references to the MW, WM, and WW categories. This pattern was visible in both the primary All→CS slice and the more restrictive CS→CS slice. These differences describe the observed composition of citation flows; they do not by themselves establish over- or under-citation, because the available pool of potentially citable papers differs substantially across gender categories and over time. The expected-citation baselines reported elsewhere in this chapter explicitly condition the comparison on selected dimensions of this compositional heterogeneity.

6.3 Topical and Temporal Citation Results

The direct-citation analyses established systematic differences between observed and expected citation counts across the four gender categories. This section examines how these differences vary across two additional dimensions of citation context: topical proximity and citation age. The topical analysis distinguishes citation relationships according to their position in the publication taxonomy, while the temporal analysis separates citations according to the elapsed time between publication of the cited and citing papers. Results are reported for both the primary All→CS citation slice and the more restrictive CS→CS slice.

6.3.1 Topical Relevance and Taxonomy-Weighted Results

The topical-relevance analysis evaluates whether the O/E differences observed in the aggregate citation network persist within increasingly local topical contexts. Four discrete citation subsets are considered: same-topic citations, same-subfield citations, cross-topic citations within the same subfield, and cross-subfield citations. The same-subfield subset includes same-topic citations, whereas the cross-topic same-subfield subset excludes them. A complementary taxonomy-weighted specification assigns larger contributions to citations between taxonomically closer papers.

Figure 6.1 presents the resulting O/E ratios for the primary All→CS citation slice. The relative ordering of the four gender categories remains stable across the topical specifications. MM papers

are observed above expectation throughout, whereas WM and WW papers remain below expectation in every comparison. MW papers are approximately at parity for same-topic citations but move further below expectation as topical distance increases.

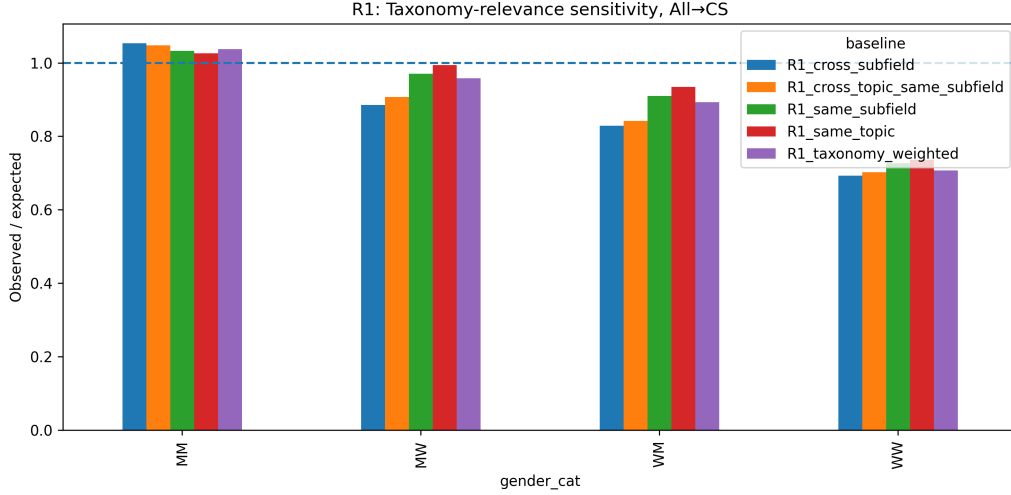


Figure 6.1: Observed-to-expected citation ratios across topical-relevance specifications in the All→CS citation slice. The dashed horizontal line denotes parity between observed and expected citations ($O/E = 1$).

The exact O/E estimates are reported in Table 6.6. The same-topic subset contains 3,630,264 citation edges. Within this most topically local comparison, MM papers have an O/E ratio of 1.026, corresponding to 2.63% more citations than expected. MW papers are close to parity at 0.994, while WM papers have an O/E ratio of 0.934. WW papers remain substantially below expectation at 0.736, corresponding to 26.37% fewer citations than expected.

Table 6.6: Observed-to-expected citation ratios across topical-relevance specifications. Same-subfield includes same-topic citations, whereas cross-topic same-subfield excludes same-topic edges.

Citation slice	Topical specification	MM	MW	WM	WW
All→CS	Same topic	1.026	0.994	0.934	0.736
	Same subfield	1.032	0.971	0.910	0.727
	Cross-topic, same subfield	1.048	0.907	0.843	0.702
	Cross-subfield	1.054	0.885	0.828	0.693
	Taxonomy-weighted	1.037	0.958	0.893	0.707
CS→CS	Same topic	1.026	0.994	0.934	0.736
	Same subfield	1.032	0.971	0.910	0.727
	Cross-topic, same subfield	1.048	0.907	0.843	0.702
	Cross-subfield	1.054	0.913	0.819	0.643
	Taxonomy-weighted	1.035	0.965	0.899	0.713

For the All→CS slice, the magnitude of the O/E differences generally increases as citation relationships become less topically local. The MM ratio rises from 1.026 among same-topic citations to 1.032 for same-subfield citations, 1.048 for cross-topic citations within the same subfield, and 1.054 for cross-subfield citations.

The reverse pattern is visible for the three remaining categories. MW papers move from near parity within the same topic ($O/E = 0.994$) to 0.885 among cross-subfield citations. For WM papers, the ratio declines from 0.934 to 0.828 across the same comparison.

The gradient is particularly clear for WW papers. Their O/E ratio decreases from 0.736 for same-topic citations to 0.727 for the inclusive same-subfield subset and 0.702 when same-topic citations are excluded. Among cross-subfield citations, the ratio falls further to 0.693. The corresponding deficit relative to expectation therefore increases from 26.37% within the same topic to 30.73% across subfield boundaries.

The taxonomy-weighted specification produces an intermediate summary of these topical patterns. In the All→CS slice, 8,054,657 citation edges receive positive relevance weight, corresponding to a total taxonomy-relevance weight of 5,934,067.5. The resulting O/E ratios are 1.037 for MM, 0.958 for MW, 0.893 for WM, and 0.707 for WW papers. Thus, weighting citation relationships according to topical proximity changes the magnitude of the estimates but preserves their ordering.

Overall, the topical analysis shows that the observed citation pattern is present even within the most topically local citation relationships. The differences are generally smaller within the same topic and larger when citations span topics or subfields, with WW papers remaining the category furthest below expectation throughout the topical specifications.

6.3.2 Citation-Lag and Long-Memory Results

The temporal analysis examines how the O/E pattern varies over the citation life cycle. As defined in the methodology, valid citation edges are first separated into five fine-grained lag buckets according to the difference between the citing and cited publication years: 0-2 years, 3-5 years, 6-10 years, 11-20 years, and more than 20 years. These intervals distinguish very recent citation activity from medium-term citation accumulation and citations to substantially older literature.

For the main temporal comparison, the five intervals are additionally aggregated into three broader citation-memory groups. The 0-2 and 3-5 year buckets form the *short* group (0-5 years), the 6-10 year bucket forms the *middle* group, and the 11-20 and 20+ year buckets form the *long* group (> 10 years).

All 8,686,237 citation edges in the All→CS analysis have valid non-negative citation lags. Of these, 3,790,516 citations (43.64%) occur within the short-memory window, 2,338,119 (26.92%) within the middle window, and 2,557,602 (29.44%) more than ten years after publication.

Figure 6.2 shows the corresponding O/E trajectories across the three broad temporal groups. MM papers remain slightly above expectation throughout, while MW and WM papers remain below one. The most pronounced temporal pattern is observed for WW papers, whose O/E ratio decreases from the short-term window through the middle period and remains lowest among long-memory citations.

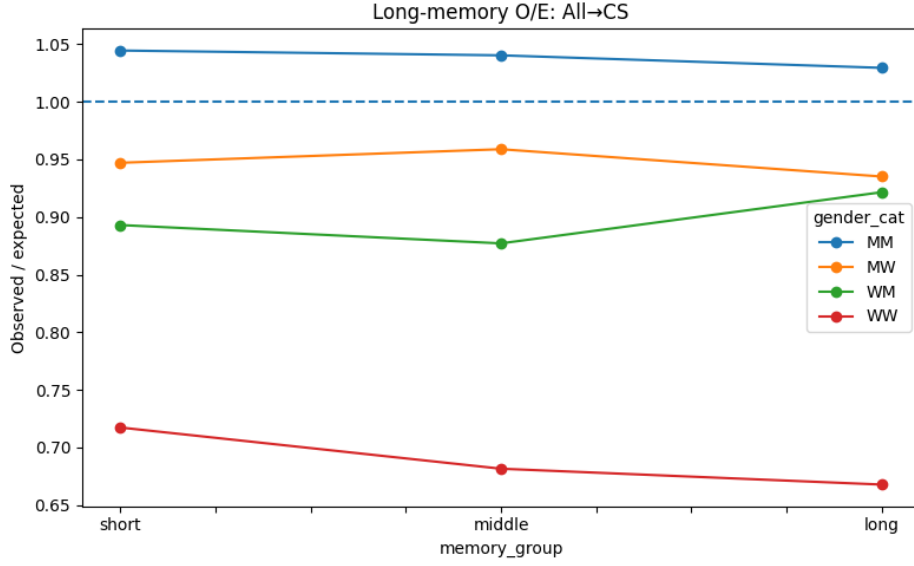


Figure 6.2: Observed-to-expected citation ratios across short (0-5 years), middle (6-10 years), and long (> 10 years) citation-memory groups in the All→CS slice. The dashed horizontal line denotes parity between observed and expected citations ($O/E = 1$).

At the broad-group level, the All→CS MM ratio decreases modestly from 1.045 in the short window to 1.040 in the middle window and 1.030 among long-memory citations. MW papers have O/E ratios of 0.947, 0.959, and 0.935, respectively, while the corresponding WM ratios are 0.893, 0.877, and 0.922.

For WW papers, the ratio is already substantially below expectation among citations occurring within five years of publication, at 0.717. It declines to 0.681 in the 6-10 year group and to 0.668 among citations with lags above ten years. The long-memory estimate therefore corresponds to 33.23% fewer citations than expected.

The five fine-grained lag buckets reveal additional variation that is partly obscured by the broader aggregation. Table 6.7 reports these estimates for the All→CS slice.

Table 6.7: Observed-to-expected citation ratios across fine-grained citation-lag buckets in the All→CS citation slice.

Citation lag	Edges	MM	MW	WM	WW
0-2 years	1,504,303	1.039	0.960	0.921	0.738
3-5 years	2,286,213	1.042	0.953	0.893	0.709
6-10 years	2,338,119	1.040	0.959	0.877	0.681
11-20 years	1,768,339	1.030	0.958	0.918	0.687
20+ years	789,263	1.040	0.771	0.833	0.611

The fine-grained results show that the MM ratio is comparatively stable across citation age, ranging from 1.030 to 1.042. No temporal bucket places MM papers below their expected citation count.

For MW papers, the ratios are also relatively stable through the first 20 years. They range from 0.953 to 0.960 across the 0-2, 3-5, 6-10, and 11-20 year windows. A substantially larger deviation appears only in the oldest citation interval, where the ratio falls to 0.771. Citations occurring more than 20 years after publication therefore contain 22.91% fewer citations to MW papers than expected.

WM papers show a non-monotonic pattern. Their O/E ratio decreases from 0.921 in the 0-2 year

window to 0.893 at 3-5 years and 0.877 at 6-10 years. It then increases to 0.918 for lags of 11-20 years before declining to 0.833 among citations occurring more than 20 years after publication.

For WW papers, the ratio declines from 0.738 in the first two years to 0.709 at 3-5 years and 0.681 at 6-10 years. It increases slightly to 0.687 in the 11-20 year interval before falling to 0.611 in the 20+ year bucket. This oldest interval therefore contains the largest WW deviation in the temporal analysis, corresponding to 38.88% fewer observed citations than expected.

The difference between the broad and fine-grained results is particularly important for the long-memory category. Aggregating the 11-20 and 20+ year intervals yields a WW O/E ratio of 0.668, whereas the individual ratios are 0.687 and 0.611, respectively. The broad long-memory estimate therefore summarizes a heterogeneous older-citation period in which the largest deviation occurs specifically among citations more than two decades after publication.

Overall, the temporal results show that the below-expectation WW pattern is not confined to the early citation window. It is present in every fine-grained lag interval and remains visible more than 20 years after publication. The fine-grained analysis also reveals that the oldest citation window differs from the preceding intervals for MW and WM papers, making citation age relevant not only for WW papers but for the wider composition of long-term citation flows.

6.3.3 All-to-CS versus CS-to-CS Contextual Results

Restricting the citing population to Computer Science produces broadly similar topical and temporal patterns. In the topical analysis, the same-topic, same-subfield, and cross-topic same-subfield subsets yield identical O/E estimates in the All→CS and CS→CS slices. Under the taxonomy metadata used in the analysis, these citation subsets therefore contain the same edges in both citation slices.

Differences between the two citing populations emerge primarily for cross-subfield citations. The All→CS cross-subfield subset contains 3,658,388 edges, compared with 1,995,651 edges under the CS→CS restriction. The MM ratio is nearly identical at 1.054 in both slices. For MW papers, the ratio increases from 0.885 to 0.913, whereas the WM ratio changes from 0.828 to 0.819. The largest difference occurs for WW papers, whose cross-subfield ratio declines from 0.693 in All→CS to 0.643 in CS→CS.

The taxonomy-weighted results are substantially more similar across citation slices. The O/E ratios are 1.037 versus 1.035 for MM, 0.958 versus 0.965 for MW, 0.893 versus 0.899 for WM, and 0.707 versus 0.713 for WW papers in All→CS and CS→CS, respectively.

The temporal analysis shows the same broad stability. The CS→CS slice contains 7,023,500 valid-lag citation edges. Of these, 3,147,998 (44.82%) belong to the short-memory group, 1,904,243 (27.11%) to the middle group, and 1,971,259 (28.07%) to the long-memory group.

Figure 6.3 presents the corresponding CS→CS trajectories. As in the primary slice, MM papers remain above expectation throughout, whereas WW papers have the lowest O/E ratio in each broad temporal group.

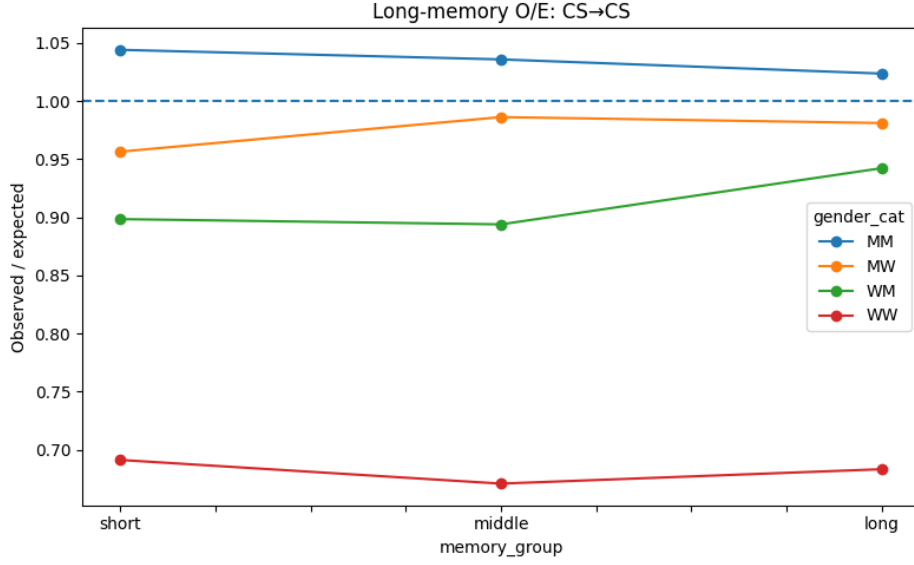


Figure 6.3: Observed-to-expected citation ratios across short (0-5 years), middle (6-10 years), and long (> 10 years) citation-memory groups in the CS→CS slice. The dashed horizontal line denotes parity between observed and expected citations ($O/E = 1$).

Table 6.8 directly compares the broad temporal estimates across citation slices.

Table 6.8: Observed-to-expected citation ratios across broad citation-memory groups in the All→CS and CS→CS citation slices.

Citation slice	Memory group	Edges	MM	MW	WM	WW
All→CS	Short (0-5)	3,790,516	1.045	0.947	0.893	0.717
	Middle (6-10)	2,338,119	1.040	0.959	0.877	0.681
	Long (> 10)	2,557,602	1.030	0.935	0.922	0.668
CS→CS	Short (0-5)	3,147,998	1.044	0.956	0.898	0.691
	Middle (6-10)	1,904,243	1.036	0.986	0.894	0.671
	Long (> 10)	1,971,259	1.024	0.981	0.942	0.683

For WW papers, the CS→CS ratio is slightly lower than the All→CS ratio in the short window (0.691 versus 0.717) and the middle window (0.671 versus 0.681). In the broad long-memory group, the ordering reverses slightly, with a CS→CS ratio of 0.683 compared with 0.668 in the primary slice. These differences are small relative to the distance of both sets of estimates from parity.

The fine-grained lag results provide a more detailed comparison. For WW papers, the CS→CS O/E ratios are 0.708, 0.685, 0.671, 0.693, and 0.646 across the 0-2, 3-5, 6-10, 11-20, and 20+ year buckets. The corresponding All→CS values are 0.738, 0.709, 0.681, 0.687, and 0.611. The within-CS WW ratios are therefore lower in the first three temporal intervals but somewhat higher in the two oldest intervals. In both slices, however, WW papers remain below expectation in every lag bucket.

For MW papers, restricting the citing population to Computer Science generally moves the temporal ratios closer to parity. This difference is especially visible among citations more than 20 years after publication, where the ratio increases from 0.771 in All→CS to 0.842 in CS→CS. The WM results are more similar across slices; in the 20+ bucket, for example, the ratios are 0.833 and 0.832. MM estimates remain highly stable across both citing populations and throughout the citation-age distribution.

Taken together, the contextual analyses show that the principal ordering of the citation results is stable across topical and temporal specifications and across the two citing populations. Topical restriction generally produces smaller differences within the most local citation relationships and larger differences across topic or subfield boundaries. The temporal analysis shows that the WW category remains below expectation from the earliest 0-2 year citation window through citations occurring more than two decades after publication. Restricting the citing population to Computer Science changes the magnitude of individual estimates but does not alter this broad empirical pattern.

6.4 Network-Based Baseline Results

The preceding baselines defined expected citation patterns using global, temporal, topical, and venue-based characteristics of the cited-paper population. The network-based baselines instead derive paper-level expectations from structurally related papers in the citation network. This section reports results for the bibliographic-coupling neighborhood baseline (N1) and the co-citation neighborhood baseline (N2), followed by a comparison of their coverage and observed-to-expected patterns.

Because both approaches require each focal paper to have at least one valid known-gender CS neighbor in the corresponding network, the network-based analyses cover subsets of the full known-gender CS population. The size and composition of these covered subsets are therefore reported explicitly alongside the O/E results.

6.4.1 Bibliographic-Coupling Neighborhood Results

The bibliographic-coupling network used by N1 contained 3,864,772 retained weighted edges among CS papers. After requiring valid known-gender neighbors, N1 produced expectations for 347,023 of the 1,081,363 known-gender CS papers, corresponding to an overall paper coverage of 32.09%. The same network-covered paper set was used for the All→CS and CS→CS citation outcomes.

Figure 6.4 shows the resulting O/E ratios for both citation slices. The relative ordering of the gender categories was highly similar across the two analyses. MM papers were observed above their neighborhood-based expectation, whereas MW, WM, and WW papers remained below expectation.

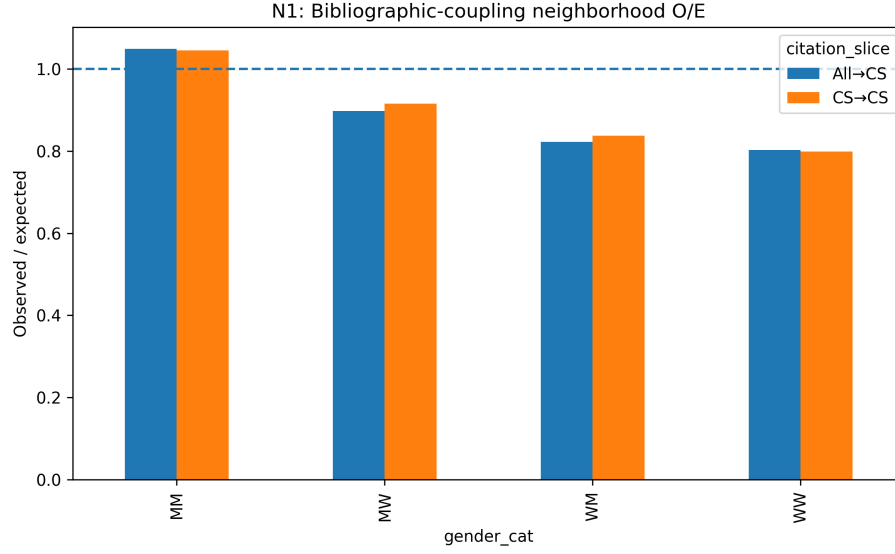


Figure 6.4: Observed-to-expected citation ratios under the N1 bibliographic-coupling neighborhood baseline for the All→CS and CS→CS citation slices. The dashed horizontal line denotes parity between observed and expected citations ($O/E = 1$).

In the primary All→CS slice, the 347,023 covered papers received 4,210,667 citations. MM papers received 3,349,437 citations compared with 3,193,405 expected under N1, producing an O/E ratio of 1.049. This corresponds to 4.89% more citations than expected from the weighted citation performance of their bibliographic-coupling neighbors.

The remaining three categories were below their N1 expectations. MW papers had an O/E ratio of 0.898, corresponding to 10.26% fewer citations than expected. The difference was larger for WM papers, with an O/E ratio of 0.822 and a deficit of 17.81%. WW papers had the lowest N1 ratio at 0.803, corresponding to 19.74% fewer observed citations than expected.

The CS→CS restriction produced a closely related pattern. The O/E ratio was 1.045 for MM, 0.915 for MW, 0.838 for WM, and 0.799 for WW papers. The largest absolute change relative to the All→CS result was therefore less than 0.02 for any gender category. In particular, the WW result remained approximately 0.80 in both slices.

Thus, under the local bibliographic-coupling neighborhood baseline, several of the deviations were smaller than under the broader direct-citation baselines, while not all categories were close to parity. Within the N1-covered population, MM papers remained above expectation and WW papers remained furthest below expectation in both citation slices.

6.4.2 Co-Citation Neighborhood Results

The co-citation network used by N2 was substantially smaller than the bibliographic-coupling network, containing 232,265 retained weighted edges. After restricting neighborhoods to valid known-gender CS papers, N2 covered 70,803 papers, corresponding to 6.55% of the known-gender CS population.

Despite this narrower coverage, the resulting O/E ratios were markedly closer to parity than under N1. Figure 6.5 shows the N2 estimates for the All→CS and CS→CS citation slices.

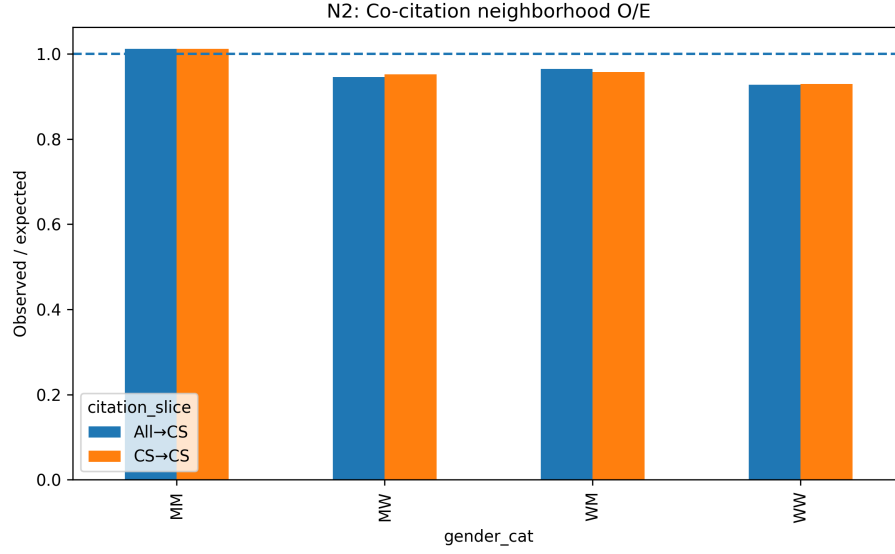


Figure 6.5: Observed-to-expected citation ratios under the N2 co-citation neighborhood baseline for the All→CS and CS→CS citation slices. The dashed horizontal line denotes parity between observed and expected citations ($O/E = 1$).

In the All→CS analysis, the N2-covered papers received 3,999,830 citations. MM papers received 3,261,757 citations compared with 3,224,040 expected, producing an O/E ratio of 1.012. The positive difference from expectation was therefore only 1.17%.

MW papers remained below expectation, with an O/E ratio of 0.945, corresponding to 5.46% fewer citations than expected. WM papers were closer still to parity, with an O/E ratio of 0.964 and a 3.57% deficit. WW papers again had the lowest ratio, although the difference was considerably smaller than under N1: their N2 O/E ratio was 0.927, corresponding to 7.27% fewer citations than expected.

The CS→CS restriction again produced only small changes. The O/E ratios were 1.012 for MM, 0.952 for MW, 0.958 for WM, and 0.929 for WW papers. The largest difference between the two citation slices was below 0.01. The ordering also remained unchanged: MM papers were slightly above expectation, while WW papers had the lowest ratio.

The co-citation-based expectations therefore produced the strongest convergence towards parity among the two network-neighborhood specifications. In particular, the WM ratio was close to one in both citation slices, while the WW deviation corresponded to approximately 7% fewer citations than expected. At the same time, these results apply to a substantially smaller subset of the known-gender CS population than the N1 estimates, making network coverage an important part of the comparison between the two baselines.

6.4.3 Network Coverage and Comparison between N1 and N2

The two network baselines differed substantially in the number of papers for which a valid neighborhood expectation could be constructed. Table 6.9 summarizes their overall and gender-specific coverage.

Table 6.9: Network coverage under the N1 bibliographic-coupling and N2 co-citation baselines. Gender-specific coverage is calculated relative to the total number of known-gender CS papers in each category.

Gender category	N1: Bibliographic coupling		N2: Co-citation	
	Covered papers	Coverage	Covered papers	Coverage
MM	255,440	32.31%	55,896	7.07%
MW	31,764	32.61%	5,580	5.73%
WM	45,395	32.95%	6,870	4.99%
WW	14,424	25.95%	2,457	4.42%
Overall	347,023	32.09%	70,803	6.55%

N1 therefore covered approximately five times as many papers as N2. Its coverage was relatively similar for MM, MW, and WM papers, ranging from 32.31% to 32.95%, but was lower for WW papers at 25.95%. The N2 network was more selective overall and displayed a stronger coverage gradient across gender categories. Coverage declined from 7.07% for MM papers to 5.73% for MW, 4.99% for WM, and 4.42% for WW papers.

The difference in network coverage is also reflected in the number of retained network edges. The bibliographic-coupling network contained 3,864,772 edges, compared with 232,265 co-citation edges. N2 therefore provided a structurally more restricted neighborhood baseline and produced valid expectations for a much smaller set of papers.

Table 6.10 directly compares the O/E estimates from the two network baselines.

Table 6.10: Comparison of observed-to-expected citation ratios under the N1 and N2 network-neighborhood baselines.

Citation slice	Baseline	MM	MW	WM	WW
All→CS	N1: Bibliographic coupling	1.049	0.898	0.822	0.803
	N2: Co-citation	1.012	0.945	0.964	0.927
CS→CS	N1: Bibliographic coupling	1.045	0.915	0.838	0.799
	N2: Co-citation	1.012	0.952	0.958	0.929

The contrast between N1 and N2 is clearest for WM and WW papers. In the All→CS slice, the WM ratio increased from 0.822 under N1 to 0.964 under N2, a difference of 0.142 in the O/E ratio. The WW ratio increased from 0.803 to 0.927, a difference of 0.125. MW papers also moved closer to parity, from 0.898 to 0.945. Conversely, the MM ratio declined from 1.049 under N1 to 1.012 under N2, moving substantially closer to one.

The same pattern was visible in the CS→CS slice. Relative to N1, the N2 ratio increased by approximately 0.038 for MW, 0.120 for WM, and 0.130 for WW papers, while the MM estimate declined from 1.045 to 1.012. N2 therefore produced estimates closer to parity for all four gender categories in both citation slices.

The distinction is particularly notable for the WW category. Under N1, WW papers received approximately 20% fewer citations than expected from their bibliographic-coupling neighborhoods in both citation slices. Under N2, the corresponding deficit was approximately 7%. The network representation used to define the local comparison set therefore materially changes the magnitude of the residual O/E difference, even though the direction and category ordering remain stable.

The comparison with the preceding baselines provides additional context. For the primary All→CS slice, the WW O/E ratio was 0.649 under the global D0 baseline, 0.695 under D1, 0.712 under D2, and 0.762 under D3. The N1 estimate increased further to 0.803, while the N2 estimate reached

0.927. Similarly, the WM ratio increased from 0.734 under D0 to 0.907 under D3 and to 0.964 under N2. The increasingly local comparison structures therefore yielded progressively smaller deviations for several categories, although the sequence should not be interpreted as a nested set of models because the baselines define different comparison sets and conditioning structures.

For MM papers, all reported network estimates remained above one, while MW, WM, and WW remained below one. The key difference between N1 and N2 was therefore not a reversal of the observed ordering, but a substantial change in magnitude. Co-citation neighborhoods produced O/E estimates much closer to parity than bibliographic-coupling neighborhoods, particularly for WM and WW papers.

These findings should be interpreted together with the substantial coverage difference between the two networks. N1 provides expectations for roughly one third of known-gender CS papers, whereas N2 covers only 6.55%. The near-parity N2 estimates therefore characterize a considerably smaller and more selectively connected subset of the corpus. Within their respective covered populations, however, both network baselines reproduce the same broad ordering across gender categories and show very similar results when the citing population is restricted from All→CS to CS→CS.

6.5 GLM-Based Expected Citation Results

The GLM framework provides a complementary paper-level approach to expected citation estimation by considering several structural and network-derived paper characteristics jointly. Five specifications are evaluated using a common structural reference model. M1 contains publication-decade, topical-stratum, venue-rank, and reference-list-size covariates. M2 augments M1 with outgoing direct-citation out-degree, M3 augments M1 with bibliographic-coupling degree and edge-weight summaries, M4 augments M1 with the corresponding co-citation features, and M5 combines the structural, direct-citation, bibliographic-coupling, and co-citation feature blocks. The network-derived count and weight features are transformed using $\log(1+x)$ before standardization, as defined in Section 4.8.

All five specifications use the same 1,081,363 known-gender Computer Science papers. Gender category is excluded from the predictor set and is used only after model fitting to aggregate observed and expected citation counts into the MM, MW, WM, and WW categories. Separate models are fitted for the All→CS and CS→CS citation outcomes.

6.5.1 Model Performance and Calibration

The All→CS outcome contains 8,686,237 observed citations, while the CS→CS outcome contains 7,023,500. For the full-data models, predicted citation counts are multiplicatively calibrated within each model and citation slice so that the total expected citation count equals the corresponding observed total. This calibration fixes the overall citation mass but does not guarantee accurate paper-level predictions or uniform calibration across different parts of the citation distribution.

Model fit is evaluated primarily using mean Poisson deviance and Poisson D^2 . The former is appropriate for the non-negative count outcome, whereas D^2 expresses the reduction in Poisson deviance relative to a constant-mean null model. Mean absolute error (MAE) is additionally reported in citation-count units, together with median absolute error as a more robust measure of the error for a typical paper. RMSE was also calculated as a secondary diagnostic but is not used as a primary comparison because of its high sensitivity to the extreme right tail of the citation distribution.

Table 6.11 reports the full-data in-sample diagnostics.

Table 6.11: Full-data in-sample fit diagnostics for the five GLM specifications. Lower values indicate better fit for Poisson deviance, MAE, and median absolute error, whereas higher Poisson D^2 indicates a larger reduction in deviance relative to the constant-mean null model.

Citation slice	Model	Poisson deviance	D^2	MAE	Median AE
All→CS	M1	26.454	0.174	9.748	5.138
	M2	26.459	0.174	9.747	5.120
	M3	26.169	0.183	9.706	4.904
	M4	13.766	0.570	6.538	3.157
	M5	13.757	0.570	6.539	3.145
CS→CS	M1	21.634	0.169	8.039	4.341
	M2	21.588	0.171	8.023	4.293
	M3	21.345	0.180	7.990	4.079
	M4	10.975	0.578	5.338	2.599
	M5	10.940	0.580	5.331	2.571

The structural M1 model produces Poisson D^2 values of 0.174 for All→CS and 0.169 for CS→CS, corresponding to a reduction of approximately 17% in Poisson deviance relative to the constant-mean benchmark.

Adding the outgoing direct-citation out-degree in M2 produces essentially no change. In the All→CS model, D^2 changes from 0.1738 to 0.1737 and mean Poisson deviance changes from 26.454 to 26.459. In the CS→CS model, the corresponding D^2 increases only slightly from 0.1687 to 0.1705. The differences in MAE and median absolute error are similarly small.

Bibliographic-coupling features in M3 provide a modest improvement. Poisson D^2 increases to 0.183 for All→CS and 0.180 for CS→CS. Relative to M1, mean Poisson deviance decreases by approximately 1.1% and 1.3%, respectively, while median absolute error also decreases in both citation slices.

The largest change occurs when co-citation features are introduced in M4. Poisson D^2 increases to 0.570 for All→CS and 0.578 for CS→CS. Relative to M1, mean Poisson deviance decreases by 47.96% and 49.27%, respectively. MAE decreases by approximately one third in both slices, while median absolute error decreases from 5.138 to 3.157 citations in All→CS and from 4.341 to 2.599 in CS→CS.

M5, which combines all available network feature blocks, performs almost identically to M4. In All→CS, D^2 changes only from 0.5701 under M4 to 0.5704 under M5. In CS→CS, the increase is from 0.5783 to 0.5797. The remaining error measures also differ only slightly between the two specifications.

Figure 6.6 highlights the separation between M1-M3 and the co-citation-based M4-M5 specifications.

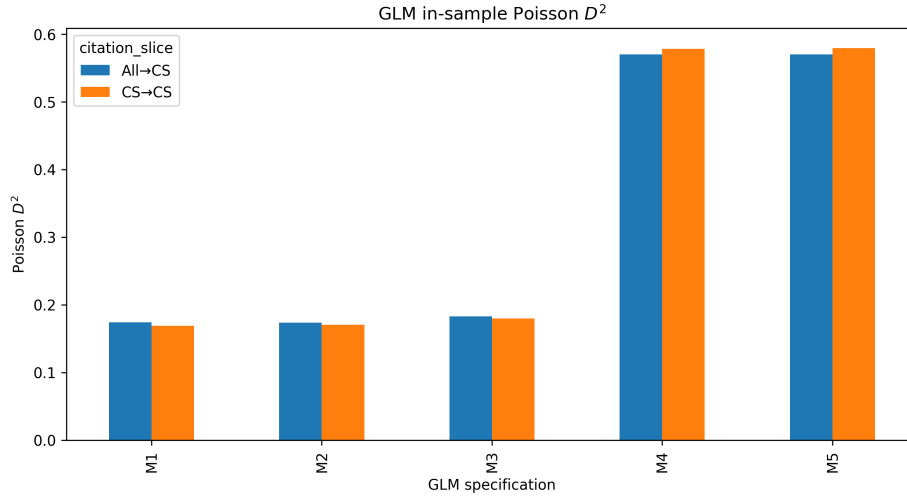


Figure 6.6: Full-data in-sample Poisson D^2 across GLM specifications M1-M5 for the All→CS and CS→CS citation outcomes. Higher values indicate a larger reduction in Poisson deviance relative to the constant-mean null model.

Because these full-data diagnostics are calculated on the same papers used for model estimation, an additional 80/20 paper-level holdout validation was performed. The same split was used for all five specifications and both citation outcomes. The training set contained 865,090 papers and the held-out test set contained 216,273 papers. Preprocessing and GLM coefficient estimation were performed using only the training papers.

Importantly, the multiplicative calibration factor was also estimated using only the training set and was subsequently applied unchanged to the test predictions. Held-out citation counts were therefore not used to recalibrate the models.

Table 6.12 reports the test-set diagnostics.

Table 6.12: Out-of-sample fit diagnostics on the held-out 20% of papers. The models and preprocessing transformations were estimated using only the training subset.

Citation slice	Model	Poisson deviance	D^2	MAE	Median AE
All→CS	M1	24.578	0.177	9.587	5.158
	M2	24.593	0.177	9.584	5.136
	M3	24.371	0.184	9.543	4.906
	M4	13.602	0.545	6.429	3.146
	M5	13.598	0.545	6.433	3.142
CS→CS	M1	20.276	0.173	7.920	4.364
	M2	20.246	0.174	7.904	4.300
	M3	20.045	0.182	7.871	4.089
	M4	10.735	0.562	5.230	2.597
	M5	10.703	0.563	5.227	2.568

The held-out results reproduce the same model ordering observed in the full-data analysis. For All→CS, test-set Poisson D^2 is 0.177 for M1, 0.177 for M2, and 0.184 for M3, compared with 0.545 for both M4 and M5. For CS→CS, the corresponding values are 0.173, 0.174, 0.182, 0.562, and 0.563.

The co-citation specifications therefore retain substantially better predictive fit on papers that were not used to estimate the regression coefficients. The test-set D^2 values for M4 and M5 are

moderately below their training values. In the All→CS validation, M4 decreases from 0.5758 on the training subset to 0.5446 on the test subset, while M5 decreases from 0.5761 to 0.5447. For CS→CS, the corresponding declines are from 0.5820 to 0.5621 for M4 and from 0.5834 to 0.5633 for M5. Despite these train-to-test differences, the separation from M1-M3 remains substantial.

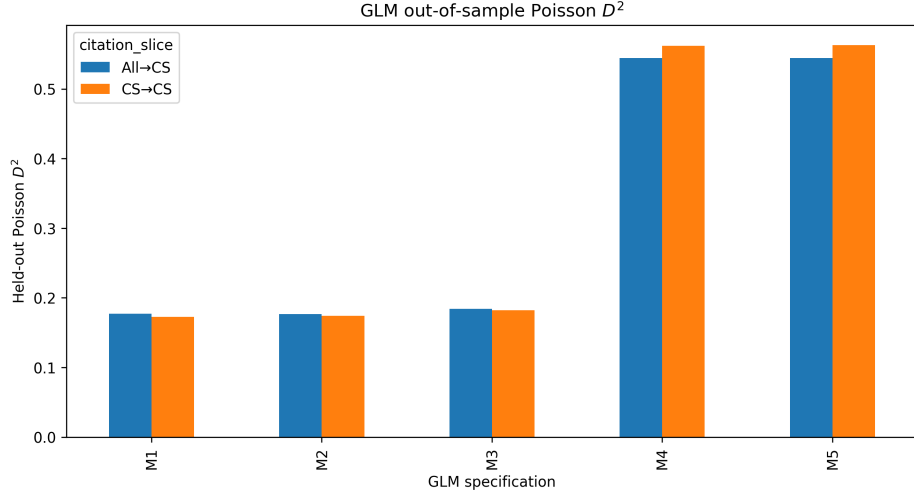


Figure 6.7: Held-out Poisson D^2 across GLM specifications M1-M5 using a common 80/20 paper-level train/test split.

The training-derived calibration also transfers more closely to the held-out data under M4 and M5. In the All→CS test set, total observed-to-expected ratios are 0.972 for M1, 0.972 for M2, 0.973 for M3, 0.988 for M4, and 0.988 for M5. For CS→CS, the corresponding values are 0.975, 0.975, 0.977, 0.990, and 0.991. Thus, without any recalibration using held-out outcomes, the aggregate expected citation totals under M4 and M5 are within approximately 1-1.2% of the observed test-set totals.

Calibration nevertheless varies across the predicted-citation distribution. Figure 6.8 shows held-out O/E ratios after test papers are divided into model-specific deciles according to their expected citation counts.

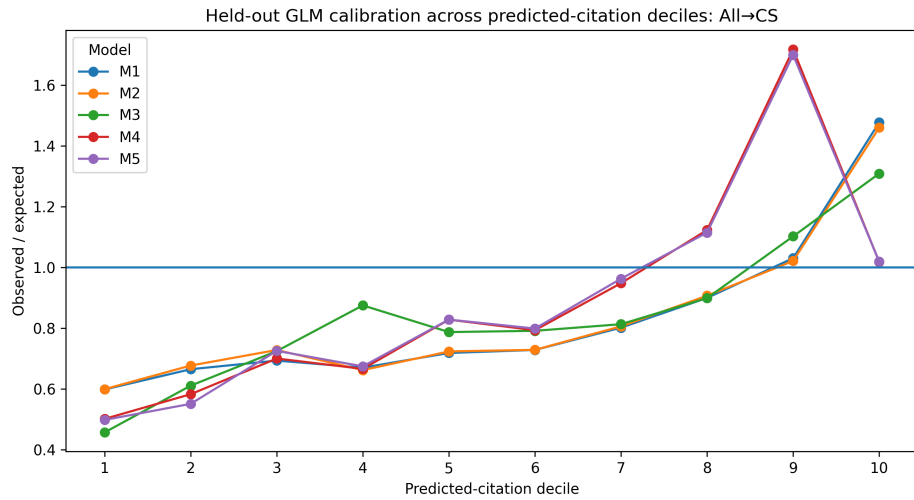


Figure 6.8: Held-out calibration across model-specific predicted-citation deciles for the All→CS outcome. The training-derived calibration factor is applied unchanged to the test predictions. The horizontal line denotes $O/E = 1$.

For all specifications, the lower predicted-citation deciles generally have O/E ratios below one, while the ratios increase towards and above parity in the upper part of the predicted distribution. M4 and M5 produce a distinctive pattern in the highest deciles: the ninth decile lies substantially above expectation, while the tenth decile returns close to parity. The holdout results therefore indicate that the models are not uniformly calibrated across the citation distribution despite their close aggregate calibration.

The holdout analysis evaluates generalization of the fitted GLM mapping to papers excluded from coefficient estimation. It does not turn the co-citation specifications into prospective models: the network features are derived from the observed citation network, and the co-citation features in M4 and M5 remain future-informed with respect to the focal paper.

6.5.2 Observed/Expected Results across GLM Specifications

The gender-aggregated O/E results show a model sequence that closely parallels the changes in paper-level fit. Figure 6.9 presents the five full-data specifications for the primary All→CS citation slice.

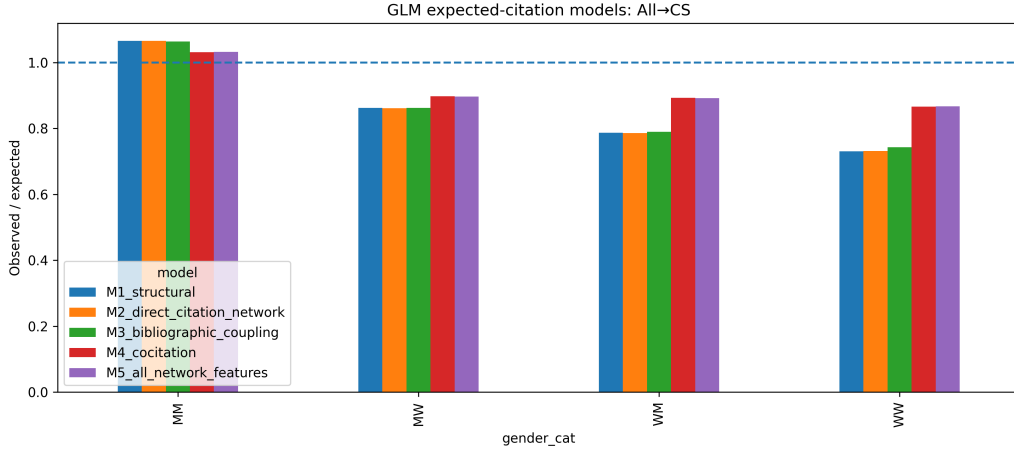


Figure 6.9: Observed-to-expected citation ratios across GLM specifications M1-M5 for the All→CS citation slice. The dashed horizontal line denotes parity between observed and expected citations ($O/E = 1$).

Under M1, the O/E ratio is 1.066 for MM papers, 0.862 for MW papers, 0.787 for WM papers, and 0.731 for WW papers. These values correspond to 6.58% more citations than expected for MM papers and deficits of 13.77%, 21.29%, and 26.90% for MW, WM, and WW papers, respectively.

Adding the outgoing direct-citation out-degree in M2 produces almost no change. The ratios are 1.066 for MM, 0.861 for MW, 0.786 for WM, and 0.732 for WW papers. The maximum absolute change from M1 is approximately 0.001.

The bibliographic-coupling model M3 also produces comparatively modest changes. Its O/E ratios are 1.064 for MM, 0.863 for MW, 0.790 for WM, and 0.743 for WW papers. The largest movement relative to M1 occurs for WW papers, whose ratio increases by approximately 0.012.

The major shift occurs under M4. The MM ratio decreases to 1.032, while the MW and WM ratios increase to 0.898 and 0.893, respectively. The WW ratio increases from 0.731 under M1 to 0.866 under M4. Its deficit relative to expectation therefore decreases from 26.90% to 13.38%.

M5 produces an almost identical aggregate pattern. The All→CS ratios are 1.032 for MM, 0.897 for MW, 0.892 for WM, and 0.867 for WW papers. Thus, WW papers remain 13.30% below the

M5 expectation, while MM papers remain 3.21% above expectation.

Table 6.13: Observed-to-expected citation ratios across the five full-data GLM specifications.

Citation slice	Model	MM	MW	WM	WW
All→CS	M1	1.0658	0.8623	0.7871	0.7310
	M2	1.0661	0.8613	0.7863	0.7316
	M3	1.0642	0.8627	0.7896	0.7431
	M4	1.0320	0.8982	0.8928	0.8662
	M5	1.0321	0.8973	0.8923	0.8670
CS→CS	M1	1.0664	0.8734	0.7859	0.7148
	M2	1.0669	0.8714	0.7843	0.7163
	M3	1.0644	0.8739	0.7890	0.7290
	M4	1.0306	0.9120	0.8970	0.8537
	M5	1.0311	0.9092	0.8957	0.8555

The CS→CS results reproduce the same sequence. Figure 6.10 shows the corresponding full-data O/E ratios.



Figure 6.10: Observed-to-expected citation ratios across GLM specifications M1-M5 for the CS→CS citation slice. The dashed horizontal line denotes parity between observed and expected citations ($O/E = 1$).

Under M1, the CS→CS ratios are 1.066 for MM, 0.873 for MW, 0.786 for WM, and 0.715 for WW papers. M2 again produces negligible changes, while M3 increases the WW ratio moderately to 0.729.

Under M4, the ratios move to 1.031 for MM, 0.912 for MW, 0.897 for WM, and 0.854 for WW papers. M5 produces corresponding values of 1.031, 0.909, 0.896, and 0.856. The WW deficit therefore decreases from 28.52% under M1 to 14.45% under M5.

The held-out gender-specific analysis reproduces the same broad pattern. MM papers remain above expectation across all five specifications, while MW, WM, and WW papers remain below expectation. As in the full-data analysis, the largest movement towards parity occurs under M4 and M5, particularly for WM and WW papers. The gender-aggregated pattern is therefore also present among papers excluded from GLM coefficient estimation.

Across both citation slices and both the full-data and holdout analyses, the direction of the gender-specific differences remains stable. Model specification primarily changes the magnitude of the

O/E differences rather than their ordering.

6.5.3 Changes in Expected Citation Estimates across Network Feature Specifications

The representation-specific specifications allow each network feature block to be compared with the common structural M1 model. M2 adds the outgoing direct-citation out-degree defined in Section 4.8; M3 adds bibliographic-coupling degree, weighted strength, mean edge weight, and maximum edge weight; M4 adds the analogous co-citation quantities; and M5 combines all three network feature blocks. The paper-level fit diagnostics and gender-aggregated O/E estimates therefore show how the fitted expected-citation mapping changes as each form of network information is introduced.

The outgoing direct-citation feature in M2 produces almost no change in either dimension relative to M1. In the full-data All→CS analysis, Poisson D^2 changes by only approximately -0.0001 relative to M1, while the WW O/E ratio changes by approximately $+0.0006$. In CS→CS, the corresponding changes are approximately $+0.0018$ in D^2 and $+0.0014$ in WW O/E.

Bibliographic-coupling features in M3 produce a small but consistent improvement. Relative to M1, full-data Poisson D^2 increases by 0.0089 for All→CS and 0.0111 for CS→CS. The WW O/E ratio increases by 0.0121 and 0.0142, respectively. The held-out D^2 values similarly increase from 0.177 to 0.184 in All→CS and from 0.173 to 0.182 in CS→CS.

The introduction of co-citation information produces a substantially larger shift. Relative to M1, M4 increases full-data Poisson D^2 by 0.3963 in All→CS and by 0.4096 in CS→CS. At the same time, the All→CS O/E ratio changes by -0.0338 for MM, $+0.0359$ for MW, $+0.1057$ for WM, and $+0.1352$ for WW papers. Each category therefore moves closer to one.

The corresponding CS→CS changes are -0.0358 for MM, $+0.0386$ for MW, $+0.1111$ for WM, and $+0.1389$ for WW papers. The largest shifts therefore occur for WM and WW papers, which also have the lowest O/E ratios under the structural M1 specification.

Table 6.14: Change in observed-to-expected ratio relative to the structural M1 specification after adding representation-specific network feature blocks.

Citation slice	Comparison	MM	MW	WM	WW
All→CS	M2 - M1	+0.0002	-0.0010	-0.0007	+0.0006
	M3 - M1	-0.0016	+0.0004	+0.0025	+0.0121
	M4 - M1	-0.0338	+0.0359	+0.1057	+0.1352
	M5 - M1	-0.0337	+0.0350	+0.1053	+0.1360
CS→CS	M2 - M1	+0.0005	-0.0020	-0.0016	+0.0014
	M3 - M1	-0.0020	+0.0005	+0.0031	+0.0142
	M4 - M1	-0.0358	+0.0386	+0.1111	+0.1389
	M5 - M1	-0.0353	+0.0358	+0.1098	+0.1407

M4 and M5 are nearly indistinguishable in both the full-data and holdout results. For All→CS, their full-data D^2 values differ by only 0.0003 and their held-out values by 0.0001. Their category-specific O/E ratios differ by approximately 0.001 or less. For CS→CS, M5 improves full-data D^2 by 0.0014 and held-out D^2 by 0.0012 relative to M4, while the gender-specific O/E changes remain small.

Within the model sequence, the co-citation feature block is therefore associated with the dominant improvement in both paper-level fit and the movement of gender-aggregated expected citation estimates towards parity. Adding direct-citation and bibliographic-coupling features on top of the co-citation specification produces little further change.

This result requires a different interpretation from the structural and backward-looking network specifications. Co-citation features are constructed from subsequent citation activity and are therefore future-informed with respect to the focal paper. M4 and M5 consequently represent retrospective expected-citation specifications rather than prospective prediction models. The hold-out results demonstrate that the mapping from these network characteristics to citation counts generalizes to papers excluded from coefficient estimation, conditional on the available network representation; they do not establish co-citation structure as a causal explanation for the citation differences.

Even under the most strongly adjusted specifications, complete parity is not reached. Under M5, the full-data All→CS ratios are 1.032 for MM, 0.897 for MW, 0.892 for WM, and 0.867 for WW papers. In CS→CS, the corresponding ratios are 1.031, 0.909, 0.896, and 0.856. The gender-aggregated citation deviations are therefore substantially smaller under the network-augmented specifications, particularly those incorporating co-citation features, while the same directional pattern is preserved across the full-data and held-out analyses.

6.6 Randomized Robustness Results

The preceding analyses derive expected citation counts from observed paper characteristics or network neighborhoods. The randomized robustness analyses provide a complementary perspective by replacing observed citation targets with alternative eligible targets under explicit randomization rules. Two families of randomization are considered. The first redraws references from temporally eligible known-gender CS papers, either uniformly or in proportion to prior citation visibility. The second restricts replacement targets to local bibliographic-coupling or co-citation neighborhoods.

The resulting O/E ratios are interpreted within the corresponding randomized opportunity sets. Because the network-conditioned procedures can replace only citation edges whose cited targets have valid network neighbors, their coverage is reported separately in Section 6.6.3.

6.6.1 Randomized Reference Drawing Results

The randomized reference-drawing analysis evaluates four combinations of temporal eligibility and drawing probability. Eligible targets are either all known-gender CS papers published before the citing year or papers within the preceding 0-10 years. Within each temporal pool, references are drawn either uniformly or with probabilities proportional to the target paper's prior citation count plus one.

All 8,686,237 All→CS and 7,023,500 CS→CS known-gender citation edges enter each randomized drawing specification. Figure 6.11 presents the four All→CS variants.

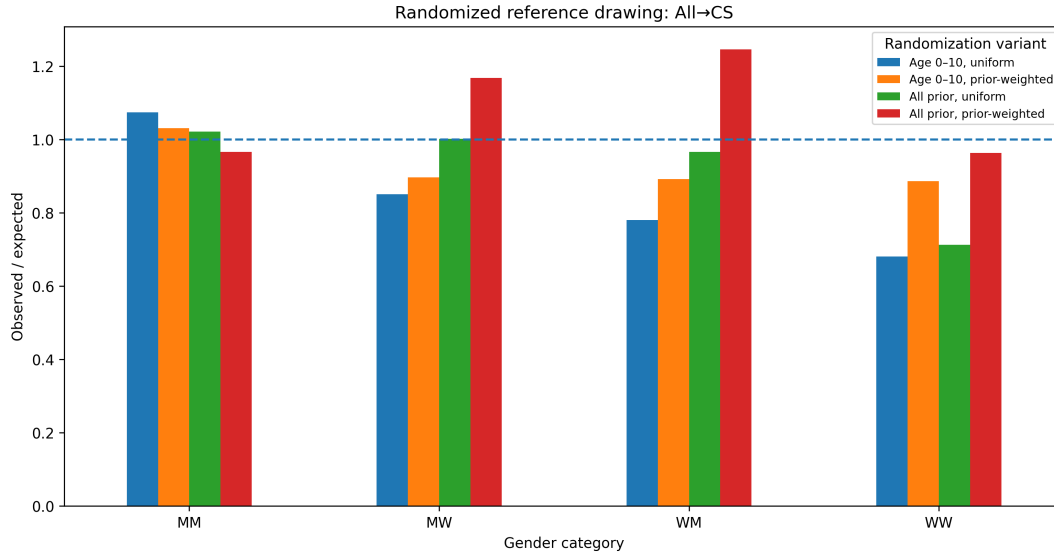


Figure 6.11: Observed-to-expected citation ratios under the four randomized reference-drawing specifications for the All→CS citation slice. The dashed horizontal line denotes parity between observed and randomized expected citations ($O/E = 1$).

Table 6.15 reports the exact O/E ratios for both citation slices.

Table 6.15: Observed-to-expected citation ratios under randomized reference drawing. Prior-weighted variants assign replacement probabilities proportional to prior citation count plus one.

Citation slice	Eligibility	Drawing scheme	MM	MW	WM	WW
All→CS	0-10 years	Uniform	1.074	0.850	0.780	0.681
	0-10 years	Prior-weighted	1.031	0.897	0.892	0.886
	All prior	Uniform	1.021	1.002	0.966	0.713
	All prior	Prior-weighted	0.966	1.168	1.246	0.963
CS→CS	0-10 years	Uniform	1.069	0.874	0.795	0.672
	0-10 years	Prior-weighted	1.027	0.921	0.907	0.872
	All prior	Uniform	1.017	1.030	0.984	0.702
	All prior	Prior-weighted	0.963	1.201	1.268	0.947

The uniform 0-10 year specification produces an ordering similar to the earlier direct-citation baselines. In All→CS, MM papers are 7.38% above the randomized expectation, whereas MW, WM, and WW papers are 14.97%, 21.98%, and 31.89% below expectation, respectively. The corresponding CS→CS ratios are very similar, with WW papers at 0.672.

Under prior-citation weighting with the same 0–10 year eligibility window, these deviations are substantially smaller. In All→CS, the MM ratio decreases from 1.074 to 1.031, while the MW, WM, and WW ratios increase from 0.850 to 0.897, from 0.780 to 0.892, and from 0.681 to 0.886, respectively. The same pattern occurs in CS→CS, where the WW ratio increases from 0.672 under uniform drawing to 0.872 under prior-weighted drawing.

The change associated with prior-citation weighting is even more pronounced when the candidate pool includes all previously published papers. Under uniform all-prior drawing, the All→CS ratios are 1.021 for MM, approximately 1.002 for MW, 0.966 for WM, and 0.713 for WW. When prior-citation weighting is introduced, the MM ratio decreases below one to 0.966, while MW and WM move above expectation to 1.168 and 1.246. The WW ratio also increases strongly, from 0.713 to 0.963, bringing it close to parity.

The corresponding CS→CS specification produces the same qualitative pattern. Under all-prior, prior-weighted drawing, the ratios are 0.963 for MM, 1.201 for MW, 1.268 for WM, and 0.947 for WW papers. Thus, this specification is the only randomized reference-drawing variant in which MM falls below expectation while MW and WM move substantially above it.

The randomized reference-drawing results are therefore sensitive to both the temporal eligibility rule and the treatment of prior citation visibility. Uniform drawing retains comparatively large below-expectation ratios for WW papers, while prior-citation weighting moves all categories closer to, or in some cases across, parity. In particular, the difference between the 0-10 year uniform and all-prior prior-weighted variants is large for every gender category. The randomized null expectation therefore depends materially on whether previously accumulated citation visibility is incorporated into the drawing process.

6.6.2 Network-Conditioned Edge Randomization Results

The network-conditioned analysis replaces cited papers only with valid known-gender CS neighbors in either the bibliographic-coupling or co-citation network. Replacement candidates are additionally restricted to the same publication decade as the original cited paper, and replacement probabilities are proportional to the corresponding network edge weights.

Figure 6.12 shows the All→CS O/E ratios for the two network representations.

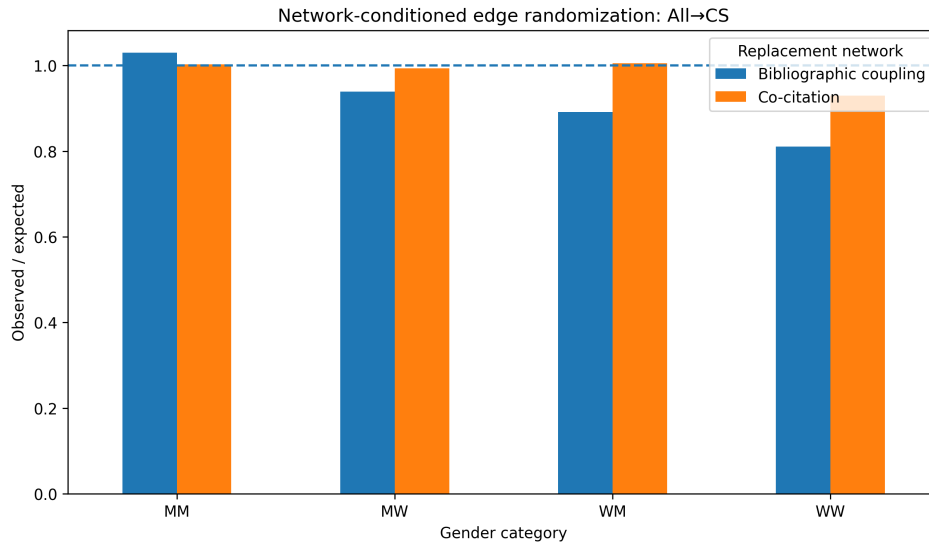


Figure 6.12: Observed-to-expected citation ratios under bibliographic-coupling and co-citation-conditioned edge randomization for the All→CS citation slice. Only replaceable citation edges are included. The dashed horizontal line denotes $O/E = 1$.

Under bibliographic-coupling-conditioned randomization, the All→CS O/E ratios are 1.030 for MM, 0.939 for MW, 0.891 for WM, and 0.811 for WW papers. The corresponding deviations are +3.04%, −6.11%, −10.86%, and −18.90%. Thus, several of the deviations are substantially smaller under the bibliographic-coupling-conditioned replacement set than under the broader randomized opportunity sets, while WW papers remain furthest below expectation.

The co-citation-conditioned results are considerably closer to parity. For All→CS, the O/E ratios are 1.003 for MM, 0.994 for MW, 1.006 for WM, and 0.930 for WW papers. MM, MW, and WM are therefore within approximately one percentage point of parity, whereas WW papers remain 7.05% below their randomized expectation.

Table 6.16: Observed-to-expected citation ratios under network-conditioned edge randomization.

Citation slice	Replacement network	MM	MW	WM	WW
All→CS	Bibliographic coupling	1.030	0.939	0.891	0.811
	Co-citation	1.003	0.994	1.006	0.930
CS→CS	Bibliographic coupling	1.028	0.951	0.899	0.821
	Co-citation	1.004	0.998	0.991	0.937

Restricting the citing population to Computer Science produces only small changes. Under bibliographic-coupling randomization, the CS→CS ratios are 1.028 for MM, 0.951 for MW, 0.899 for WM, and 0.821 for WW papers. Under co-citation randomization, the ratios are 1.004, 0.998, 0.991, and 0.937, respectively. The relative ordering therefore remains similar across the two citation slices, with the co-citation-conditioned specification consistently producing estimates closer to one.

An additional sensitivity analysis examined whether the bibliographic-coupling replacement probabilities changed when network similarity was multiplied by prior citation visibility. This specification retains the same replacement edges and same-decade restriction but weights candidate papers according to

$$w_{ij}^* = w_{ij}^{BC} (H(j, t_i) + 1),$$

where $H(j, t_i)$ denotes citations accumulated by candidate j before the citing year of citation edge i .

Table 6.17: Sensitivity of bibliographic-coupling edge randomization to prior-citation weighting.

Citation slice	BC replacement weighting	MM	MW	WM	WW
All→CS	Network weight only	1.030	0.939	0.891	0.811
	Network $\times$ prior citation	1.015	0.990	0.949	0.829
CS→CS	Network weight only	1.028	0.951	0.899	0.821
	Network $\times$ prior citation	1.013	0.992	0.955	0.853

Prior-citation weighting moves the bibliographic-coupling estimates closer to parity for all four categories. In All→CS, the MW ratio increases from 0.939 to 0.990 and the WM ratio from 0.891 to 0.949, while the MM ratio decreases from 1.030 to 1.015. WW also moves towards one, although more modestly, from 0.811 to 0.829.

The same pattern occurs in CS→CS. The WW ratio increases from 0.821 to 0.853, while MW reaches 0.992 and WM 0.955. The prior-citation-weighted bibliographic-coupling specification produces smaller residual differences than the unweighted variant, while the WW deviation remains larger than under the co-citation-conditioned specification.

Across the network-conditioned variants, co-citation randomization produces the smallest gender-category deviations from parity. This result is consistent with the earlier N2 and GLM analyses, in which co-citation information also generated substantially smaller O/E differences than bibliographic-coupling information. The comparison describes how the estimates change across network-conditioned opportunity sets and does not imply a causal role for co-citation structure.

6.6.3 Coverage of Replaceable Edges

Unlike the randomized reference-drawing analysis, network-conditioned randomization cannot be applied to every observed citation edge. A citation is replaceable only when its cited target has at

least one valid known-gender, same-decade neighbor in the corresponding network. Consequently, the network-randomization results refer to subsets of the citation graph whose size differs substantially between bibliographic coupling and co-citation.

Table 6.18 reports both edge coverage and target coverage. Edge coverage measures the fraction of known-gender citation edges whose cited targets can be replaced, whereas target coverage measures the fraction of distinct cited known-gender CS papers with at least one valid replacement set.

Table 6.18: Coverage of network-conditioned edge randomization.

Citation slice	Network	Replaceable edges	Edge coverage	Replaceable targets	Target coverage
All→CS	Bibliographic coupling	3,796,895	43.71%	218,903	30.92%
	Co-citation	3,272,004	37.67%	59,058	8.34%
CS→CS	Bibliographic coupling	3,187,042	45.38%	208,574	32.30%
	Co-citation	2,810,712	40.02%	59,010	9.14%

Bibliographic-coupling randomization has the broader target coverage. In the All→CS slice, 3,796,895 citation edges are replaceable, representing 43.71% of all known-gender citation edges. These edges refer to 218,903 distinct cited papers, corresponding to 30.92% target coverage. In CS→CS, edge coverage increases slightly to 45.38%, while target coverage reaches 32.30%.

Co-citation randomization has lower coverage, particularly at the target level. In All→CS, 3,272,004 edges are replaceable, giving 37.67% edge coverage, but they correspond to only 59,058 distinct cited targets, or 8.34% of all cited known-gender CS papers. In CS→CS, the corresponding values are 2,810,712 replaceable edges, 40.02% edge coverage, 59,010 replaceable targets, and 9.14% target coverage.

Figure 6.13 visualizes the contrast between edge and target coverage.

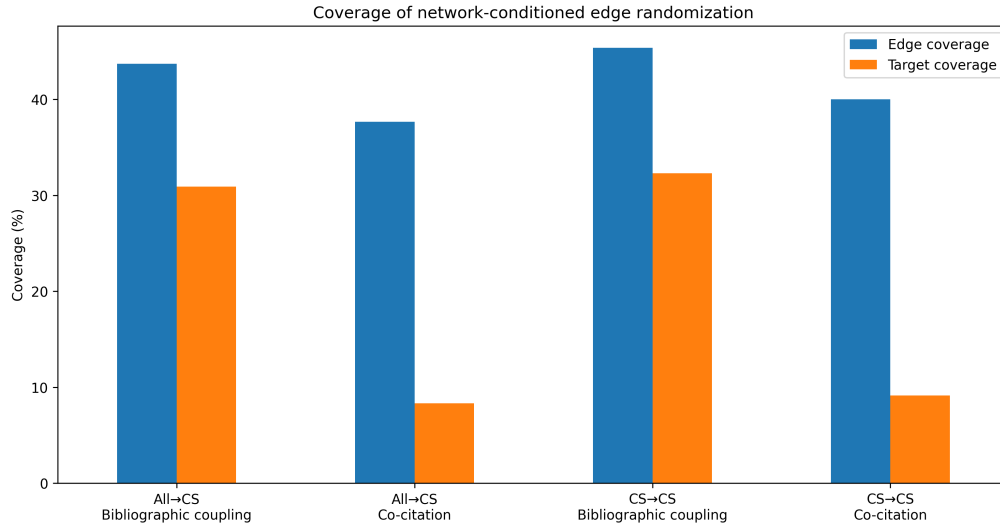


Figure 6.13: Edge and target coverage of bibliographic-coupling and co-citation-conditioned citation randomization. Coverage is reported separately for the All→CS and CS→CS citation slices.

The difference between edge and target coverage is particularly pronounced for co-citation. Although approximately 38-40% of citation edges are replaceable, fewer than 10% of distinct cited targets have a valid co-citation replacement set. This indicates that the covered targets receive disproportionately many citations relative to the broader cited paper population.

The near-parity co-citation randomization results must therefore be interpreted within this restricted coverage. They describe a relatively small subset of cited papers that are sufficiently embedded in the co-citation network to support same-decade replacement. Bibliographic coupling covers a substantially larger target population but produces larger residual O/E differences, particularly for WM and WW papers.

Overall, the randomized robustness analyses reproduce the main finding that the magnitude of gender-category citation differences depends strongly on how the expected citation opportunity set is constructed. Uniform temporal randomization retains relatively large WW deficits, whereas the deviations are substantially smaller under prior-citation-weighted drawing. They are smaller still under the network-conditioned specifications, with co-citation-based replacement producing ratios closest to parity. However, the latter results apply to a considerably more restricted subset of cited papers, making coverage an essential part of their interpretation.

6.7 Cross-Baseline Robustness

The preceding sections evaluate citation imbalance under several distinct definitions of expected citation opportunity. To assess the extent to which the resulting patterns depend on a particular specification, the main baseline results are brought together in a unified cross-baseline comparison.

The comparison contains 19 specifications drawn from six baseline families: four direct structural baselines (D0-D3), the taxonomy-weighted topical relevance baseline R1, two network-neighborhood baselines (N1-N2), five GLM specifications (M1-M5), four randomized reference-drawing variants, and three network-conditioned randomization variants. The latter consist of bibliographic-coupling randomization, co-citation randomization, and the additional bibliographic-coupling randomization with prior-citation weighting.

Detailed topical subcategories and citation-lag buckets are not included as separate rows in this unified comparison, because they represent within-family contextual subdivisions rather than additional general baseline specifications. The cross-baseline comparison is therefore intended as a compact summary of the principal expected-citation constructions examined in the chapter.

The purpose of this synthesis is not to identify a single preferred baseline or a single baseline-independent effect size. Instead, it evaluates whether the direction of the gender-specific deviations remains stable when expected citation opportunity is defined using substantively different comparison structures.

6.7.1 Unified Observed/Expected Comparison across Baselines

Figure 6.14 presents the complete All→CS comparison. The color scale is centered at $O/E = 1$, such that values below and above parity are visually separated, while the numerical value of each cell is shown directly.

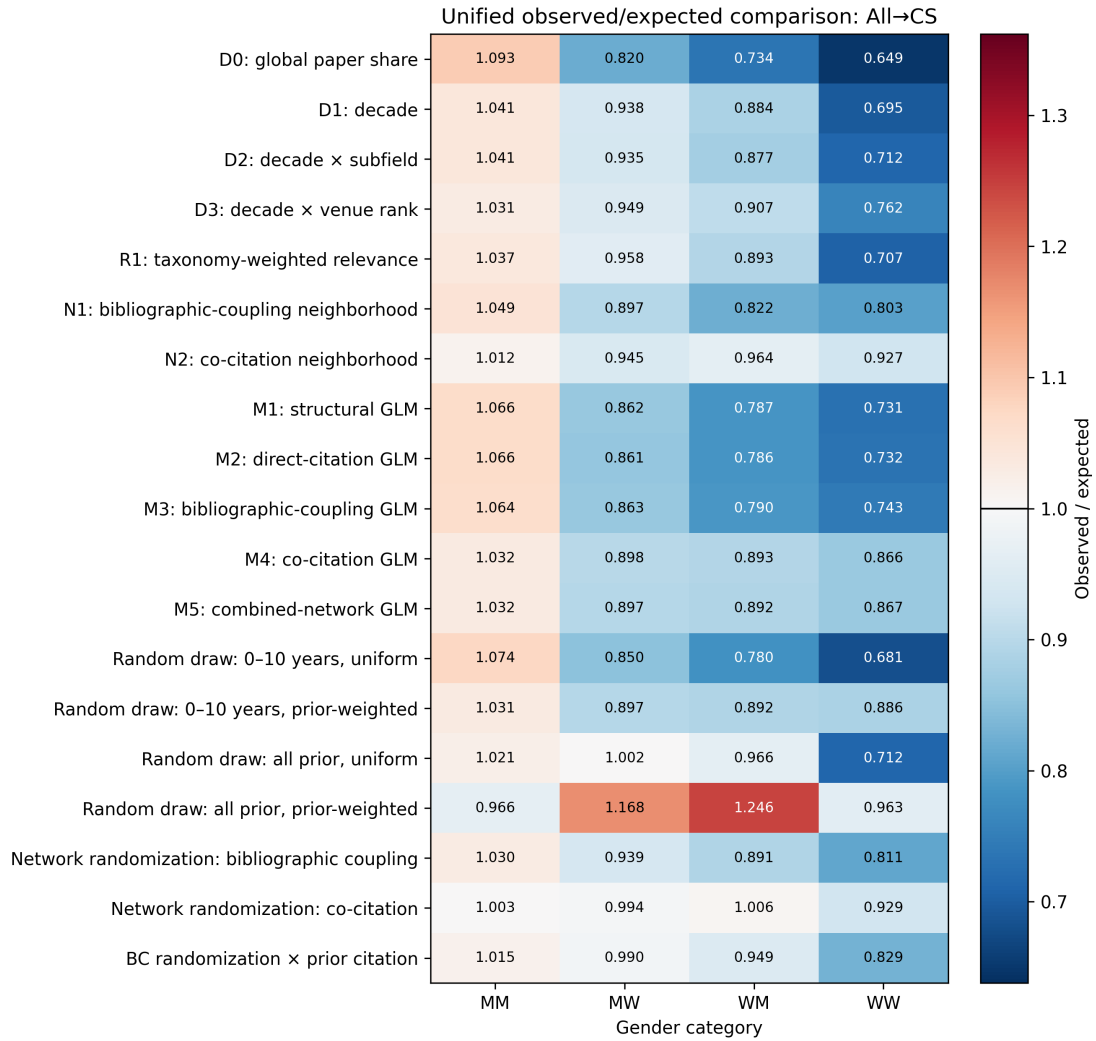


Figure 6.14: Unified observed-to-expected citation ratios across 19 baseline specifications for the All→CS citation slice. The diverging color scale is centered at parity ($O/E = 1$), and exact O/E ratios are shown within each cell.

Several broad patterns recur across the specifications. MM papers are above expectation under 18 of the 19 All→CS specifications. Their O/E ratios range from 0.966 to 1.093, with a median of 1.032. The only specification placing MM below expectation is the all-prior, prior-citation-weighted randomized reference draw.

MW and WM papers show the opposite dominant direction. Both categories are below expectation under 17 of the 19 All→CS specifications. For MW papers, O/E ranges from 0.820 under D0 to 1.168 under all-prior, prior-weighted randomization, with a median of 0.935. For WM papers, the range is wider, from 0.734 under D0 to 1.246 under the same randomized specification, with a median of 0.892.

WW papers show the strongest directional consistency. Their O/E ratio is below one under all 19 specifications. The values range from 0.649 under D0 to 0.963 under all-prior, prior-weighted randomized drawing, with a median of 0.762. No All→CS specification places the WW category within one percentage point of parity.

The specifications producing values closest to parity differ across the gender categories. Co-

citation-based specifications generally produce smaller deviations than the corresponding structural or bibliographic-coupling specifications. For example, the N2 neighborhood baseline produces All→CS ratios of 1.012, 0.945, 0.964, and 0.927 for MM, MW, WM, and WW, respectively. Network-conditioned co-citation randomization produces corresponding ratios of 1.003, 0.994, 1.006, and 0.929. Under the GLM framework, the co-citation-based M4 and combined M5 specifications similarly move all four categories closer to parity relative to M1-M3.

This movement towards parity should not be interpreted as evidence that co-citation provides a superior baseline. Co-citation has substantially lower coverage and is future-informed, so it defines a different and more selective comparison population.

The most pronounced departure from the otherwise dominant directional pattern occurs under the all-prior, prior-citation-weighted randomized reference draw. Under this specification, the All→CS O/E ratios are 0.966 for MM, 1.168 for MW, 1.246 for WM, and 0.963 for WW. Thus, MM crosses below one and MW and WM cross above one, whereas WW remains below expectation.

Restricting the citing population to Computer Science produces a highly similar cross-baseline pattern, as shown in Figure 6.15.

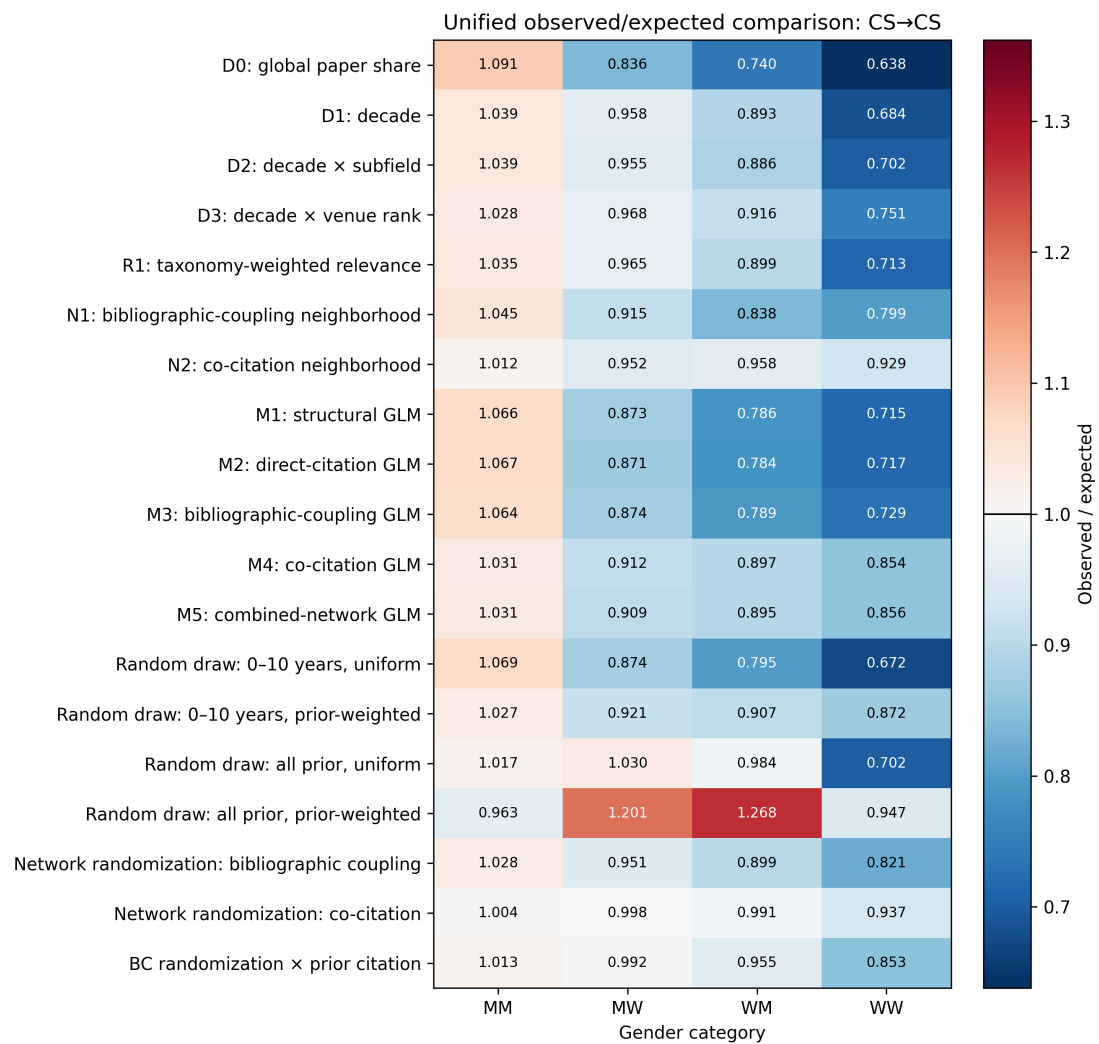


Figure 6.15: Unified observed-to-expected citation ratios across 19 baseline specifications for the CS→CS citation slice. The color scale is identical to Figure 6.14, allowing direct visual comparison between citation slices.

For CS→CS, MM is again above expectation under 18 of 19 specifications, MW is below expectation under 17 of 19, and WW is below expectation under all 19. WM is below expectation under 18 of the 19 CS→CS specifications, with only the all-prior, prior-citation-weighted randomized draw moving it above one.

The absolute differences between corresponding All→CS and CS→CS O/E ratios are generally small. Across the 19 specifications, the mean absolute difference between citation slices is 0.002 for MM, 0.015 for MW, 0.009 for WM, and 0.011 for WW. The maximum difference observed for any individual specification is 0.005 for MM, 0.033 for MW, 0.023 for WM, and 0.024 for WW. The cross-baseline ordering and dominant directions therefore remain largely unchanged when the citing population is restricted to CS.

6.7.2 Directional Stability and Magnitude Differences

Although the majority of specifications produce the same direction for each gender category, the magnitude of the O/E deviations varies substantially across baseline constructions. Table 6.19 summarizes the number of specifications below and above parity together with the minimum, median, and maximum O/E ratios.

The specification counts are descriptive summaries of robustness rather than independent statistical tests. Several specifications are derived from related data structures or share common conditioning variables, and the 19 estimates should therefore not be interpreted as 19 independent replications of the same hypothesis.

Table 6.19: Directional stability and magnitude of O/E ratios across the 19 principal baseline specifications. Minimum and maximum values describe the observed cross-baseline range rather than statistical confidence intervals.

Citation slice	Gender	$N(O/E < 1)$	$N(O/E > 1)$	Minimum	Median	Maximum
All→CS	MM	1	18	0.966	1.032	1.093
	MW	17	2	0.820	0.935	1.168
	WM	17	2	0.734	0.892	1.246
	WW	19	0	0.649	0.762	0.963
CS→CS	MM	1	18	0.963	1.031	1.091
	MW	17	2	0.836	0.951	1.201
	WM	18	1	0.740	0.897	1.268
	WW	19	0	0.638	0.751	0.947

The MM category has the narrowest cross-baseline range. Its O/E values span 0.966-1.093 in All→CS and 0.963-1.091 in CS→CS. The corresponding medians are 1.032 and 1.031. Consequently, changes in expected-citation construction affect the magnitude of the MM deviation but rarely change its direction.

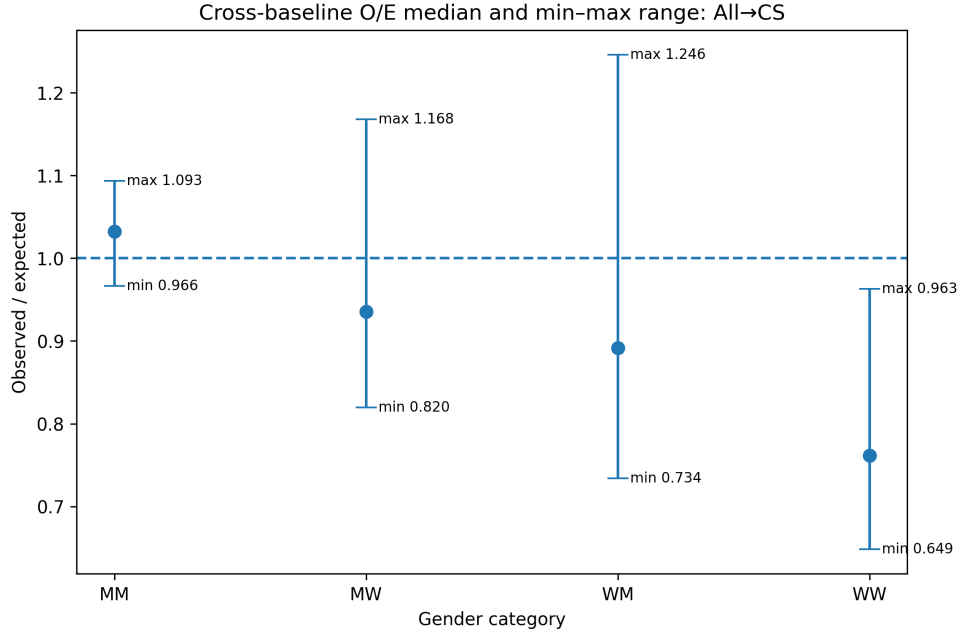
MW shows substantially greater specification sensitivity. Its All→CS range is 0.820-1.168 and its CS→CS range is 0.836-1.201. Nevertheless, the median remains below one in both citation slices, at 0.935 and 0.951, respectively. Seventeen of the 19 specifications place MW below expectation in each slice.

WM displays the widest cross-baseline range. In All→CS, its O/E ratio varies from 0.734 to 1.246, producing a total range of 0.511. In CS→CS, the range is slightly wider at 0.740-1.268. Despite these large extrema, the medians remain below one at 0.892 and 0.897. The large upper endpoints arise from the all-prior, prior-citation-weighted randomized drawing specification rather than from a general shift across the baseline set.

WW differs from MW and WM in that its entire cross-baseline range remains below parity. Its

All→CS values span 0.649-0.963 and its CS→CS values span 0.638-0.947. Even the maximum value is therefore below one in both slices. At the same time, the magnitude of the WW deficit varies considerably: the implied shortfall ranges from approximately 3.7% to 35.1% in All→CS and from approximately 5.3% to 36.2% in CS→CS.

Figure 6.16 summarizes these differences visually for the primary All→CS analysis. The marker denotes the median across specifications and the whiskers denote the observed minimum-to-maximum range.



Marker = median across specifications; whiskers = observed min-max range (descriptive, not confidence intervals).

Figure 6.16: Median and minimum-to-maximum range of O/E ratios across the 19 principal All→CS specifications. The marker denotes the median across specifications and the whiskers denote the descriptive cross-baseline range; the whiskers are not confidence intervals.

The contrast between median and range is particularly important for MW and WM. Their medians remain clearly below one even though individual randomized specifications cross above parity. The same distinction is visible for MM in the opposite direction: the median remains above one despite one specification falling below parity. WW is the only category for which both the median and the complete observed range remain below one.

The same pattern is also visible at the baseline-family level. Six families are represented in the comparison: direct structural, topical relevance, network neighborhood, GLM, randomized reference drawing, and network-conditioned randomization. In All→CS, five of the six families place MM uniformly above one, while the randomized reference-drawing family crosses parity. Five families place MW uniformly below one, again with randomized reference drawing crossing parity. For WM, four families remain uniformly below one, while randomized reference drawing and network-conditioned randomization cross parity. For WW, all six families remain entirely below one.

The CS→CS family-level results are still more directionally consistent for WM: five of the six families remain entirely below one, with only randomized reference drawing crossing parity. For WW, all six families again remain below expectation.

6.7.3 Summary of Evidence by Gender Category

The cross-baseline analysis separates two aspects of robustness: directional stability and stability of magnitude. Table 6.20 summarizes the resulting pattern by gender category.

Table 6.20: Cross-baseline summary by gender category. Specification counts refer to the 19 principal baselines; family counts refer to six broader baseline families.

Gender category	Directional pattern	Magnitude pattern
MM	Above expectation in 18/19 specifications in both citation slices; five of six baseline families are uniformly above one.	Narrow cross-baseline range; median O/E is approximately 1.03 in both slices.
MW	Below expectation in 17/19 specifications in both slices; five of six baseline families are uniformly below one.	Median O/E remains below one, but the range crosses parity under selected randomized reference-drawing variants.
WM	Below expectation in 17/19 All→CS and 18/19 CS→CS specifications.	Widest cross-baseline range; median remains approximately 0.89, while selected randomized specifications produce O/E ratios above parity.
WW	Below expectation in all 19 specifications and all six baseline families in both citation slices.	Median O/E is approximately 0.76 in All→CS and 0.75 in CS→CS; even the maximum specification remains below one.

For MM papers, the cross-baseline pattern is highly stable in direction and comparatively narrow in magnitude. The All→CS median of 1.032 and CS→CS median of 1.031 are close to the values produced by several of the more strongly adjusted specifications. The single directional exception is the all-prior, prior-citation-weighted randomized draw.

MW papers exhibit a predominantly below-expectation pattern but greater dependence on baseline construction. The category is below one under 17 of 19 specifications in both citation slices and under five of the six baseline families. Its strongest departures above parity occur under the all-prior, prior-citation-weighted randomized draw, where O/E reaches 1.168 in All→CS and 1.201 in CS→CS.

WM papers show a similar dominant direction but still greater magnitude variation. Their O/E ratios remain below one under 17 of 19 All→CS specifications and 18 of 19 CS→CS specifications. The highest ratios, 1.246 and 1.268, again occur under all-prior, prior-citation-weighted randomization. In contrast, co-citation-conditioned specifications tend to place WM much closer to parity, including an All→CS ratio of 1.006 under network-conditioned co-citation randomization.

WW papers provide the most directionally stable result across the baseline set. All 19 specifications produce O/E below one in both citation slices, and all six baseline families remain uniformly below one. The magnitude nevertheless depends strongly on the opportunity set. The lowest ratios are 0.649 for All→CS and 0.638 for CS→CS under D0, whereas the highest values are 0.963 and 0.947 under the all-prior, prior-citation-weighted randomized draw. Co-citation-based neighborhood, GLM, and randomization specifications also move WW substantially closer to parity than the least-adjusted structural baselines, but none of the principal specifications reaches or exceeds one.

Taken together, the cross-baseline comparison shows substantially greater stability in direction than in magnitude. WW papers remain below expectation under all 19 principal specifications and all six baseline families in both citation slices, making this the most directionally robust result in the analysis. MM papers are above expectation under nearly all specifications, whereas MW

and WM papers are below expectation under the large majority of specifications but cross parity under selected randomized null models, particularly those incorporating prior citation visibility. The results therefore should not be summarized as a single common effect shared equally by all gender categories. Rather, the main robustness conclusion is that the direction of the WW deviation is stable across a wide range of definitions of citation opportunity, while its magnitude, and the results for the remaining categories, are more sensitive to baseline specification. The close correspondence between the All→CS and CS→CS slices further indicates that this qualitative cross-baseline pattern persists when the citing population is restricted to Computer Science.

6.8 Results Summary

Across the analyses, the observed citation distribution differs systematically from the expectations generated by a wide range of structural, topical, network-based, regression-based, and randomized baselines. The magnitude of these differences depends substantially on how expected citation opportunity is defined, but several directional patterns recur across specifications.

Under the direct structural baselines D0-D3, MM papers consistently receive more citations than expected, whereas MW, WM, and WW papers remain below expectation. Under the decade-conditioned D1 baseline, the deviations are substantially smaller than under the global D0 baseline, while the subfield- and venue-conditioned specifications produce smaller additional changes. In the primary All→CS analysis, the WW O/E ratio increases from 0.649 under D0 to 0.762 under D3, but remains below parity. Restricting the citing population to Computer Science produces very similar results.

The contextual analyses show that the same broad pattern persists across topical and temporal partitions. Citation ratios for female-represented categories generally move closer to parity for more closely matched topical contexts, although they remain below expectation in the principal taxonomy-weighted comparison. Citation-lag results similarly show that the magnitude of the deviation varies across citation ages. In particular, the oldest citation bucket exhibits some of the largest below-expectation ratios for WW papers, whereas broader lag groups partially average over this heterogeneity.

Network-based expectations substantially change the magnitude of the observed differences. The WW deviation is smaller under the bibliographic-coupling neighborhood baseline N1 than under the direct structural baselines, while the co-citation-based N2 estimates are considerably closer to parity for all four categories. The same distinction appears in the GLM analysis. Structural, direct-citation, and bibliographic-coupling specifications M1-M3 produce similar gender-aggregated O/E ratios, whereas M4, which incorporates co-citation features, shows a large improvement in paper-level model fit and substantially smaller gender-category deviations. M5, which combines all network feature blocks, produces results very close to M4.

The improvement in M4 and M5 is not confined to in-sample model fit. In the 80/20 paper-level holdout validation, the large increase in Poisson D^2 relative to M1-M3 remains present for papers excluded from coefficient estimation. At the same time, the direction of the gender-aggregated O/E differences is preserved: MM remains above expectation and MW, WM, and WW remain below expectation under the principal GLM specifications. The observed differences are therefore smaller under the co-citation models, while the same directional pattern remains. Because co-citation features are future-informed, these models remain retrospective expected-citation specifications rather than prospective forecasting models.

The randomized robustness analyses show that expected-citation construction can also alter the direction of the results for some categories. Uniform temporal reference drawing retains relatively large below-expectation ratios for MW, WM, and especially WW papers. Incorporating prior citation visibility moves the randomized expectations substantially closer to the observed citation distribution. Under the all-prior, prior-weighted specification, MM moves below parity while

MW and WM move above it. WW, however, remains below expectation. Network-conditioned randomization produces still smaller deviations, particularly when replacement candidates are defined through co-citation, although these estimates apply to a substantially more restricted set of replaceable citation targets.

The cross-baseline comparison consolidates these results across 19 principal specifications. In All→CS, MM is above expectation under 18 of 19 specifications, MW and WM are below expectation under 17 of 19, and WW is below expectation under all 19. The corresponding CS→CS results are nearly identical: MM is above expectation under 18 of 19 specifications, MW is below expectation under 17 of 19, WM under 18 of 19, and WW under all 19. Median O/E ratios across the All→CS specifications are 1.032 for MM, 0.935 for MW, 0.892 for WM, and 0.762 for WW.

The strongest directional stability is therefore observed for WW papers: their O/E ratio remains below one across every principal baseline and both citation slices. MM also shows high directional stability, with only one randomized specification reversing its usual above-expectation position. MW and WM show greater sensitivity to the construction of the null model, particularly under prior-citation-weighted randomized drawing. Overall, the results indicate that baseline choice materially affects the estimated magnitude of citation imbalance and can alter the direction for selected categories, while the dominant cross-baseline ordering remains stable across the majority of specifications and across both citation slices.

7. Discussion

7.1 Interpretation of the Main Findings

The central empirical finding of this thesis is that the measured gender citation imbalance in Computer Science is simultaneously *directionally stable* and *magnitude-sensitive*. Across most specifications, MM papers receive more citations than expected, while categories containing at least one woman receive fewer. This pattern is strongest for WW papers, which remain below expectation under every principal specification considered in the cross-baseline analysis. At the same time, the size of these deviations changes substantially as the definition of expected citation opportunity becomes more specific. The results therefore do not support interpreting citation imbalance as a single fixed numerical quantity. Instead, the estimated imbalance depends on which dimensions of the citation environment are treated as relevant for constructing the expectation.

This distinction is important for interpreting the contribution of the multi-baseline design. A large deviation under a simple paper-share baseline does not necessarily persist at the same magnitude after accounting for publication period, topical composition, venue structure, prior visibility, or citation-derived network position. Conversely, movement towards parity under a more structured baseline does not by itself imply that the original deviation was spurious. Rather, the different baselines answer different questions about what constitutes a comparable citation opportunity. The most informative conclusion is therefore obtained from the pattern across specifications rather than from any single O/E estimate.

7.1.1 Direction and Magnitude of Citation Imbalance

The distinction between direction and magnitude is clearest in the direct citation results. Under the global D0 baseline, the citation distribution deviates substantially from the underlying distribution of known-gender Computer Science papers. In the All→CS slice, MM papers receive approximately 9% more citations than expected, whereas MW, WM, and WW papers receive approximately 18%, 27%, and 35% fewer, respectively. These are the largest deviations among the principal direct structural baselines.

A substantial part of this difference is reduced once publication period is taken into account. The movement from D0 to D1 is particularly important because the gender composition of Computer Science changes strongly over time. Older publication cohorts contain much larger MM shares than recent cohorts, while older papers have also had substantially more time to accumulate citations. Comparing all papers against one global gender distribution therefore combines citation reception with major temporal differences in the available publication pool. The smaller deviations under D1 show that the estimated magnitude is substantially sensitive to temporal conditioning.

Further conditioning produces more modest changes. Subfield conditioning in D2 leaves the ratios relatively close to D1, while conditioning on venue-rank strata in D3 moves MW, WM, and WW somewhat further towards parity. This suggests that broad temporal composition is more consequential for the direct baseline estimates than the additional subfield or venue-rank adjustment considered separately here. Nevertheless, none of these direct specifications changes the ordering of the four categories: MM remains above expectation, while MW, WM, and WW remain below, with WW consistently exhibiting the largest negative deviation.

The cross-baseline analysis reinforces the importance of separating directional stability from numerical stability. Across the 19 principal specifications, the median All→CS O/E ratio is 1.032 for MM, 0.935 for MW, 0.892 for WM, and 0.762 for WW. However, the corresponding ranges are considerably wider for the non-MM categories. In particular, some randomized reference-drawing specifications move MW and WM above parity and MM below parity. Their direction is therefore not invariant to the assumed null citation process.

WW behaves differently. Its O/E ratio varies substantially in magnitude, ranging from 0.649 to 0.963 in All→CS, but it never crosses parity in any of the 19 principal specifications. The same is

true in CS→CS, where the range is 0.638-0.947. This makes the WW result qualitatively stronger than a claim based only on its comparatively low O/E ratio under D0 or D3. The important feature is not that one particular baseline produces a large deficit, but that substantially different definitions of citation opportunity continue to place WW citation reception below expectation.

The MW and WM findings require more qualification. Both categories are below expectation under the large majority of specifications, but their estimates are more responsive to how prior citation visibility is incorporated into the null model. In the broad all-prior, prior-citation-weighted reference-drawing specification, both move above parity. This indicates that part of their observed citation pattern is sensitive to cumulative visibility: when papers that were already highly cited are assigned greater expected probability of being selected, the resulting expected gender distribution changes enough to alter the direction of the comparison. For these categories, it is therefore more appropriate to describe the evidence as predominantly, rather than universally, below expectation.

The results should consequently not be summarized as a single common “female citation effect” across MW, WM, and WW. The three categories exhibit different magnitudes and different levels of specification sensitivity. In particular, the persistent WW result contrasts with the greater movement of MW and WM under richer or visibility-weighted expectations. The first-last author categorization therefore reveals heterogeneity that would be obscured if all papers containing women were aggregated into one group.

7.1.2 Robustness across Baselines and Citation Slices

The comparison across baseline families shows that robustness in this thesis should be understood as consistency under alternative definitions of expected citation opportunity, rather than agreement on one numerical estimate. Structural baselines, topical specifications, local network neighborhoods, GLMs, and randomized citation models preserve different properties of the observed citation system. It is therefore expected that they produce different O/E magnitudes. What is more informative is whether changing these assumptions also changes the dominant direction of the result.

For MM and especially WW, the dominant direction is highly stable. MM is above expectation in 18 of 19 principal specifications in each citation slice, while WW is below expectation in all 19. MW and WM also show a clear dominant direction, but with exceptions under selected randomized nulls. The cross-baseline comparison therefore supports a graded interpretation of robustness: strongest for WW, high for MM, and weaker for MW and WM.

The randomized results are particularly useful for understanding why a multi-baseline approach is necessary. Under uniform reference drawing, substantial deviations remain. Once candidate targets are weighted by prior citation accumulation, several ratios move much closer to parity, and under the broadest all-prior specification the usual direction for MM, MW, and WM is reversed. Prior visibility therefore represents an important modeling choice rather than a minor technical adjustment. A null model in which every temporally eligible paper is equally likely to be cited produces a different expectation from one in which already visible papers are more likely to be selected.

This does not imply that the prior-citation-weighted specification is the uniquely correct baseline. Giving greater probability to already cited papers explicitly incorporates cumulative advantage into the expected citation process. Whether this should be treated as part of legitimate citation opportunity or as part of the process through which citation inequalities accumulate is itself an interpretive choice. The contrast between uniform and prior-weighted randomization therefore illustrates why expected citations cannot be treated as an assumption-free quantity.

The similarity between All→CS and CS→CS provides a different form of robustness. Across the 19 principal specifications, the mean absolute difference between the two slices is only approximately 0.002 for MM, 0.015 for MW, 0.009 for WM, and 0.011 for WW. Restricting the citing popula-

tion to Computer Science therefore changes the magnitude only modestly and rarely changes the qualitative interpretation.

This similarity indicates that the main citation-reception pattern is not specific to one definition of the citing population. Citations entering Computer Science from other fields alter the total citation volume and some category-specific ratios, but the dominant O/E pattern remains closely reproduced within CS→CS. This is particularly important because the All→CS slice includes heterogeneous external citation contexts, while CS→CS imposes a stricter disciplinary boundary. Agreement between them shows that the main findings are not highly sensitive to whether cross-field citing papers are retained.

At the same time, agreement across slices should not be interpreted as independence from all corpus-definition choices. Both slices are constructed from the same underlying filtered dataset, use the same gender operationalization, and share many of the same cited Computer Science papers. Their similarity therefore addresses one particular robustness dimension: the definition of the citing population. It does not remove limitations associated with gender coverage, corpus-internal citations, taxonomy classification, or network availability, which are considered separately in Section 7.5.

7.1.3 Temporal, Topical, and Network-Based Interpretation

The temporal and topical analyses show that citation imbalance is embedded in a strongly structured citation environment. Citations are concentrated among recent and topically related papers, but neither temporal nor topical conditioning eliminates the main gender-category differences. Instead, these dimensions primarily change their magnitude.

The citation-lag results indicate that below-expectation citation reception for WW papers is not confined to the first years after publication. WW remains below expectation in every fine-grained lag group in both citation slices. Moreover, the lowest All→CS ratio occurs for citations more than twenty years after publication, where the WW O/E ratio is 0.611. The corresponding CS→CS value is 0.646. These results indicate that the aggregate WW deviation is not solely a short-term phenomenon associated with early citation accumulation.

The lag pattern should not, however, be interpreted as a simple monotonic decline with citation age. O/E ratios fluctuate across the intermediate lag groups, and different gender categories show different trajectories. Consequently, the long-memory analysis is better interpreted as evidence that the imbalance remains observable across multiple citation-age regimes, with particularly large WW deviations in the oldest citation relationships, rather than as evidence of one uniform aging mechanism.

Topical proximity produces another systematic variation. For MW, WM, and WW, same-topic citation relationships generally yield ratios closer to parity than more distant cross-topic or cross-subfield relationships. In All→CS, for example, the WW ratio is 0.736 among same-topic citations but 0.693 among cross-subfield citations. For MW the corresponding values are 0.994 and 0.885, and for WM 0.934 and 0.828. The taxonomy-weighted specification falls between these more localized and more distant comparisons.

One interpretation of this pattern is that restricting the comparison to closer intellectual contexts removes some heterogeneity that is present in broader citation flows. Papers competing for recognition within the same topic are more similar in substantive content than papers connected across subfields, and their relevant opportunity sets are therefore narrower. However, topical proximity does not remove the WW deviation, even among same-topic citations. The topical results therefore show that the estimated magnitude varies across levels of topical proximity, while the WW deviation remains below expectation even among same-topic citations.

The strongest changes in magnitude arise when citation-derived network information is incor-

porated. Bibliographic coupling and co-citation provide two distinct definitions of local scholarly context. Bibliographic coupling groups papers according to shared references and therefore approximates similarity in their intellectual foundations. Co-citation instead groups papers according to how later literature cites them together and therefore captures subsequent community association.

The bibliographic-coupling N1 baseline reduces some of the direct-baseline deviations but preserves a substantial ordering across categories. In All→CS, the WW O/E ratio remains 0.803 under N1. Incorporating prior citation visibility into bibliographic-coupling-conditioned edge randomization moves several categories closer to parity, but WW remains below expectation. This shows that the deviations are smaller under the bibliographic-coupling comparison than under several of the broader structural baselines, while a substantial WW deviation remains.

Co-citation produces a markedly different result. N2 yields All→CS O/E ratios of 1.012, 0.945, 0.964, and 0.927 for MM, MW, WM, and WW, respectively, substantially closer to parity than N1. The same pattern appears in the GLM analysis: adding bibliographic-coupling features in M3 produces only a modest improvement over the structural models, whereas introducing co-citation features in M4 produces a large increase in paper-level predictive fit and simultaneously moves the gender-aggregated O/E ratios much closer to one. Adding the remaining network blocks in M5 produces little additional change. Co-citation-conditioned edge randomization similarly generates the smallest deviations among the principal network-conditioned randomization variants.

Taken together, these results show that later community association is highly informative about citation reception. Papers that occupy similar positions in co-citation space also tend to have substantially more similar citation outcomes than papers matched only through structural metadata or shared reference foundations. Co-citation-conditioned specifications therefore produce substantially better paper-level fit and smaller gender-aggregated O/E deviations than the structural or bibliographic-coupling specifications.

This result requires careful interpretation. Co-citation is constructed from later citing activity and is therefore partly a representation of the same citation system whose outcomes are being analyzed. It is not information that would have been fully available when a paper was published or when earlier citations were made. Consequently, the movement towards parity under co-citation conditioning cannot be interpreted as showing that co-citation structure causally “explains away” citation imbalance. Instead, it shows that once papers are compared within the intellectual associations produced by later citation behavior, the remaining gender-category differences in citation reception are considerably smaller.

This distinction also explains why the bibliographic-coupling and co-citation results should not be ranked simply according to which produces an O/E ratio closest to one. They condition on different information and refer to different covered paper populations. Bibliographic coupling provides a backward-looking comparison based on references available at publication, whereas co-citation provides a retrospective comparison based on subsequent scholarly recognition. The difference between them is therefore substantively informative: the estimated magnitude of gender citation imbalance depends not only on how much structural information is incorporated, but also on which stage of the citation process that information represents.

Overall, the temporal, topical, and network analyses support a common interpretation. Citation imbalance is not independent of scholarly context: its estimated magnitude changes when papers are compared within more specific temporal, topical, visibility, or network-defined opportunity sets. However, these adjustments do not affect all gender categories equally, and no single contextual specification eliminates the WW below-expectation pattern. The results therefore point to a layered rather than uniform structure of citation imbalance, in which part of the aggregate difference is associated with observable citation opportunity and network position, while a residual directional difference remains across substantially different comparison models.

7.2 Gender Operationalization and Interpretation

The gender variable used throughout this thesis is an operational measure constructed from name-based inference for the first and last authors of each paper. It should therefore be interpreted as a paper-level classification of inferred author-gender configurations rather than as a direct measurement of gender identity. This distinction is important because the empirical findings refer to differences in citation outcomes across the categories MM, MW, WM, and WW as constructed by the inference procedure, not to all possible forms of gender representation within research teams.

The operationalization was chosen because it provides a consistent paper-level variable that can be applied across more than one million Computer Science papers and across all citation-derived representations used in the thesis. At the same time, it necessarily simplifies both authorship structure and gender. The interpretation of the results must therefore remain conditional on this operational definition.

7.2.1 Name-Inferred First-Last Author Gender Categories

The use of first- and last-author gender produces four known-gender categories: MM, MW, WM, and WW. This representation has the practical advantage of being compact and consistent across direct citations, topical analyses, bibliographic coupling, co-citation, citation-lag analyses, GLM specifications, and randomized robustness checks. It also avoids reducing a paper to a single author while retaining information about two prominent authorship positions.

However, these categories should not be interpreted as descriptions of the gender composition of the full author team. A paper classified as MM, for example, may contain women among its intermediate authors, while a WW paper may contain men elsewhere in the author list. The categories identify only the inferred genders of the first and last listed authors. For single-author papers, the same individual occupies both positions and the resulting paper category therefore reflects a single inferred author.

This distinction matters when interpreting the citation results. The finding that WW papers remain below expected citation reception across all principal baselines does not imply that every paper containing women is systematically undercited, nor does the MM category represent exclusively male research teams. The estimand is narrower: citation reception is compared across papers grouped according to the inferred gender configuration of their first and last authors.

The interpretation of first and last authorship also varies across scholarly contexts. In some research communities, first authorship is closely associated with the primary contribution and last authorship with senior supervision or group leadership. In others, alphabetical ordering, equal-contribution practices, or different disciplinary conventions weaken these interpretations. Computer Science itself is heterogeneous in this respect. The first-last classification should therefore be understood primarily as a reproducible paper-level grouping variable rather than as a universal representation of contributor hierarchy.

The empirical results nevertheless suggest that preserving the distinction between MM, MW, WM, and WW is informative. The three categories containing at least one inferred woman do not behave identically across the robustness analyses. In particular, WW exhibits substantially greater directional stability than MW and WM, which are more sensitive to the construction of the expected-citation baseline. Aggregating all papers containing at least one woman into a single category would therefore conceal meaningful heterogeneity observed in the results.

The known-gender restriction is also integral to the interpretation of these categories. Of the 1,542,691 Computer Science papers in the final analytical corpus, 1,081,363, or 70.10%, could be assigned to one of the four known-gender categories. Papers for which either the first- or last-author gender could not be assigned confidently were excluded from gender-specific O/E

aggregation rather than treated as a fifth substantive category. The reported gender-category results consequently describe the known-gender subset of the Computer Science corpus rather than the full set of papers.

7.2.2 Interpretive Limits of Binary Gender Inference

The gender labels used in this thesis are inferred from names and should not be interpreted as self-identified gender. The inference procedure maps author names to a probability-like score and assigns labels using the thresholds $p \geq 0.7$ for W, $p \leq 0.3$ for M, and an unknown label for intermediate or unresolved cases. This operationalization makes large-scale analysis possible, but it introduces both conceptual and measurement limitations.

First, the procedure imposes a binary representation of gender. Authors whose identities do not conform to the male-female distinction used by the inference system cannot be represented directly, and the resulting classification should therefore not be interpreted as a complete model of gender identity. The categories used in this thesis are best understood as name-inferred binary classifications employed for bibliometric comparison.

Second, the reliability of name-based gender inference is unlikely to be uniform across linguistic and cultural naming contexts. The corpus is not restricted to particular countries, while the same inference procedure and probability thresholds are applied globally. Names differ substantially in how strongly they encode gender, how consistently they are represented in reference databases, and how they are affected by transliteration or naming conventions. A threshold such as 0.7 may therefore correspond to different effective levels of classification reliability across different parts of the corpus.

This issue is particularly important because the threshold should not be interpreted as a uniformly validated confidence level for every naming context represented in the dataset. Some naming systems may produce more ambiguous assignments or a greater probability of misclassification than others. Because the present analysis does not perform country-specific or language-specific validation of the inference procedure, it cannot determine whether the same nominal threshold yields comparable classification accuracy across all author populations.

Third, uncertainty in gender inference can affect the analysis through two distinct channels. Authors whose names cannot be classified with sufficient confidence become part of the unknown-gender population and are excluded from gender-specific aggregation. Authors who receive a confident but incorrect classification remain in the known-gender sample and may contribute to misclassification between MM, MW, WM, and WW. The latter is particularly important because coverage statistics alone cannot reveal classification errors among names that did receive a label.

The unknown-gender population should also not be assumed to be missing at random. Gender-classification coverage varies across publication periods and may additionally vary with language, region, naming convention, metadata quality, venue, and research area. Within Computer Science, complete first-last gender coverage declines from higher levels in earlier decades to 61.88% for papers published between 2020 and 2024. Consequently, the known-gender subset may differ systematically from papers for which gender could not be inferred.

These considerations affect the scope of the conclusions rather than changing the meaning of the reported O/E calculations. The results establish citation patterns among papers that can be assigned to the four inferred first-last author categories under the chosen procedure. They do not establish that the same numerical patterns would necessarily be obtained under self-reported gender data, alternative inference thresholds, country-specific inference models, or a non-binary gender representation.

For this reason, the terminology used throughout the thesis deliberately refers to *inferred* gender categories rather than treating the labels as direct observations of author identity. The cross-

baseline robustness of the citation results addresses sensitivity to alternative definitions of expected citation opportunity, but it does not remove uncertainty introduced by the gender measurement itself. The consequences of incomplete coverage, cross-cultural classification reliability, and the absence of a threshold-sensitivity analysis are considered further in Section 7.5.1.

7.3 Comparison with Related Work

The findings of this thesis are broadly consistent with previous evidence of gender-related differences in citation reception, while extending that literature in two directions. First, the analysis evaluates a substantially broader set of expected-citation specifications within Computer Science, rather than relying on a single reference model. Second, it uses bibliographic coupling and co-citation not only as descriptive bibliometric representations, but as alternative structures for defining local citation expectations and testing the robustness of gender-category citation differences.

The comparison with prior studies should nevertheless be made primarily in terms of qualitative patterns rather than exact effect sizes. Earlier studies differ in their publication populations, time periods, gender-inference procedures, citation windows, reference models, and definitions of expected citation opportunity. An over- or under-citation estimate from one study is therefore not numerically equivalent to an O/E ratio from another. The most informative comparison concerns whether similar directional patterns and forms of contextual heterogeneity appear under these different empirical designs.

7.3.1 Relation to Prior Gender Citation Imbalance Studies

The dominant direction of the results is consistent with a broader literature reporting lower citation reception for research associated with women and higher citation reception for research associated with men. Such patterns have previously been reported in political science, astronomy, neuroscience, and physics (5; 6; 8; 9; 10). The present findings therefore place Computer Science within a broader cross-disciplinary pattern, while also showing that the magnitude of the difference depends strongly on how expected citation opportunity is defined.

The neuroscience study of Dworkin et al. (9) provides a particularly close point of comparison because it also classifies papers according to the inferred genders of their first and last authors. They found that papers with men in both positions appeared in reference lists more often than expected, whereas papers with women in these authorship positions were underrepresented. They additionally found that citation behaviour differed according to the gender composition of the citing authors and examined the role of local coauthorship-network composition.

The results of the present thesis reproduce the same broad directional asymmetry under most specifications. MM papers are generally above expectation, while WW papers are consistently below expectation. The latter finding is particularly stable: WW remains below expectation under all 19 principal specifications in both the All→CS and CS→CS citation slices. MW and WM also remain below expectation under most specifications, although they are more sensitive than WW to the construction of the expected-citation model.

The comparison with Dworkin et al. also highlights an important distinction. The present study does not interpret the descriptive citing-side gender-flow matrices as direct evidence of gender homophily. Although WW citing papers allocate a larger proportion of their known-gender references to WW papers than MM citing papers do, the available citation pools differ across publication periods, topics, and other structural dimensions. The present analysis therefore separates such descriptive citing-side patterns from the expected-citation comparisons used to quantify citation reception.

The findings are also consistent with the physics analysis of Teich et al. (10), who reported over-citation of papers authored by men and under-citation of papers authored by women relative to

expectation. An important contribution of their study was to show that the magnitude of the citation imbalance varied according to who was citing, where the citation occurred, and the citing authors' proximity to the cited research. The present thesis similarly finds substantial contextual heterogeneity rather than one uniform citation imbalance.

The topical results provide one particularly relevant comparison. In the present analysis, citations between papers in the same topic generally produce smaller deviations from parity for MW, WM, and WW than citations crossing larger topical distances. For example, the All→CS WW O/E ratio is 0.736 for same-topic citations and 0.693 for cross-subfield citations. This is qualitatively compatible with the broader observation by Teich et al. that citation imbalance varies with scholarly proximity. However, the measures are not equivalent: the present thesis operationalizes proximity through the OpenAlex taxonomy, whereas the physics study uses different measures of familiarity and proximity. The similarity should therefore be interpreted as evidence that citation imbalance is context-dependent, not as a replication of the same underlying mechanism.

The closest discipline-specific comparison is the recent study by Nakajima et al. (12), which extends their earlier conference-focused analysis (11) to both conference and journal publications in Computer Science. Using direct-citation reference models, they found that papers with a woman as either the first or last author received fewer citations than expected, with the imbalance being more pronounced for conference papers and particularly for papers published at top-tier conferences. Their analysis further showed that citation homophily with respect to country of affiliation, research topic, and venue rank was strongly associated with the measured imbalance, whereas additionally preserving heterogeneity in the number of citations received per paper produced comparatively little change. They also identified associations between gendered citing patterns and characteristics of the citing papers, including author prominence and the local coauthorship networks of the first and last authors.

The results of the present thesis are directionally consistent with these findings despite using substantially different definitions of citation opportunity. WW papers remain below expectation under all 19 principal specifications in both citation slices, while MM papers are above expectation under nearly all specifications. Importantly, this dominant pattern is not restricted to direct-citation reference models. It also remains visible when scholarly comparability is defined through bibliographic-coupling and co-citation neighborhoods, as well as under topical, temporal, GLM-based, and randomized specifications.

The present findings can therefore be interpreted as a cross-representation confirmation of the broad directional pattern reported by Nakajima et al. (12). At the same time, the multi-representation analysis qualifies any interpretation of this pattern as a single, baseline-independent citation gap. Bibliographic-coupling specifications preserve comparatively larger deviations from parity, whereas co-citation specifications produce substantially smaller deviations, even though WW papers remain below expectation. The agreement with the recent direct-citation literature is therefore strongest in the direction of the imbalance rather than in its numerical magnitude.

This distinction is important because the different approaches condition on different aspects of the citation system. Nakajima et al. (12) vary the structural properties preserved by direct-citation reference models, whereas the present thesis additionally changes the representation through which scholarly similarity and citation opportunity are defined. The persistence of the dominant WW and MM patterns under these non-direct citation representations indicates that the directional result is not specific to a single direct-citation reference model, while the movement of the observed/expected ratios across representations shows that its estimated magnitude remains representation-dependent.

The present study also extends the comparison across structural contexts. Conditioning on publication decade substantially reduces the differences observed under the global paper-share baseline, while additional subfield, venue-rank, and topical specifications alter their magnitude further. This supports a central point shared with earlier gender-citation research: aggregate citation differences

cannot be separated from the structure of the available literature without an explicit model of expected citation opportunity.

At the same time, the randomized reference-drawing results further qualify any interpretation of a single discipline-level citation-gap estimate. Sufficiently different assumptions about prior visibility can reverse the usual direction for MM, MW, and WM. Consequently, the agreement with previous studies is strongest at the level of the dominant cross-specification pattern, especially the persistent below-expectation position of WW papers, rather than at the level of one fixed numerical estimate.

A further distinction concerns temporal analysis. Previous studies have examined changes in citation imbalance across calendar time (9; 11; 12). The long-memory analysis in this thesis addresses a different temporal question: whether the imbalance differs with the age of the citation relationship itself. The finding that WW papers remain below expectation across all citation-lag groups therefore complements, rather than directly replicates, evidence about historical changes in citing behaviour. It indicates that the WW deviation is observable in both short-term and long-term citation reception, including citations occurring more than twenty years after publication.

7.3.2 Relation to Randomized and Network-Based Citation Baselines

The methodological contribution of the present thesis is also related to two partly separate strands of prior work: randomized reference models for gender citation imbalance and comparative research on citation-derived network representations.

The use of reference models follows the general logic adopted in earlier citation-imbalance studies, particularly the recent Computer Science analysis of Nakajima et al. (12). Their framework compares observed citation patterns with randomized citation networks under three increasingly structured reference models: random-draws, homophilic-draws, and preferential-draws. These models preserve the number of citations made by each paper while progressively incorporating homophily in country of affiliation, research topic, and venue rank, and heterogeneity in the number of citations received per paper. As in the present thesis, the resulting definition of over- or under-citation therefore depends directly on which properties of the empirical citation system are retained in the reference model.

The randomized reference-drawing results in this thesis demonstrate the importance of this choice particularly clearly. When citation targets are drawn uniformly from temporally eligible papers, substantial deviations from parity remain. When candidate papers are weighted according to citations already accumulated before the citing year, several of these deviations become substantially smaller. Under the broad all-prior, prior-citation-weighted specification, the usual direction is even reversed for MM, MW, and WM.

This sensitivity is consistent with the broader motivation for reference models in citation-network research: a null model is informative only relative to the structural properties it preserves. In the present setting, prior citation accumulation is especially consequential because it represents cumulative visibility. A baseline that treats all temporally eligible papers as equally likely citation targets asks a different question from one that allows previously visible papers to be more likely targets. The randomized results therefore extend prior reference-model approaches by showing empirically how strongly the estimated gender imbalance can depend on this choice.

The present randomization framework is not a direct replication of the reference models used by Nakajima et al. (12). Their random-draws, homophilic-draws, and preferential-draws models vary the extent to which properties of the direct citation network are preserved, particularly citation homophily and heterogeneity in received citation counts. The robustness analyses in this thesis instead emphasize temporal eligibility and prior citation visibility, and additionally introduce replacement sets defined through local bibliographic-coupling and co-citation neighborhoods. Author-overlap self-citations and time-aware prior-coauthor citations are removed before these analyses are performed. The two approaches are therefore complementary: Nakajima et

al. vary the structural properties preserved within direct citation reference models, whereas the present thesis also varies the citation-derived representation used to define plausible alternative targets and scholarly comparability.

The network-neighborhood analyses extend this comparison beyond direct citation reference models. Bibliographic coupling and co-citation have long been established as distinct representations of scholarly relatedness. Boyack and Klavans (15) demonstrated that direct citation, bibliographic coupling, and co-citation recover different representations of research fronts, while Kleminski et al. (16) showed within Computer Science that the three network structures provide complementary and partly non-overlapping information for scientific-topic identification.

The present results are consistent with the broader implication of these studies that citation-derived representations should not be expected to be interchangeable. Here, however, the consequence of representation choice is examined for disparity measurement rather than science mapping or topic identification. Bibliographic-coupling and co-citation neighborhoods produce markedly different expected-citation estimates even though both are derived from the same underlying citation system.

Bibliographic coupling preserves relatively substantial residual differences. Under the N1 neighborhood baseline, the All→CS O/E ratio is 0.803 for WW, compared with 0.822 for WM and 0.898 for MW. A similar pattern appears under bibliographic-coupling-conditioned edge randomization. These results indicate that comparison with papers sharing similar reference foundations narrows some of the aggregate differences but leaves a substantial WW deviation.

Co-citation produces considerably smaller residual differences. Under N2, the All→CS WW ratio rises to 0.927, while MW and WM reach 0.945 and 0.964. Co-citation-conditioned edge randomization brings the gender categories closer to parity still. The GLM comparison leads to the same broad pattern: bibliographic-coupling features provide only a modest improvement over the structural specification, whereas adding co-citation features produces a large increase in paper-level predictive fit and substantially reduces the gender-aggregated O/E deviations.

This difference between bibliographic coupling and co-citation provides an empirical extension of the comparative bibliometric literature. Boyack and Klavans (15) and Kleminski et al. (16) demonstrate that different citation relations recover different aspects of scientific structure. The present analysis shows that the same representational choice can also materially affect the estimated magnitude of a demographic citation disparity. In that sense, network representation is not merely a technical preprocessing decision; it contributes directly to the definition of the comparison population against which imbalance is evaluated.

The results should not, however, be interpreted as establishing co-citation as a superior baseline because it produces O/E ratios closer to one. Co-citation and bibliographic coupling encode substantively different information. Bibliographic coupling is based on shared prior references, whereas co-citation is generated by later citation behaviour and is therefore future-informed. Moreover, the retained co-citation network covers a much smaller and more selective subset of the Computer Science corpus. Its near-parity estimates refer to papers sufficiently established in subsequent citation activity to obtain usable co-citation neighborhoods.

This distinction is especially important in relation to the existing bibliometric literature. A representation that performs well for recovering research fronts or identifying topics is not automatically the most appropriate representation of citation opportunity for disparity analysis. The objective of this thesis is therefore deliberately different from ranking direct citation, bibliographic coupling, and co-citation according to one criterion. Their value lies in defining substantively different comparison structures and in revealing which parts of the citation-imbalance conclusion remain stable when those structures change.

Overall, the relation to prior work is therefore one of both replication and extension. The dominant direct-citation pattern is consistent with previous gender-citation studies in neuroscience, physics,

and Computer Science, particularly in the overrepresentation of MM and underrepresentation of WW relative to expected citation rates. At the same time, the multi-baseline results show that the magnitude of this pattern is strongly conditional on the expected-citation model. By combining structural normalization, taxonomy-based relevance, bibliographic coupling, co-citation, GLM expectations, and randomized reference models within one common O/E framework, the present thesis shifts the comparison from asking only whether gender citation imbalance is detectable to asking how robust that conclusion is to alternative representations of citation opportunity.

7.4 Methodological Implications

Beyond the substantive findings on gender citation imbalance, the analyses illustrate several broader methodological issues in the measurement of citation disparities. Most importantly, an “expected” number of citations is not an observed quantity and is not uniquely determined by the data. It is produced by a comparison model that specifies which papers are treated as plausible alternatives, which structural characteristics are preserved, and which aspects of the observed citation system are allowed to influence the expectation.

The empirical results show that these choices are consequential. Some baseline families preserve relatively large deviations between observed and expected citation reception, whereas others move the same gender categories considerably closer to parity. In selected randomized specifications, the direction of the deviation also changes for some categories. This does not make the expected-citation framework unstable or uninformative. Rather, it shows that the interpretation of citation imbalance must remain tied to the comparison structure under which the expectation was constructed.

A second implication concerns the role of increasingly rich information. Adding more conditioning variables or network features does not automatically produce a more valid estimate of citation imbalance. Richer baselines can provide more locally comparable expectations, but they may also restrict the covered population, incorporate information that is temporally downstream of the citation event, or condition on variables that are themselves part of the process through which scholarly recognition develops. Baseline complexity therefore creates trade-offs between comparability, coverage, temporal interpretability, and proximity to the observed outcome.

7.4.1 What Expected Citations Mean in Practice

In this thesis, expected citations should be interpreted as *baseline-specific reference quantities*. They describe the citation reception implied by a particular definition of comparable citation opportunity. They do not represent the number of citations that a paper “should” receive in a normative sense, nor do they constitute a unique counterfactual citation count.

This distinction can be illustrated using the direct baselines. Under D0, expected citations are determined by the overall gender composition of known-gender Computer Science papers. The corresponding question is therefore whether citations are distributed in proportion to the overall availability of the four paper categories. D1 asks a narrower question by defining availability within publication decades. D2 conditions additionally on subfield, while D3 conditions on venue-rank strata instead. Each baseline changes the reference population and therefore changes what an O/E ratio of one represents.

The substantial movement from D0 to D1 illustrates why this matters in practice. The Computer Science publication population changes strongly over time, while older papers have had longer exposure to citation accumulation. A global expectation combines these two processes. Once publication period is included in the reference structure, part of the original deviation is absorbed into the expectation. The resulting O/E ratio answers a different question: not whether a category receives citations in proportion to its overall corpus share, but whether it receives citations in proportion to its availability within temporally comparable publication cohorts.

The same logic applies to the topical specifications. A same-topic analysis defines citation opportunity more narrowly than a corpus-wide or same-subfield comparison. The taxonomy-weighted specification introduces a graded definition in which citation relationships contribute differently according to topical proximity. Expected citation reception is therefore conditional not only on which papers exist, but also on which citation relationships are treated as more substantively relevant.

Network-neighborhood baselines narrow the comparison in another way. Under N1, the expected citation count of a paper is derived from papers with similar reference foundations. Under N2, it is derived from papers that subsequent literature tends to cite together. These expectations are not simply more detailed versions of D0–D3. They represent qualitatively different notions of comparability. A paper may be similar to another according to publication period and subfield while occupying a very different local position in bibliographic-coupling or co-citation space.

The randomized baselines make the dependence on the reference model even more explicit. Uniform reference drawing assumes equal probability among eligible candidate papers, whereas prior-citation-weighted drawing assigns greater probability to papers that were already more visible at the time of the citation. Both are internally coherent null models, but they encode different views of citation opportunity. The first asks what citation patterns would look like under temporal eligibility alone; the second additionally treats cumulative visibility as part of the expected citation process.

This distinction raises an important interpretive question. Prior visibility can reasonably be treated as part of citation opportunity because highly visible papers are more likely to be encountered and cited. At the same time, prior visibility is itself partly the result of earlier citation accumulation and may already contain disparities generated earlier in the citation process. Conditioning on it can therefore reduce a disparity that would remain visible under a baseline treating earlier citation accumulation as part of the phenomenon being studied.

Expected citations should consequently be interpreted relative to an explicit statement of what the baseline preserves. An O/E ratio below one under a decade-conditioned model means that the observed citation count is below the expectation implied by decade-specific paper availability. An O/E ratio below one under a network-conditioned model has a different meaning: the observed citation count is below the expectation implied by local citation-derived comparability. Neither statement is meaningful without specifying the corresponding reference structure.

This also explains why the cross-baseline analysis is central to the methodology. Rather than treating one baseline as the definitive estimate, the thesis evaluates which conclusions survive when the meaning of expected citation opportunity changes. Robustness is therefore assessed through consistency across explicitly different comparison models rather than through repeated estimation of one supposedly model-free quantity.

7.4.2 Trade-Offs between Baseline Families

The baseline families used in this thesis differ not only in statistical form but also in the information they preserve, the population they cover, and the interpretation they support. No single family dominates along all of these dimensions.

The direct structural baselines have the clearest interpretation and the broadest coverage. Their comparison sets are defined through observable metadata such as publication period, subfield, and venue rank. This makes the source of adjustment transparent: movement in the O/E ratio can be related directly to the structural characteristic introduced by the baseline. However, discrete stratification is necessarily coarse. Papers within the same decade and subfield may still differ substantially in topic, scholarly visibility, reference foundations, or network position.

Increasing the number of stratification variables would not solve this problem indefinitely. Highly

crossed comparison cells quickly become small or empty, especially for less common gender categories and fragmented variables such as exact topic or venue identity. Structural baselines therefore face a trade-off between local comparability and statistical support. Coarser cells retain broad coverage but leave more within-cell heterogeneity, while finer cells increase similarity at the cost of sparsity.

The topical baselines provide a different compromise. They condition directly on the relationship between the citing and cited papers rather than only on the characteristics of the cited paper. This is useful when the substantive relevance of a citation is expected to depend on topical proximity. The OpenAlex hierarchy additionally provides a scalable and interpretable measure across a very large corpus. However, taxonomy-based proximity remains a proxy for substantive relatedness. Papers assigned to the same topic need not be equally relevant to one another, and cross-topic citations can still represent important intellectual relationships.

Bibliographic coupling provides a more paper-specific notion of similarity. Because coupling is derived from overlap in reference lists, it can distinguish papers that occupy similar research areas but build on different bodies of prior work. It also has an attractive temporal interpretation because the relevant reference information is fixed when the papers are published. These features make bibliographic coupling well suited to retrospective comparisons that remain grounded in information associated with the papers' research foundations.

Its main cost is coverage. Constructing meaningful local neighborhoods requires sufficient reference information and sufficiently strong overlap with other papers. The retained bibliographic-coupling representation therefore covers only part of the known-gender Computer Science corpus. Moreover, the top- K construction deliberately retains strong local relationships rather than all possible weak similarities. N1 consequently defines a more selective estimand than the direct structural baselines.

Co-citation creates an even stronger trade-off. Empirically, co-citation information is highly informative about citation outcomes. Both N2 and the co-citation GLM specifications produce substantially smaller residual gender-category deviations, and co-citation features provide a large increase in paper-level predictive fit relative to structural and bibliographic-coupling specifications. This makes co-citation a powerful retrospective representation of how papers are positioned within the recognized structure of the literature.

At the same time, co-citation is the most restrictive and temporally problematic representation used in the thesis. A paper requires sufficient later citation activity before a meaningful co-citation neighborhood can emerge. Coverage is therefore much lower than for bibliographic coupling and is especially limited for recent papers. More fundamentally, the representation is future-informed: the same later citation activity used to define co-citation relationships is related to the citation reception being modeled.

For this reason, the strong performance of co-citation-based models should not be interpreted as evidence that co-citation provides the universally superior definition of expected citation opportunity. A baseline may fit the observed citation distribution more closely because it incorporates information that is closely related to the realized outcome. This can be useful for retrospective robustness analysis while making the same representation unsuitable for a prospective expectation available at publication time.

The GLM specifications illustrate a related distinction between flexibility and interpretability. Regression models can combine several continuous and categorical predictors without creating the sparse cells produced by highly crossed stratification. They also generate expected citation counts for the full modeled paper population, including papers without a valid local network-neighborhood expectation when network features can be represented as zero.

However, the resulting expectation is less transparent than a simple paper-share baseline. Its value depends jointly on model specification, regularization, transformations, feature encoding, and the assumed Poisson mean structure. The strong improvement produced by co-citation features again

shows that predictive fit and substantive interpretation are not identical criteria. A model can provide an accurate retrospective description of citation counts without representing a plausible prospective citation process.

Randomized baselines occupy another position in this trade-off. Their main advantage is conceptual explicitness: they define an alternative citation process and specify exactly which properties of the observed system are preserved during replacement. This makes them useful for examining sensitivity to assumptions such as temporal eligibility, cumulative visibility, or local network similarity.

Their results can nevertheless be highly sensitive to the candidate pool and replacement weights. The contrast between uniform and prior-weighted reference drawing demonstrates that different reasonable null processes can produce materially different gender expectations. Network-conditioned randomization additionally inherits the coverage restrictions of the underlying citation-derived representation.

The methodological implication is therefore that baseline families should be treated as complementary rather than ordered from “weak” to “strong” or from “simple” to “correct”. Structural baselines provide broad and transparent comparisons; topical baselines incorporate citation relevance; bibliographic coupling provides backward-looking local similarity; co-citation provides a powerful but future-informed retrospective representation; GLMs combine multiple dimensions within one predictive framework; and randomized baselines make assumptions about alternative citation selection explicit.

A multi-baseline analysis is valuable precisely because these strengths and limitations differ. Agreement across baseline families is informative because it shows that a conclusion does not depend entirely on one particular definition of comparison. Disagreement is equally informative because it identifies which assumptions about citation opportunity materially affect the estimated imbalance.

7.4.3 Descriptive Rather than Causal Interpretation

The observed/expected framework used in this thesis is descriptive and should not be interpreted as estimating a causal effect of author gender on citation reception. An O/E ratio below one establishes that the observed number of citations to a gender category is lower than the number implied by a specified baseline. It does not identify why this deviation occurs.

Several conceptually distinct processes could contribute to the same observed pattern. These include differences in visibility, collaboration networks, research-topic selection, venue access, seniority, institutional resources, citation conventions, prior cumulative advantage, or citation decisions made by individual authors. The available paper-level bibliographic data do not permit these mechanisms to be separated causally.

Conditioning on observable characteristics does not change this limitation. For example, a decade-and-subfield baseline adjusts the comparison for temporal and broad topical composition, but the remaining O/E deviation cannot automatically be interpreted as discrimination within otherwise identical papers. Unobserved differences may remain within the comparison groups, and the conditioning variables themselves can be associated with earlier stages of the process through which scholarly inequalities develop.

The same issue applies to the GLM specifications. Gender is deliberately excluded from the prediction models, so the gender-aggregated O/E ratios describe whether realized citation counts differ from predictions based on non-gender features. This design prevents gender from mechanically determining the expected counts, but it does not turn the residual group difference into a causal gender coefficient. The predictors can be correlated with unobserved paper characteristics, and some network variables may themselves reflect earlier citation outcomes.

Co-citation requires particular caution in this respect. Because it is constructed from later citing

activity, conditioning on co-citation structure uses information generated downstream of publication and partly downstream of citation accumulation itself. The movement of O/E ratios towards parity under N2, M4, M5, or co-citation-conditioned randomization therefore does not show that co-citation structure causally explains the original disparity. It shows that papers with similar realized positions in later co-citation space have more similar citation outcomes.

Prior-citation weighting raises a related issue. When prior citation accumulation is included in the expected citation process, part of the observed difference is absorbed into a visibility-based expectation. This demonstrates sensitivity to cumulative advantage but cannot determine whether the earlier citation accumulation should be interpreted as a confounder, a mediator, or part of the disparity itself. Distinguishing these roles would require an explicit causal model and stronger assumptions about the temporal relationships between gender, visibility, scholarly position, and citation decisions.

The language used to describe the results should therefore remain consistent with the estimation. Terms such as *below expectation*, *above expectation*, *citation imbalance*, and *observed-versus-expected deviation* accurately describe the analyses. Stronger statements such as “gender causes under-citation” or that a particular structural factor “explains” a fixed percentage of the disparity would not be supported by the present design.

This descriptive interpretation does not make the results uninformative. Establishing that a gender-category difference remains visible across substantially different definitions of citation opportunity is itself an important empirical finding. In particular, directional stability across structural, topical, network-based, model-based, and randomized specifications shows which patterns are difficult to remove through alternative descriptive normalizations. Conversely, movement towards parity identifies dimensions of the citation environment with which the aggregate disparity is strongly associated.

The appropriate methodological conclusion is therefore not that the analysis isolates the causal effect of gender, but that it characterizes the robustness of gender-associated citation disparities under multiple representations of scholarly opportunity. Causal explanations for these disparities remain a separate research problem requiring designs that can distinguish competing mechanisms more directly.

7.5 Limitations

The results of this thesis are based on a large citation corpus and are evaluated across multiple expected-citation specifications, but several limitations constrain their interpretation. These limitations arise from three main sources: the operationalization and coverage of author gender, the use of taxonomy and sparse citation-derived network representations, and the scope and filtering of the underlying bibliographic data.

Importantly, the multi-baseline robustness analysis does not remove these limitations. Agreement across baselines shows that a result is not specific to one particular definition of expected citation opportunity, but the baselines are constructed from the same underlying corpus and share several measurement assumptions. Robustness across models should therefore be distinguished from robustness to errors or omissions in the underlying data.

7.5.1 Gender Inference, Missingness, and Coverage

The most direct limitation concerns the measurement of gender. Gender is not observed from self-reported information but inferred from author names and reduced to binary male and female labels. Paper-level categories are then constructed from the inferred genders of the first and last authors. As discussed in Section 7.2, these categories provide a scalable operationalization

for bibliometric analysis but should not be interpreted as direct measurements of author gender identity or of the gender composition of the complete author team.

The inference procedure uses a common probability threshold across the full dataset, assigning W when $p \geq 0.7$, M when $p \leq 0.3$, and U otherwise. The corpus is not restricted to particular countries or linguistic contexts. This creates an important source of uncertainty because the effective reliability of name-based inference is unlikely to be constant across naming systems. Given names differ in how consistently they encode gender, and transliteration, name order, romanization, and unequal representation in the underlying inference data can further affect classification performance. A threshold of 0.7 should therefore be regarded as a pragmatic operational choice rather than as a uniformly validated confidence level across all author populations represented in the corpus.

This limitation can affect the analysis in two ways. First, names that cannot be classified with sufficient confidence are assigned to the unknown category and excluded from gender-specific O/E calculations. Second, a name may receive a sufficiently confident but incorrect label and remain in the known-gender sample. Coverage diagnostics can quantify the first problem but cannot detect the second. Consequently, the reported known-gender coverage does not constitute an estimate of classification accuracy.

The extent of missing gender information is substantial. Of the 1,542,691 Computer Science papers in the final analytical corpus, 1,081,363, or 70.10%, have known inferred gender for both first and last authors. Almost 30% of Computer Science papers therefore do not enter the principal gender-specific analyses. Moreover, this coverage is not temporally stable. Complete first–last-author gender coverage declines to 69.32% for papers from the 2010s and to 61.88% for papers published between 2020 and 2024.

Unknown assignments should not be assumed to be missing at random. Their frequency may depend on geographical, linguistic, temporal, disciplinary, and metadata-related characteristics. If these characteristics are also associated with citation behaviour, restricting the analysis to papers with known inferred gender can alter the composition of the analytical population. The reported O/E ratios should therefore be interpreted as estimates for the known-gender Computer Science subset rather than automatically generalized to all Computer Science publications.

A related limitation is that no country-specific or language-specific validation of the gender-inference procedure was conducted. In particular, the present thesis does not quantify whether the selected threshold produces different error rates across naming contexts in which name-based gender inference is known to be more difficult. Similarly, no alternative-threshold sensitivity analysis was performed. The effect of changing the 0.7/0.3 decision thresholds on the final O/E estimates is therefore unknown. These issues are particularly relevant because a more permissive threshold can increase analytical coverage while simultaneously increasing the possibility of misclassification.

The first–last-author representation introduces an additional form of simplification. Intermediate authors are not incorporated into the paper-level gender category, and the interpretation of first and last authorship differs across fields and collaboration conventions. The MM, MW, WM, and WW categories should therefore be interpreted narrowly as inferred first–last author configurations rather than complete representations of research-team gender composition.

Taken together, these limitations mean that the strong cross-baseline directional stability observed for some categories, particularly WW, should not be interpreted as eliminating uncertainty associated with gender measurement. The robustness analysis varies the expected-citation model while holding the gender operationalization fixed. Errors or systematic missingness in that operationalization can therefore propagate across all baseline families.

7.5.2 Taxonomy, Network Coverage, and Future-Informed Co-Citation

A second group of limitations concerns the representations used to define topical and network-based similarity. The topical analyses rely on the OpenAlex taxonomy rather than on direct semantic comparison of full paper content. This provides a standardized and scalable hierarchy across the corpus, but taxonomy membership remains an imperfect proxy for substantive citation relevance.

The analysis primarily uses the OpenAlex primary topic and its associated subfield, field, and domain hierarchy. Papers can address multiple research problems simultaneously, and two papers assigned to different primary topics may nevertheless be closely related. Conversely, papers sharing a broad taxonomy category need not be direct intellectual substitutes. Classification errors or changes in the taxonomy can therefore affect both the structural subfield baselines and the edge-level topical-relevance analyses.

The taxonomy-weighted specification introduces an additional modeling choice. Weights of 1.00, 0.75, 0.50, and 0.25 are assigned to progressively more distant shared taxonomy levels. These values provide an interpretable graded ordering, but they are not empirically estimated measures of citation relevance. No systematic sensitivity analysis over alternative weighting schemes is included. The taxonomy-weighted O/E ratios should therefore be understood as results under this particular operationalization of hierarchical proximity rather than as uniquely determined relevance-adjusted estimates.

The bibliographic-coupling and co-citation representations introduce more substantial coverage constraints. Both networks are intentionally sparse top- K approximations and require sufficient observed reference or citation information before a usable neighborhood can be constructed. In the network EDA, the retained bibliographic-coupling representation contains nodes for approximately 36% of known-gender Computer Science papers, while the co-citation representation covers fewer than 8%. The effective coverage of the N1 and N2 expected-citation baselines is smaller still: N1 produces valid expectations for 347,023 papers, or 32.09% of the known-gender CS population, whereas N2 covers 70,803 papers, or only 6.55%.

Coverage is also unequal across gender categories. Under N1, valid baseline coverage ranges from approximately 32–33% for MM, MW, and WM papers to 25.95% for WW papers. Under N2, coverage declines from 7.07% for MM to 4.42% for WW. The network-baseline results therefore refer to selected subpopulations whose composition differs from the complete known-gender Computer Science corpus. In particular, an O/E ratio closer to parity under N2 cannot be interpreted as if it had been estimated on the same population as a corpus-wide direct baseline.

The same issue affects network-conditioned edge randomization. Bibliographic coupling provides valid replacement sets for approximately 31–32% of distinct cited targets, whereas co-citation provides them for only about 8–9%. At the same time, the corresponding co-citation edge coverage is much higher, at approximately 38–40%. This discrepancy indicates that the relatively small set of co-citation-covered targets receives a disproportionate share of observed citations. Network-conditioned results are consequently weighted toward papers with sufficient citation activity to obtain usable local neighborhoods.

The sparse network construction itself also introduces approximation. Full pairwise bibliographic-coupling and co-citation construction is computationally infeasible at this corpus scale, so candidate pairs are generated using MinHash locality-sensitive hashing and only a limited number of strong neighbors are retained. The resulting networks deliberately omit many weak relationships. Choices such as minimum activity and overlap thresholds, MinHash signature length, LSH thresholds, and top- K neighborhood sizes therefore influence which local comparisons are available. These parameters were selected as pragmatic settings for constructing computationally tractable sparse representations rather than through an exhaustive hyperparameter optimization, so alternative sparsification settings could change network coverage and local neighborhood composition.

These choices make the analysis tractable but mean that N1, N2, and the associated network features represent the retained sparse neighborhoods rather than complete bibliographic-coupling or co-citation graphs.

Co-citation requires an additional temporal qualification. Unlike bibliographic coupling, which is derived from reference lists fixed at publication, co-citation relationships emerge through later citing activity. A paper can obtain a strong co-citation neighborhood only after subsequent literature has repeatedly cited it together with other work. Co-citation is therefore future-informed relative to earlier citation events.

This future-informed character is visible directly in network coverage. Co-citation coverage falls sharply for recent publication cohorts because recent papers have had less time to accumulate shared citers. Consequently, N2, M4, M5, and co-citation-conditioned randomization should be interpreted as retrospective analyses of citation structure rather than as citation expectations that could have been constructed at publication time.

The strong predictive performance of co-citation features must be interpreted in the same light. Co-citation structure is produced partly by the citation system whose outcomes are being modeled, so its ability to account for citation-count heterogeneity is not directly comparable to that of backward-looking variables available at publication. The movement of O/E ratios towards parity under co-citation conditioning therefore does not show that a prospective model would have predicted those citation outcomes or that co-citation structure causally accounts for the observed differences.

These limitations do not make the citation-derived network analyses uninformative. Rather, they constrain their estimands. Bibliographic coupling provides a broader backward-looking comparison based on shared research foundations, while co-citation provides a more selective retrospective comparison based on later scholarly association. Their disagreement is therefore informative, but their numerical estimates should not be compared without simultaneously considering coverage and temporal interpretation.

7.5.3 Scope of the Dataset and Filtering Choices

The final group of limitations concerns the scope of the bibliographic corpus and the filtering choices used to construct the analytical citation graph. Although the dataset contains several million publications and citation relationships, it is not a complete representation of all scholarly literature or all references made by the included papers.

Most importantly, the analysis uses corpus-internal citation links. A reference contributes a citation edge only when the cited paper can also be resolved as a paper in the underlying dataset. Citations to publications outside the corpus are therefore absent. As a result, the in-degree, out-degree, citation-lag, bibliographic-coupling, and co-citation quantities used in the thesis represent citation activity observed within the corpus rather than complete bibliographic histories.

This limitation can affect papers unevenly. Coverage of external literature may vary across publication periods, disciplines, venues, languages, and document types. A paper with few observed internal references is not necessarily a paper with a short original bibliography. Similarly, an apparently uncited paper in the analytical graph may have received citations from works that are not represented in the dataset. The resulting O/E estimates are therefore conditional on the observed citation system.

The definition of Computer Science is also dataset-dependent. Papers are classified as Computer Science according to the field associated with their OpenAlex primary topic. This provides a reproducible and scalable field definition but cannot capture every interdisciplinary publication equally well. Some work relevant to Computer Science may be assigned primarily to another field, while papers assigned to Computer Science may participate strongly in citation communities

outside the field. The use of both All→CS and CS→CS citation slices partly addresses the latter issue, but it does not remove classification uncertainty in the field boundary itself.

The observation period creates additional right-censoring for recent papers. The corpus extends through 2024, meaning that papers near the end of the observation window have had substantially less time to accumulate citations than older papers. Temporal conditioning and citation-lag analyses account for this issue in several baseline families, but the available data cannot reveal future citations that have not yet occurred. This is especially consequential for co-citation, whose network structure requires subsequent citation events to form.

Venue information also has unequal quality. Venue name and type are available throughout the retained Computer Science corpus, but recognized venue-rank information is available for only 57.02% of Computer Science papers. Papers without a recognized rank are retained rather than removed, avoiding a ranked-only selection of the corpus, but the resulting unavailable-rank category combines heterogeneous publications for which external rank information could not be assigned. Venue-rank conditioning should therefore be interpreted as a coarse structural adjustment rather than a complete measure of venue prestige or visibility.

The citation graph is additionally affected by deliberate filtering choices. Duplicate edges and paper-identifier self-links are removed, followed by author-overlap self-citations and citations originating from prior coauthors of the cited paper’s authors. The latter filter is time-aware, using only collaboration relationships established before the citing paper was published. Papers that become isolated after citation filtering are then removed from the final analytical universe.

These filters are intended to reduce direct self-citation and citation relationships arising from established collaboration networks, but they are necessarily operational definitions. A citation between prior collaborators can still represent a substantively independent scholarly citation, while a citation without recorded author overlap or prior coauthorship can arise through other forms of social proximity not represented in the data. Furthermore, the accuracy of these filters depends on the completeness and correctness of author identifiers and collaboration histories.

No separate before-and-after sensitivity analysis was performed for the self-citation and prior-coauthor filters. All principal results are computed on the final filtered graph, so the thesis cannot quantify how much the reported O/E ratios would change if these edges were retained or if an alternative collaboration-based exclusion rule were used. The filtering choices should therefore be understood as part of the definition of the analytical citation system rather than as empirically validated corrections for all forms of social dependence.

Removing papers that become isolated after edge filtering introduces a related selection. The final analytical corpus excludes papers with no remaining connection to the filtered internal citation graph. This makes the subsequent network analyses well defined, but the results do not describe publications that remain completely disconnected after the filtering procedure. The final estimand is therefore conditional on participation in the retained citation network.

Finally, several relevant characteristics of papers and authors are not observed or not incorporated into the expected-citation models. Examples include institutional prestige, academic seniority, geographic location, funding, paper quality, detailed semantic content, and broader social networks. The GLM and stratified baselines condition on a substantial set of observable structural characteristics, but residual O/E differences may still be associated with omitted variables. This is another reason why the remaining deviations should not be interpreted causally.

Overall, the limitations define the population and comparison structures to which the conclusions apply. The thesis provides evidence about gender-associated citation imbalance among known-gender Computer Science papers represented in the final filtered, corpus-internal citation graph. The consistency of several directional patterns across multiple baselines strengthens confidence that they are not artifacts of one particular expected-citation specification, but it does not eliminate uncertainty arising from gender inference, taxonomy assignment, selective network coverage,

corpus boundaries, or filtering decisions.

7.6 Future Work

The limitations and robustness results of this thesis suggest several directions for extending the analysis. The most important concern the measurement of author gender and the construction of richer models of citation opportunity. Future work should therefore focus both on improving the underlying author metadata and on developing citation models that more explicitly represent the temporal, semantic, and network processes through which scholarly attention accumulates.

7.6.1 Improved Author and Gender Data

The most immediate extension would be to improve the quality and coverage of author-gender information. The present analysis relies on name-based inference and requires both the first and last author to receive sufficiently confident binary labels. This provides a scalable operationalization but excludes almost 30% of Computer Science papers from the main gender-specific analyses and cannot represent gender identities outside the inferred male–female classification.

Future work could reduce this limitation by combining name-based inference with additional author-level metadata. Where available and ethically appropriate, self-reported gender information would provide a substantially stronger basis for classification than inference from names alone. Persistent author identifiers and curated researcher profiles could also help distinguish authors more reliably and support the integration of information from multiple bibliographic sources.

A particularly important extension would be to evaluate gender-inference performance across linguistic and geographic naming contexts. The current procedure applies the same probability thresholds across a globally distributed corpus, even though classification reliability may differ substantially between naming systems. Future analyses could therefore report inference coverage and, where validation data are available, classification accuracy by country, language, or naming context. This would help determine whether the known-gender analytical subset systematically underrepresents particular author populations.

Sensitivity to the inference threshold should also be examined directly. Repeating the principal O/E analyses under more conservative thresholds would quantify the trade-off between classification coverage and confidence. If the main directional findings remain stable when the known-gender sample is restricted to higher-confidence assignments, this would provide an additional form of robustness that is not assessed in the present thesis.

More generally, uncertainty in gender inference could be propagated into the citation analysis rather than resolved entirely through hard classification. Instead of assigning every author deterministically to M, W, or U, future models could retain probabilistic gender information and propagate this uncertainty when aggregating observed and expected citation counts. Such an approach would avoid treating assignments immediately above and below a fixed threshold as equally certain.

The first–last-author operationalization could also be extended. Future work could examine the full author team, alternative authorship positions, or discipline-specific authorship conventions. This would make it possible to distinguish whether the patterns observed for MM, MW, WM, and WW papers also appear under broader measures of team composition. Such analyses would be particularly useful in Computer Science, where authorship order and team size vary across subfields and publication cultures.

7.6.2 Richer Models of Citation Behavior

A second direction concerns the modeling of citation opportunity itself. The results of this thesis show that the estimated magnitude of citation imbalance depends strongly on which dimensions of the citation environment are included in the expected-citation model. Future work could therefore move from a set of comparative baselines toward richer models that represent several components of citation behavior simultaneously.

One extension would be to improve the representation of topical similarity. The present analysis uses the OpenAlex taxonomy because it is scalable, interpretable, and consistently available across the corpus. However, taxonomy assignments provide only a coarse proxy for substantive similarity. Future work could combine hierarchical taxonomy information with semantic representations derived from titles, abstracts, or full text. Multilingual embedding models could be particularly useful for constructing continuous measures of similarity while reducing some of the limitations associated with language-specific text processing.

Temporal modeling could also be made more explicit. The citation-lag analysis used here compares citation outcomes within predefined lag buckets, while the randomized reference-drawing models use temporally eligible candidate sets. A richer approach could model citation hazard or citation intensity continuously over time, allowing the probability of receiving a citation to depend jointly on paper age, prior visibility, topic, venue, and network position. Such models could distinguish more precisely between early citation accumulation, sustained recognition, and long-term scientific memory.

The strong contribution of co-citation features to predictive fit also motivates further investigation of temporally valid network representations. The present co-citation network is intentionally interpreted retrospectively because it uses later citation activity. Future research could construct time-indexed co-citation networks using only information available before each focal citation event. Although substantially more computationally demanding, such a design would make it possible to compare retrospective and prospectively available co-citation structure directly.

Bibliographic-coupling models could likewise be extended beyond the sparse top- K neighborhoods used here. Alternative similarity measures, normalized overlap scores, adaptive neighborhood sizes, or approximate graph methods could test how sensitive the resulting expectations are to the network construction parameters. Similar sensitivity analyses could be performed for the MinHash and LSH thresholds used to construct the sparse network representations.

A further extension would be to model cumulative visibility more explicitly. The randomized results show that weighting citation opportunities by prior citation accumulation materially changes the expected gender distribution. Future models could separate several forms of visibility, including prior citations, venue exposure, author reputation, institutional visibility, and network centrality. Modeling these processes jointly could clarify which components of cumulative advantage are most strongly associated with the movement of O/E ratios across baseline families.

The count models could also be generalized. The Poisson GLMs used in this thesis provide a scalable common framework, but citation counts are strongly overdispersed and contain a substantial mass at zero. Negative-binomial, hurdle, or zero-inflated specifications could therefore be compared with the Poisson models. More flexible machine-learning models could additionally be used as predictive benchmarks, provided that their improved fit is evaluated separately from the interpretability of the resulting expected-citation estimates.

Another useful extension would be to quantify uncertainty around the cross-baseline results. The present analysis focuses primarily on descriptive O/E ratios and directional robustness. Bootstrap procedures, repeated randomization, or other resampling approaches could provide uncertainty intervals around individual baseline estimates and help distinguish substantively meaningful differences from variation induced by sampling or randomized reference construction.

Finally, future work could move beyond descriptive robustness toward explicit causal questions. Doing so would require a different research design from the one used here. Temporal information about authors, institutions, collaborations, visibility, and citation exposure could support causal models that distinguish confounding variables from mediating mechanisms. Such work could investigate not only whether citation outcomes differ across inferred gender categories, but which processes contribute to the emergence and persistence of those differences.

Taken together, these extensions would broaden the present framework in two directions. Improved author data would reduce uncertainty in the demographic measurement itself, while richer temporal, semantic, and network models would refine the definition of citation opportunity. Both are necessary for moving from robust descriptive quantification toward a more complete understanding of how gender-associated citation differences arise and evolve within Computer Science.

8. Conclusion

8.1 Answers to the Research Questions

This thesis examined whether gender citation imbalance in Computer Science can be quantified consistently across multiple definitions of citation opportunity and multiple citation-derived representations. The empirical results show that the answer depends on distinguishing robustness of *direction* from invariance of *magnitude*. The estimated size of the imbalance changes substantially across baselines, but several directional patterns remain stable across structural, topical, network-based, regression-based, and randomized specifications. This section summarizes the answers to the main research question and the seven subquestions.

8.1.1 Main Research Question

The main research question was:

Can gender citation imbalance in Computer Science be robustly quantified using multiple citation-derived network representations and topical similarity signals?

The results support a qualified affirmative answer. Gender citation imbalance can be quantified robustly across multiple citation-derived representations, but the robustness is primarily directional rather than numerical. There is no single observed/expected ratio that can be interpreted as an invariant estimate of the magnitude of citation imbalance. Instead, the estimated deviation from expectation depends on how citation opportunity is defined, including the temporal, topical, venue, network-neighborhood, prior-visibility, and model-based information incorporated into the baseline.

The strongest cross-baseline consistency is observed for WW papers. Across the 19 principal specifications included in the final cross-baseline comparison, WW papers remain below expectation in every specification in both the All→CS and CS→CS citation slices. In the primary All→CS slice, their O/E ratios range from 0.649 under the global paper-share baseline to 0.963 under the randomized reference-drawing specification closest to parity, with a median of 0.762 across the 19 specifications. The corresponding CS→CS range is 0.638-0.947, with a median of 0.751.

The other categories also exhibit substantial directional stability, although with greater sensitivity to the null specification. MM papers are above expectation in 18 of 19 principal specifications in each citation slice. MW papers are below expectation in 17 of 19 specifications in both slices. WM papers are below expectation in 17 of 19 All→CS specifications and 18 of 19 CS→CS specifications. The exceptions are concentrated in randomized specifications that incorporate prior citation visibility, showing that conclusions for these categories are more dependent on the construction of expected citation opportunity.

The multi-representation design therefore leads to two complementary conclusions. First, the direction of the deviation is highly stable for some gender categories, particularly WW, and remains visible under direct, topical, bibliographic-coupling, co-citation, GLM, and randomized specifications. Second, the magnitude of the deviation is not invariant: conditioning on richer structural or network information can move O/E ratios substantially towards parity and, for some categories, can reverse their direction. Gender citation imbalance in Computer Science can therefore be robustly quantified as a pattern of cross-baseline deviations from expected citation reception, but not as one universal numerical effect independent of the chosen citation representation or expected-citation model.

8.1.2 Subquestions

SQ1: Operationalization feasibility. The first subquestion asked whether author gender could be operationalized at the paper level using first- and last-author categories, and what pro-

portion of the Computer Science corpus would be excluded because of gender-inference limitations.

The paper-level operationalization was feasible for a substantial majority, but not for the complete Computer Science corpus. Of the 1,542,691 Computer Science papers in the final analytical corpus, 1,081,363 papers, or 70.10%, had known gender classifications for both the first and last author and could therefore be assigned to MM, MW, WM, or WW. The remaining 29.90% could not enter the gender-specific O/E analyses. Coverage also varied over time, falling to 61.88% for papers published between 2020 and 2024. The operationalization is therefore sufficiently complete for large-scale analysis, but all gender-specific findings apply to the known-gender subset rather than to the full Computer Science literature.

SQ2: Direct citation evidence. The second subquestion asked whether gender citation imbalance could be identified in the direct citation setting after accounting for temporal, venue, and topical structure, both in citation reception and in descriptive citing-side patterns.

The direct citation results show a persistent difference between observed and expected citation reception. Under the unconditioned D0 baseline in the All→CS slice, the O/E ratios were 1.093 for MM, 0.820 for MW, 0.734 for WM, and 0.649 for WW. Conditioning on publication decade and additional structural characteristics substantially reduced these differences. Under D3, which conditions on publication decade and venue-rank stratum, the corresponding ratios were 1.031, 0.949, 0.907, and 0.762. Thus, the deviations were substantially smaller under the more structured direct baselines, while their direction remained unchanged.

The CS→CS analysis produced the same qualitative pattern. Under D3, the ratios were 1.028 for MM, 0.968 for MW, 0.916 for WM, and 0.751 for WW. The direct-citation result therefore persists when the citing population is restricted to Computer Science.

Citing-side reference composition also differed across first-last author gender categories. However, these row-normalized flows do not control for the gender composition of the available citation opportunity set and are therefore descriptive rather than evidence of gender-specific citation preference. Taken together, the direct-citation analyses identify persistent observed-versus-expected differences in citation reception while separating these results from unconditional differences in citing-side composition.

SQ3: Topical relevance sensitivity. The third subquestion asked how the measurement of citation imbalance changes when citations are restricted or weighted according to topical relevance in the OpenAlex taxonomy.

Topical conditioning changes the magnitude of the estimated imbalance but does not remove the main directional pattern. In the All→CS analysis, the WW O/E ratio ranges from 0.736 for same-topic citations to 0.693 for cross-subfield citations. Under the taxonomy-weighted specification, which gives progressively lower weight to more distant citation relationships, the WW ratio is 0.707. The corresponding taxonomy-weighted ratios for MM, MW, and WM are 1.037, 0.958, and 0.893.

Results are closely reproduced in the CS→CS slice. The taxonomy-weighted ratios are 1.035 for MM, 0.965 for MW, 0.899 for WM, and 0.713 for WW. Topical relevance therefore matters for the estimated magnitude of the deviation, but restricting attention to topically closer citation relationships does not eliminate the below-expectation citation reception of the female-represented categories, particularly WW.

SQ4: Co-citation consistency. The fourth subquestion asked whether gender citation imbalance could also be identified within co-citation neighborhoods representing later community association between papers and potentially capturing aspects of canon formation.

The co-citation results identify substantially smaller deviations from parity than most structural and bibliographic-coupling specifications, but the direction remains below expectation for WW. Under the N2 co-citation neighborhood baseline, the All→CS O/E ratios are 1.012 for MM, 0.945 for MW, 0.964 for WM, and 0.927 for WW. The corresponding CS→CS values are 1.012, 0.952, 0.958, and 0.929.

The GLM and network-randomization analyses show a similar movement toward parity when co-citation information is introduced. The co-citation-conditioned edge randomization produces All→CS ratios of 1.003, 0.994, 1.006, and 0.930, placing all four categories considerably closer to parity than under bibliographic-coupling randomization.

These results must be interpreted conditional on the substantially more selective coverage of the co-citation representation. N2 provides valid neighborhood expectations for 70,803 known-gender CS papers, or 6.55% of the known-gender CS population, and co-citation is future-informed because its neighborhoods depend on later citing activity. The co-citation results therefore support the presence of a residual WW deviation within the covered subset, while also showing that its estimated magnitude is substantially smaller under this representation.

SQ5: Bibliographic-coupling consistency. The fifth subquestion asked whether gender citation imbalance could be identified within bibliographic-coupling neighborhoods representing shared research foundations.

The bibliographic-coupling results preserve the main directional pattern more strongly than the co-citation results. Under N1, the All→CS O/E ratios are 1.049 for MM, 0.898 for MW, 0.822 for WM, and 0.803 for WW. The CS→CS values are similarly 1.045, 0.915, 0.838, and 0.799. The baseline covers 347,023 known-gender CS papers, corresponding to 32.09% of the known-gender CS population.

Bibliographic-coupling-conditioned edge randomization produces the same broad ordering, with All→CS ratios of 1.030, 0.939, 0.891, and 0.811. Incorporating prior citation visibility into the replacement weights moves these estimates closer to parity, particularly for MW and WM, but WW remains below expectation at 0.829. The bibliographic-coupling analyses therefore show that the principal directional differences remain visible when citation opportunity is defined relative to papers sharing similar reference foundations, although their magnitude depends on whether prior visibility is also incorporated.

SQ6: Long-memory consistency. The sixth subquestion asked whether gender citation imbalance remains visible across citation-lag buckets corresponding to shorter- and longer-term citation attention.

The results show persistent directional differences across citation ages, but the magnitude varies with citation lag. In the broad All→CS groups, WW O/E ratios are 0.717 for citations with a lag of 0-5 years, 0.681 for 6-10 years, and 0.668 for citations more than ten years old. In the finer analysis, WW remains below expectation in every lag bucket, with ratios of 0.738 for 0-2 years, 0.709 for 3-5 years, 0.681 for 6-10 years, 0.687 for 11-20 years, and 0.611 for citations more than twenty years old.

The CS→CS analysis produces the same overall conclusion. WW ratios remain below one across all fine-grained lag groups, ranging from 0.708 in the 0-2-year bucket to 0.646 in the 20+ year bucket. Long-memory citation imbalance can therefore be identified across citation ages rather than being confined to a single period after publication, although the estimated magnitude differs between lag groups.

SQ7: Robustness across methods. The seventh subquestion asked how stable the estimated citation imbalance is across citation slices, network representations, expected-citation baselines, GLM specifications, and randomized robustness checks.

The final cross-baseline comparison shows substantial directional stability together with meaningful magnitude sensitivity. Across 19 principal specifications in All→CS, MM is above expectation in 18, MW is below expectation in 17, WM is below expectation in 17, and WW is below expectation in all 19. In CS→CS, the corresponding counts are 18 of 19 above expectation for MM, 17 of 19 below for MW, 18 of 19 below for WM, and 19 of 19 below for WW.

The definition of the citing population has comparatively little effect on these conclusions. Across the 19 specifications, the mean absolute difference between the All→CS and CS→CS O/E ratios is 0.002 for MM, 0.015 for MW, 0.009 for WM, and 0.011 for WW.

Baseline construction has a much larger effect on magnitude. Structural baselines and bibliographic-coupling specifications generally produce larger deviations from parity, while co-citation neighborhood, co-citation GLM, and co-citation-conditioned randomization specifications produce substantially smaller deviations. The randomized reference-drawing analysis additionally shows that prior-citation weighting can reverse the usual direction for MM, MW, and WM under the broad all-prior candidate pool. These reversals indicate that the findings for these categories are sensitive to the assumptions used to construct the null citation process.

WW differs from the other categories in this respect. Although its estimated O/E ratio changes considerably across specifications, it remains below expectation under every principal baseline in both citation slices. The robustness analysis therefore supports strong directional stability for WW, high but not complete directional stability for MM, and greater specification sensitivity for MW and WM. More generally, robustness across methods should be interpreted as consistency of the dominant direction across alternative expected-citation models rather than equality of the estimated O/E magnitude.

8.2 Contributions

This thesis contributes to the study of gender citation imbalance in Computer Science by combining a broad empirical analysis with a multi-representation framework for defining expected citation opportunity. Rather than treating citation imbalance as a quantity that can be estimated from one normalization alone, the thesis evaluates how strongly the resulting conclusions depend on the structural, topical, temporal, and network information incorporated into the comparison.

First, the thesis provides a large-scale empirical analysis of gender citation imbalance across a multi-million-paper bibliographic corpus, with the substantive analysis focused on Computer Science papers. The analysis distinguishes citations received from the broader observed literature (All→CS) from citations originating specifically within Computer Science (CS→CS), allowing the principal citation-reception patterns to be evaluated under both a broad and a discipline-restricted definition of the citing population. This large-scale setting also provides the basis for evaluating whether the same directional conclusions persist across multiple citation-derived representations and expected-citation baselines.

Second, the thesis develops a common observed/expected framework in which multiple baseline families can be compared using the same gender-aggregated quantity. The framework includes global and structurally conditioned direct citation baselines, taxonomy-based topical specifications, bibliographic-coupling and co-citation neighborhood expectations, regression-based expected citation models, citation-lag analyses, and randomized reference models. This makes it possible to distinguish robustness of the *direction* of citation imbalance from stability of its numerical magnitude.

Third, the thesis extends the use of established citation-derived network representations to the

measurement of citation disparity. Bibliographic coupling and co-citation have previously been used primarily to characterize scientific structure, research fronts, and topical organization. In this thesis, they are additionally used as alternative definitions of local scholarly comparability. The resulting analyses show that the choice of network representation materially affects the estimated magnitude of citation imbalance: bibliographic-coupling specifications preserve comparatively larger residual deviations, whereas co-citation-based specifications generally move the observed/expected ratios substantially closer to parity. This demonstrates that network representation is consequential not only for science mapping but also for the construction of expected citation baselines.

Fourth, the thesis incorporates topical and temporal context beyond standard paper-level structural controls. The OpenAlex hierarchy is used both to separate citation relationships by topical proximity and to construct a graded taxonomy-weighted relevance specification. Citation-lag analysis further distinguishes short-, medium-, and long-term citation reception. These analyses show that citation imbalance varies across scholarly contexts while remaining directionally persistent for some gender categories.

Fifth, the thesis evaluates expected citation estimates under several alternative robustness mechanisms rather than relying solely on deterministic normalization. Randomized reference drawing tests the consequences of temporal eligibility and prior citation visibility, while network-conditioned edge randomization evaluates alternative targets within local bibliographic-coupling and co-citation neighborhoods. These analyses reveal that prior visibility and candidate-set construction can materially alter expected gender composition and, for some categories, even reverse the direction of the observed/expected comparison. The resulting cross-baseline analysis therefore provides a direct empirical assessment of which conclusions are stable and which are specification-sensitive.

Sixth, the model-based analysis shows how citation-derived network information contributes to paper-level expected citation estimates. Structural, direct-citation, bibliographic-coupling, and co-citation feature blocks are compared within a common regularized Poisson framework, with gender excluded from model fitting. The substantial improvement obtained after introducing co-citation features, together with the corresponding holdout validation, demonstrates the strong retrospective information contained in later community association while also highlighting the interpretive distinction between predictive fit and prospective citation opportunity.

Finally, the thesis provides a scalable and reproducible workflow for constructing and evaluating citation-derived representations on a multi-million-paper corpus. The pipeline includes citation-edge filtering, time-aware removal of prior-coauthor citations, sparse MinHash-LSH construction of bibliographic-coupling and co-citation neighborhoods, coverage diagnostics, common observed/expected aggregation, model evaluation, and cross-baseline comparison. Coverage is treated as part of the estimand rather than as a purely computational issue, which is particularly important for the substantially more selective co-citation representation.

Taken together, these contributions shift the analysis from asking only whether gender citation imbalance can be detected in Computer Science to asking how robust that conclusion is to alternative definitions of citation opportunity. The central contribution is therefore both empirical and methodological: the thesis identifies persistent gender-associated citation differences while showing explicitly that their estimated magnitude is conditional on the representation and baseline used to define what citations would be expected.

8.3 Final Remarks

This thesis examined gender citation imbalance in Computer Science through multiple definitions of scholarly similarity and expected citation opportunity. The results show that citation imbalance cannot be summarized adequately by a single raw citation comparison or one universal normalized effect size. Its estimated magnitude depends on the temporal, topical, structural, and network

information incorporated into the reference model.

At the same time, this dependence does not imply that the results are arbitrary. Several directional patterns remain stable across substantially different specifications, with the strongest consistency observed for WW papers. The persistence of these patterns across direct, topical, network-based, regression-based, and randomized analyses indicates that the main conclusions are not confined to one particular representation of the citation system.

More broadly, the thesis illustrates that the measurement of citation disparities is inseparable from the definition of what constitutes a reasonable citation opportunity. Different baselines preserve different properties of scholarly communication and therefore answer different versions of the same empirical question. Treating these alternatives comparatively, rather than selecting one baseline as uniquely correct, provides a more transparent account of both the robustness and the uncertainty of the resulting conclusions.

The main contribution of this work is therefore not a single numerical estimate of gender citation imbalance, but a framework for evaluating how stable such estimates are across alternative representations of scholarly context. Within the scope and limitations of the available data, the analysis shows that gender-associated differences in citation reception remain visible across multiple views of the Computer Science citation system, while also demonstrating that their magnitude is conditional on how expected citation opportunity is defined.

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A. Appendix

A.1 Code and Reproducibility

The code used to construct the analytical datasets, citation-derived network representations, expected-citation baselines, robustness analyses, and figures reported in this thesis is available in the accompanying GitHub repository:

`https://github.com/BlazejNowak123/citation-gender-imbalance`

The repository contains the final preprocessing and analysis notebooks together with the code required to reproduce the reported tables and figures from the processed data.